



EM481

Systems Simulation



EM481 SYSTEMS SIMULATION

ABET eCOURSEBOOK

AY2026

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UNITED STATES MILITARY ACADEMY
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SECTION I

COURSE ASSESSMENTS



UNITED STATES MILITARY ACADEMY
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TAB A AT26-1 COURSE ASSESSMENT



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

MADN-SE

15 December 2025

MEMORANDUM FOR COL Brendan Thompson, Engineering Management Program
Director, Department of Systems Engineering, United States Military Academy

SUBJECT: AT26-1 Course Assessment – EM481 Systems Simulation

1. The attached report summarizes the EM481 assessment and curriculum change actions for Academic Term (AT) 26-1.
2. POC for this action is the undersigned at thomas.lainis@westpoint.edu.

TOM LAINIS
LTC, SF
SE385 Course Director

Encls

1. Course Assessment
2. SDS Program Student Outcomes Crosswalk
3. Course Syllabus & Calendar
4. Course End Feedback (TBP)

DISTRIBUTION:

1 each EM481 Instructors, Program Director, and Department Head

Enclosure 1: Course Assessment**1. Historic Course Offerings and Enrollments.****a. Course offerings by Academic Term.**

AY17		AY18		AY19		AY20		AY21	
17-1	17-2	18-1	18-2	21-1	21-1	20-1	20-2	21-1	21-2
X	X	X	X	X	X	X	X	X	X
AY22		AY23		AY24		AY25		AY26	
22-1	22-2	23-1	23-2	22-1	22-2	23-1	23-2	26-1	26-2
X	X	X	X	X	X	X	X	X	P

b. Enrollments & Course Average.

Academic Term	EM481 Enrollment	EM481 Course Average
16-2	55	85.91%
17-1	25	85.59%
17-2	42	83.74%
18-1	30	84.26%
18-2	48	83.04%
19-1	62	89.61%
19-2	26	88.07%
20-1	50	86.40%
20-2	26	86.42%
21-1	62	89.41%
21-2	38	89.98%
22-1	53	87.05%
22-2	27	87.38%
23-1	55	89.12%
23-2	42	88.23%
24-1	56	86.83%
24-2	48	87.36%
25-1	61	85.96%
25-2	50	84.70%
26-1	75	87.73%
26-2	87**	

****Current Enrollment, as of 16 December 2025**

2. Assessment of Course Goals.

a. **Course Objectives and Student Outcomes.** The assessment values are based on a 0-4 scale. The Course Director's Assessment uses 4-point GPA increments. Appendix A contains the program, course, and graded event mapping used to derive the mapped averages scores.

Course Objective		Mapped Graded Event AVG	Course Director's Assessment
1	Understand the process of conducting a simulation study	3.73	3.33
2	Understand the application of closed-form analytical queuing methods when conducting a simulation study.	3.92	3.33
3	Demonstrate the ability to generate and test sequences of pseudo-random numbers and selected classes of random variates.	3.72	2.33
4	Collect, analyze, and determine the appropriate probability distribution for data used in a simulation study (input modeling analysis).	3.55	3.33
5	Demonstrate comprehension of the simulation modeling process and be able to construct a simulation using course software.	3.64	3.67
6	Verify and validate simulation models.	3.71	3.00
7	Evaluate and interpret the quantitative output of simulation experiments for both single system and multiple systems (output analysis).	3.56	3.33
8	Demonstrate the ability to design, conduct, and analyze simulation experiments using a simulation package.	3.64	3.33
9	Prepare and brief a written analysis of a design problem supported by simulation results	3.70	3.33

b. This assessment was based on dialogue with the other instructor throughout the semester, Cadet performance on graded events, and anecdotal feedback from Cadets. Unfortunately, the students' feedback was not available until after the course closeout.

3. Assessment of Course Content.

a. **Body of Knowledge.** The course objectives adequately cover the body of knowledge. This course teaches the tools and methods necessary to be proficient in discrete-event simulation theory and application. The course is split evenly between theory and application.

b. **Quality of the Assessment Mechanisms.** The end of course survey and the department's major-graded-event content approval process provide an adequate assessment tool for an impartial and effective assessment of the course objectives. However, new course-specific survey questions should be developed to address any gaps in the USMA and Department survey questions.

4. Assessment of Course Processes.

a. **Textbook.** The textbook sufficiently covers the material outlined in the course and lesson objectives.

(1) **Recommendation.** Continue using "*Simulation Using ProModel*" textbook. Remove "*Decision Making in Systems Engineering and Management, 2nd edition*" from the Instruction Memorandum, as it is no longer used in this course. Consider a supplement when discussing Welch's Confidence Interval Test and the Paired-t Test. The textbook description is slightly confusing and gives the Cadets a poor understanding of when they can use a Paired-t test.

b. **Software.** ProModel is a great tool to learn and apply discrete event simulation concepts. The coding is fairly straight-forward, allowing Cadets to quickly build simulations. Furthermore, the output analysis and statistical analysis packages that are associated with ProModel make data analysis and output analysis easier for the user.

(1) **Recommendation.** No Change.

c. **Graded Requirements.** We assessed Cadets via 13 graded events (ten individual and six group). Shown in the table below, the individual graded events comprised 80% of the total course points while the group assignments made up the balance.

Assignment	Type	Points	% of Overall Grade
Problem Set #1	Individual	80	8%
In-Class Quiz #1	Individual	30	3.0%
WPR #1	Individual	160	16.0%

Project #1A	Individual	160	16.0%
Project #1B			
Project #1C			
In-Class Quiz #2	Individual	30	3.0%
Graded Lab	Individual	150	15.0%
Project #2A	Group	200	20.0%
Project #2B			
Project #2C			
In Class Quiz #3	Individual	25	2.5%
WPR #2	Individual	165	16.5%
Total:		1000	100.0%

(1) **Recommendations.**

(a) **Problem Set.** No change.

(b) **Quizzes.** Sustain the format of the quizzes. Consider moving quizzes 2 and 3 from Canvas to Gradescope if short answer questions are going to be incorporated.

(c) **WPRs.** No Change.

(d) **Course Projects.** Sustain both projects. Consider removing the EXSUM from Project #2. This change may require updating Course Objective #8. Potential change: "Prepare a detailed analysis of a design problem supported by simulation results."

5. Assessment of Changes to the Current Academic Term (based on previous Course Assessments).

a. **Course Objectives.** No changes from AT25-2.

b. **Textbook.** No changes from AT25-2.

c. **Pedagogy.** No changes from AT25-2.

d. **Curriculum.**

(1) **Change.** In AT25-2, there was a separate lesson for Random Number & Random Variate Generation (Lesson 13). For AT26-1, this lesson was merged into the two prior lessons (Simulation Modeling I and II, Lessons 11-12). A second Output Viewer lesson was then added to the syllabus, which focused on using Output Viewer results to identify issues in a model and troubleshoot simulation logic.

(a) **Recommendation:** Sustain this change.

6. Proposed Changes for Next Academic Term.

a. **Update Lesson Examples.** Make small adjustments to lessons 25-29 to incorporate the “911 Emergency” and “Pallet” Scenarios. Since Cadets will already be familiar with these examples, they should be able to better understand the Output Analysis on single and multiple alternatives.

b. **Entity Flow Diagrams (EFD).** Present a more complicated EFD during the EFD Lesson and require Cadets to build a more complicated EFD for homework. This should improve the quality of the submitted EFDs for Project 1A.

Enclosure 2: EM Program Student Outcomes Crosswalk

		An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	An ability to communicate effectively with a range of audiences.	An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.
Course Objectives		1	2	3	4	5	6	7
1	Understand the process of conducting a simulation study	X	X		X		X	X
2	Understand the application of closed-form analytical queuing methods when conducting a simulation study.	X	X				X	X
3	Demonstrate the ability to generate and test sequences of pseudo-random numbers and selected classes of random variates.	X						X
4	Collect, analyze, and determine the appropriate probability distribution for data used in a simulation study (input modeling analysis).	X						X
5	Demonstrate comprehension of the simulation modeling process and be able to construct a simulation using course software.	X	X				X	X
6	Verify and validate simulation models.	X	X	X	X		X	X
7	Evaluate and interpret the quantitative output of simulation experiments for both single system and multiple systems (output analysis).	X				X	X	X
8	Demonstrate the ability to design, conduct, and analyze simulation experiments using a simulation package.	X	X		X	X	X	X
9	Prepare and brief a written analysis of a design problem supported by simulation results		X	X				X

ID	Course Objectives	Graded event(s) or question(s) on graded event(s) that map to the course objective								
		PS1	Quiz 1	Quiz 2	Quiz 3	WPR 1	WPR 2	Graded Lab	Project #1	Project #2
1	Understand the process of conducting a simulation study	X	X			X		X	X	X
2	Understand the application of closed-form analytical queuing methods when conducting a simulation study.	X	X			X				
3	Demonstrate the ability to generate and test sequences of pseudo-random numbers and selected classes of random variates.			X		X	X			
4	Collect, analyze, and determine the appropriate probability distribution for data used in a simulation study (input modeling analysis).							X	X	X
5	Demonstrate comprehension of the simulation modeling process and be able to construct a simulation using course software.			X	X			X	X	X
6	Verify and validate simulation models.				X		X			X
7	Evaluate and interpret the quantitative output of simulation experiments for both single system and multiple systems (output analysis).						X	X	X	X
8	Demonstrate the ability to design, conduct, and analyze simulation experiments using a simulation package.			X	X			X	X	X
9	Prepare and brief a written analysis of a design problem supported by simulation results								X	X

Course OBIs		PS1	Quiz 1	Quiz 2	Quiz 3	WPR 1	WPR 2	Graded Lab	Project #1	Project #2	0	0	0	0	0	0	0	AVG	CD Assessment
1	Understand the process of conducting a simulation study	4	4	3.766666667	3.24	3.61875	3.781818182											3.73	3.33
2	Understand the application of closed-form analytical queuing methods when conducting a simulation study.	4	4	3.766666667														3.92	3.33
3	Demonstrate the ability to generate and test sequences of pseudo-random numbers and selected classes of random variates.			3.766666667				3.606666667	3.8									3.72	2.33
4	Collect, analyze, and determine the appropriate probability distribution for data used in a simulation study (input modeling analysis).				3.24	3.61875	3.781818182											3.55	3.33
5	Demonstrate comprehension of the simulation modeling process and be able to construct a simulation using course software.				3.24	3.61875	3.781818182		3.8	3.74								3.64	3.67
6	Verify and validate simulation models.						3.781818182	3.606666667		3.74								3.71	3.00
7	Evaluate and interpret the quantitative output of simulation experiments for both single system and multiple systems (output analysis).				3.24	3.61875	3.781818182	3.606666667										3.56	3.33
8	Demonstrate the ability to design, conduct, and analyze simulation experiments using a simulation package.				3.24	3.61875	3.781818182		3.8	3.74								3.64	3.33
9	Prepare and brief a written analysis of a design problem supported by simulation results					3.61875	3.781818182											3.70	3.33

MADN-SE
SUBJECT: AT 26-1 Course Assessment – EM481 Systems Simulation

Enclosure 3: AT26-1 Syllabus, v2

EM481: Systems Simulation AT 26-1,v2						
LSN	Lesson Name	Reading	Problems	Lesson Objectives	Notes	
Intro	1 Introduction to Simulation	1.1-1.6 1.8-1.9 1.11	1.1 1.3	1. Knowledge: Define Simulation 2. Knowledge: Describe the purpose of simulation 3. Knowledge: Describe how to conduct a simulation study		
	2 Simulation Basics	2.1-2.7 3.1-3.2 4.1-4.2	3.1	1. Knowledge: Define System and Processing System 2. Knowledge: Describe what drives complexity 3. Knowledge: Describe the characteristics of a discrete event simulation 4. Knowledge: Define system elements and the types of simulation		
Technical	3 Random Behavior & Statistics	3.3 5.8.1-5.8.2 App A	5.5 5.12	1. Skill: Calculate basic descriptive statistics from a sample dataset 2. Knowledge: Describe how distributions model random behavior 3. Knowledge: Describe the difference between discrete and continuous random variables 4. Knowledge: Describe the characteristics of a Poisson Process 5. Knowledge: Describe the key properties of the Poisson and exponential distributions	PS#1 Assigned: 25 August	
	4 Modeling a System & Data Collection	5.1-5.12 (Skip 5.7.1, 5.7.2, 5.8.3) 6.1-6.3 6.4.1-6.4.4		1. Knowledge: Know the 6 Steps for gathering data 2. Knowledge: Understand the guidelines for gathering data, types of data, sources of data, what to do in the absence of data 3. Skill: Draw an effective and useful Entity Flow Diagram (EFD), using EM481 Format		
	5 Queuing Theory I	2.9-2.10	2.16 2.17	1. Knowledge: Describe a queue and utilization 2. Knowledge: Describe the purpose and limitations of analytical queuing models 3. Skill: Define a queue using Kendall's notation 4. Skill: Calculate evaluation measures of MM/1 queues 5. Knowledge: Describe Little's Law and general queuing relationships		
	6 Queuing Theory II	Handout	Handout	1. Skill: Calculate evaluation measures of G/G/1 and G/G/S queuing models	Install ProModel	
	7 Testing Input Data I *Quiz 1*	5.7.1-5.7.2		1. Skill: Determine the critical values for a statistical tests 2. Skill: Interpret a p-value 3. Knowledge: Define independence and homogeneity (identical distribution) 4. Knowledge: Describe the importance of testing for independence and homogeneity 5. Skill: Use scatter plots, runs tests, and autocorrelation to test for independence 6. Skill: Conduct visual test for homogeneity	In-Class Quiz #1	
	8 Testing Input Data II	5.8.3	5.19	1. Knowledge: Describe the importance of fitting a distribution to data 2. Skill: Determine distribution suitability using the Chi-Square Goodness of Fit Test 3. Knowledge: Describe the differences between the Kolmogorov-Smirnov (KS) and Anderson Darling (AD) Goodness of Fit Tests	PS#1 Due: 9 September	
	9 Data Analysis using Stat:Fit	5.10-5.11	L5 Appendix A	1. Knowledge: Describe how bounded and boundless distributions affect simulation 2. Skill: Use Stat:Fit to test for independence and homogeneity 3. Skill: Model discrete distributions using continuous distributions	Project#1 Assigned: 11 September	
	10 WPR 1			1. Task: Demonstrate mastery of course concepts from lessons 1-9.		
	11 Simulation Modeling I (LEAP)	Watch video!	L1 L2.1	1. Skill: Navigate the ProModel environment to build a discrete event simulation 2. Knowledge: Describe the LEAP Model Building Process 3. Skill: Use the Logic Builder to build commands in ProModel 4. Skill: Implement the ProModel command WAIT		
	12 Simulation Modeling II (LEAP & RNG)	3.4-3.6 8.2.1	3.6 3.8	1. Skill: Use the Logic Builder to build commands in ProModel 2. Skill: Define Random Number Generator (RNG) and random variate 3. Knowledge: Describe the desirable properties of a RNG 4. Skill: Generate RNs using a linear congruential generator (LCG) and determine if the LCG will have a full cycle 5. Skill: Generate random numbers using the Inverse Transformation Method		
Software-Assisted	13 Variables and Output Viewer			1. Knowledge: Describe the difference between simple average and time average statistics 2. Skill: Assign local and global variables in ProModel 3. Skill: Implement the ProModel commands REAL and INT 4. Skill: Interpret results in Output Viewer	Project #1A Due: 25 September	
	14 Attributes, Logic, & Routing		L4.4 L4.11.1 L6.1-6.2	1. Skill: Create attributes and user distributions 2. Skill: Assign attributes a value from a user distribution 3. Skill: Implement routing using attributes and logic 4. Skill: Implement the ProModel commands: CLOCK, IF THEN ELSE, ROUTE, and GRAPHIC 5. Skill: Use attributes and logic to gather statistics		
	15 Resource and Path Networks		L4.3 L12.2	1. Skill: Create and assign resources 2. Skill: Create and integrate path networks into a simulation 3. Skill: Implement the ProModel commands: MOVE FOR, MOVE WITH, GET, FREE, and USE		
	16 Output Viewer I			1. Skill: Use logic commands, variables, and attributes to generate required output data 2. Skill: Interpret results in Output Viewer		
	17 Batching		L4.7-4.9 L6.10 L6.12	1. Skill: Create package files and import graphics 2. Skill: Implement the ProModel commands: GROUP, UNGROUP, COMBINE, and ACCUM	Project #1B Due: 9 October	
	18 Attaching		L4.7-4.9 L6.10 L6.12	1. Skill: Implement the ProModel commands: JOIN, LOAD, UNLOAD, and ACCUM		
	19 Output Viewer II *Quiz 2*			1. Skill: Apply Output Viewer-created tables, plots, and reports to troubleshoot logic 2. Skill: Apply Output Viewer to answer situation-specific questions.	In-Class Quiz #2	
	20 ADV Routing Logic			1. Skill: Create logic that allows multiple conditions to be met 2. Skill: Create IF THEN statements specific to complex scenarios 3. Skill: Execute temporary batching and attaching commands across multiple locations	Project #1C Due: 20 October	
	21 Priorities, Preemption, & Pull		L6.7-6.8	1. Knowledge: Describe the purpose and use of priorities and preemption 2. Skill: Identify "pull systems" and model them in ProModel 3. Skill: Implement the ProModel commands: SEND, and ORDER	Project#2 Assigned: 23 October	
	22 Practice Lab			1. Skill: Use Excel and STAT:FIT to clean and analyze a given dataset, and identify a best fit distribution to represent the data in a simulation 2. Skill: Given a scenario, create a functional DES model in ProModel 3. Skill: Analyze output data to draw conclusions on the performance of a model		
	23 Graded Lab			1. Task: Demonstrate mastery of course concepts from lessons 11-22		
	24 Loops and Termination Logic		L4.11.2 L8.2-8.2.1	1. Skill: End simulations using defined criteria 2. Skill: Model balking and jockeying customer behavior 3. Skill: Implement the ProModel commands: DO WHILE, DO UNTIL, WHILE DO, and STOP	Project #2A Due: 6 November	
	25 Verification, Validation, Macros, & Scenarios		L7.2-7.3 L11.1-11.2	1. Knowledge: Describe the difference and purpose of validation & verification 2. Skill: Use validation and verification techniques 3. Skill: Implement ProModel commands: TRACE, PAUSE, and DEBUG 4. Skill: Simulate multiple scenarios in ProModel using Macros and Scenario features 5. Skill: Evaluate the output from multiple scenarios from ProModel Output Viewer		
Analysis	26 Output Analysis I *Quiz 3*	8.2.2-8.3		1. Skill: Construct confidence intervals using simulation results 2. Skill: Determine the appropriate number of replications to achieve desired error 3. Knowledge: Describe the implications of the Central Limit Theorem and replications in simulation	In-Class Quiz #3	
	27 Output Analysis II	8.4 8.6	L8.3	1. Knowledge: Describe the difference between terminating and non-terminating simulations 2. Knowledge: Describe the purpose of warm-up periods and their effect on non-terminating simulations 3. Skill: Determine the length of a warm-up period using visual methods and Welch Moving Average Method 4. Skill: Create independent observations from a non-terminating simulation using replications and the batch method		
	28 Output Analysis III	9.1-9.3	9.1a	1. Knowledge: Describe the purpose of hypothesis testing in comparing two simulation alternatives 2. Skill: Use Welch Confidence Interval to compare two alternatives 3. Skill: Use Paired-t Confidence Interval to compare two alternatives		
	29 Output Analysis IV	9.4-9.5		1. Skill: Use ANOVA to analyze simulation output of multiple alternatives 2. Skill: Use Fisher's Least Significant Difference Method to analyze simulation output of multiple alternatives	Project #2B Due: 20 November	
	30 Presentations I			1. Task: Deliver a group project brief that meets all required deliverables	Project #2C Due: 3 December	
	31 Presentations II & Course Review			1. Task: Deliver a group project brief that meets all required deliverables 2. Task: Conduct Course Review		
	32 WPR 2			1. Task: Demonstrate mastery of course concepts from lessons 11-31		

EM481: Systems Simulation							
Course Calendar,v2							
	Sunday	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY
August	3	4	5	6	7	8	9
	10	11 Reorgy Week	1-4 Reorgy Week	2-6 Reorgy Week	1-5 Reorgy Week	2-7 Reorgy Week	16
	17	1-1 Introduction to Simulation	2-1 Introduction to Simulation	1-2 Simulation Basics	2-2 Simulation Basics	2-3 Random Behavior & Statistics	B Ring Weekend
	24	1-3 Random Behavior & Statistics	2-4 Modeling a System & Data Collection	1-4 Modeling a System & Data Collection	2-5 Queueing Theory I	1-5 Queueing Theory I	B 30
	31	1 Labor Day (No Class)	2-6 Queueing Theory II	1-6 Queueing Theory II	2-7 Testing Input Data I *Quiz 1*	1-7 Testing Input Data I *Quiz 1*	B Kansas State (Away)
September	7	1-8 Testing Input Data II	2-8 Testing Input Data II	1-9 Data Analysis using Stat::Fit	2-9 Data Analysis using Stat::Fit	2-10 X-Hour	A 13
	14	1-10 WPR 1	2-11 WPR 1	1-11 X-Hour	2-12 Simulation Modeling I (LEAP)	1-12 Simulation Modeling I (LEAP)	F North Texas
	21	1-13 Simulation Modeling II (LEAP & RNG)	2-13 Simulation Modeling II (LEAP & RNG)	1-14 Variables and Output Viewer	2-14 Variables and Output Viewer	2-15 X-Hour	B 27
	28	1-15 Attributes, Logic, & Routing	2-16 Attributes, Logic, & Routing	1-16 Resource and Path Networks	2-17 Resource and Path Networks	1-17 X-Hour	B UAB (Away)
October	5	1-18 Output Viewer I	2-18 Output Viewer I	1-19 Batching	2-19 Batching	2-20 X-Hour	F/B Charlotte / Family Weekend
	12	1-13 Columbus Day (No Class)	1-20 Attaching	2-21 Attaching	1-21 Output Viewer II *Quiz 2*	2-22 Output Viewer II *Quiz 2*	B Tulane (Away)
	19	1-22 X-Hour	2-23 ADV Routing Logic	1-23 ADV Routing Logic	2-24 Priorities, Preemption, & Pull	1-24 Priorities, Preemption, & Pull	A 25
	26	1-25 Practice Lab	2-25 Practice Lab	1-26 Graded Lab	2-26 Graded Lab	2-27 Loops and Termination Logic	B Air Force (Away)
November	2	1-27 Loops and Termination Logic	2-28 Verification, Validation, Macros, & Scenarios	1-28 Verification, Validation, Macros, & Scenarios	2-29 Output Analysis I *Quiz 3*	1-29 Output Analysis I *Quiz 3*	F Temple
	9	1-30 Veteran's Weekend (No Class)	2-11 Veteran's Day (No Class)	2-30 No Class	1-30 Output Analysis II	2-31 Output Analysis II	A 15
	16	1-31 Output Analysis III	2-32 Output Analysis III	1-32 Output Analysis IV	2-33 Output Analysis IV	2-34 X-Hour	F Tulsa
	23	1-33 No Class	2-35 No Class	1-34 No Class	2-27 Thanksgiving (No Class)	2-28 Thanksgiving Weekend (No Class)	B UTSA (Away)
	30	1 No Class Day	1-35 No Class	2-36 No Class	1-36 Presentations I	2-37 Presentations I	A*/B 6
December	7	1-37 Presentations II & Course Review	2-38 Presentations II & Course Review	1-38 WPR 2	2-39 WPR 2	1-39 X-Hour	F Army-Navy (MD)
	14	1-40 Reading Day	2-16 TEE Week	1-17 TEE Week	2-18 TEE Week	1-19 TEE Week	2-20 TEE Week



UNITED STATES MILITARY ACADEMY
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TAB B

AT25-2 COURSE ASSESSMENT



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

12 May 2025

MEMORANDUM FOR COL Brandon Thompson, Engineering Management Program
Director, Department of Systems Engineering

SUBJECT: AT25-2 Course Assessment – EM481, Project Management

1. The attached report summarizes the EM481 assessment and curriculum change actions for Academic Term 2026-1.
2. EM481 Systems Simulation is an introductory, discrete-event simulation course that consists of a thoughtful blend of simulation theory, quantitative and analytical concepts necessary for simulation, and the development of simulation models using ProModel software.
3. I recommend the following primary changes for improvement to the course:
 - a. Recommend adding RNG to LSN 2 as it is very short.
 - b. Recommend adding mandatory Homework Assignments, specifically in ProModel, to force more reps on students.
 - c. Use the Graded Lab as an example for LSN 29 – Output Analysis I and eliminate the McDonalds example
 - i. – Start with the Graded Lab, run 5 replications, use those means to calculate a sample mean and standard deviation. Use that data to create a 95% CI, then show in ProModel what the 95% CI is. Visually show them this.
 - d. Break up Batching & Attaching Entities I and II to two separate and distinct lessons; Batching Entities (1 lesson) and Attaching Entities (1 lesson). Eliminate the Skittles example.
 - e. In ADV Output Viewer – specifically cover the “current value” for variables. Every single student missed that question in WPR 2.
 - f. POC for this action is the undersigned at james.macgibbon@westpoint.edu.

JAMES E. MACGIBBON
MAJ, EN/FA49
EM481 Course Director

Encls

1. Course Analysis
2. Lesson Objectives / Syllabus
3. EM481 Course Objective Assessment
4. EM Program Student Outcomes Crosswalk
5. Course End Feedback

Enclosure 1: Course Analysis – EM481 Systems Simulation

Section I – Course Description

1. Redbook Description.

a. Cadets learn and explore discrete event simulation techniques and tools used to analyze and improve complex systems. Applications include operations, transportation, manufacturing and logistics systems. Topics include functional modeling with functional flow diagrams, simulation theory, the modeling process, input data analysis, generation of random numbers, verification and validation of simulation models, experimental design, output analysis, and application using simulation software. The course concepts provide cadets the tools to evaluate military and civilian systems. Emphasis is placed on using simulation in the Systems Decision Process (SDP). Cadets demonstrate proficiency and develop communication skills through design projects and briefings. Cadets spend several lessons in a computer lab environment.

b. **Lessons.** 35 @ 75 min (2.50 Att/wk) LABS: 0 @ 0 min

c. **Special Requirements.**

d. **Offered** – Every Semester

2. Enrollment.

Academic Term	Enrollment
25-2	50
25-1	61
24-2	48
24-1	56
23-2	42
23-1	55
22-2	27
22-1	53
21-2	38
21-1	62
20-2	26
20-1	50
19-2	26
19-1	62
18-2	48
18-1	30

3. Current Course Objectives.

- (1) Understand the process of conducting a simulation study.
- (2) Understand the application of closed-form analytical queuing methods when conducting a simulation study.
- (3) Demonstrate the ability to generate and test sequences of pseudo-random numbers and selected classes of random variates.
- (4) Collect, analyze, and determine the appropriate probability distribution for data used in a simulation study (input modeling analysis).
- (5) Demonstrate comprehension of the simulation modeling process and be able to construct a simulation using course software.
- (6) Verify and validate simulation models.
- (7) Evaluate and interpret the quantitative output of simulation experiments for both single system and multiple systems (output analysis).
- (8) Demonstrate the ability to design, conduct, and analyze simulation experiments using a simulation package.
- (9) Prepare and brief a written analysis of a design problem supported by simulation results.

4. Textbooks.

- 1) Harrell, C., Biman, G., Gladwin, B. & Bowden, R. (2022). *Simulation Using ProModel, 4th edition*. San Diego, CA: Cognella Academic Publishing.
- 2) Parnell, G., Driscoll, P. & Henderson, D. (Eds.). (2011). *Decision Making in Systems Engineering and Management, 2nd edition*. Hoboken, NJ: John Wiley and Sons, Inc.

5. **Course Strategy.** Through this course, cadets will learn and explore discrete event simulation techniques used to analyze and improve complex systems. Applications for discrete event simulation include manufacturing, transportation, logistics, and service systems. Topics include the simulation study process, queuing theory, input data analysis, generation and testing of random numbers, simulation construction, verification and validation of simulation models, and output analysis. Cadets will demonstrate proficiency of lesson objectives while developing communication skills through graded events, a comprehensive course project, and an interactive classroom and lab environment. Upon completion of the course, cadets will be able to simulate and analyze moderately complex civilian and military systems.

6. **Graded Requirements.**

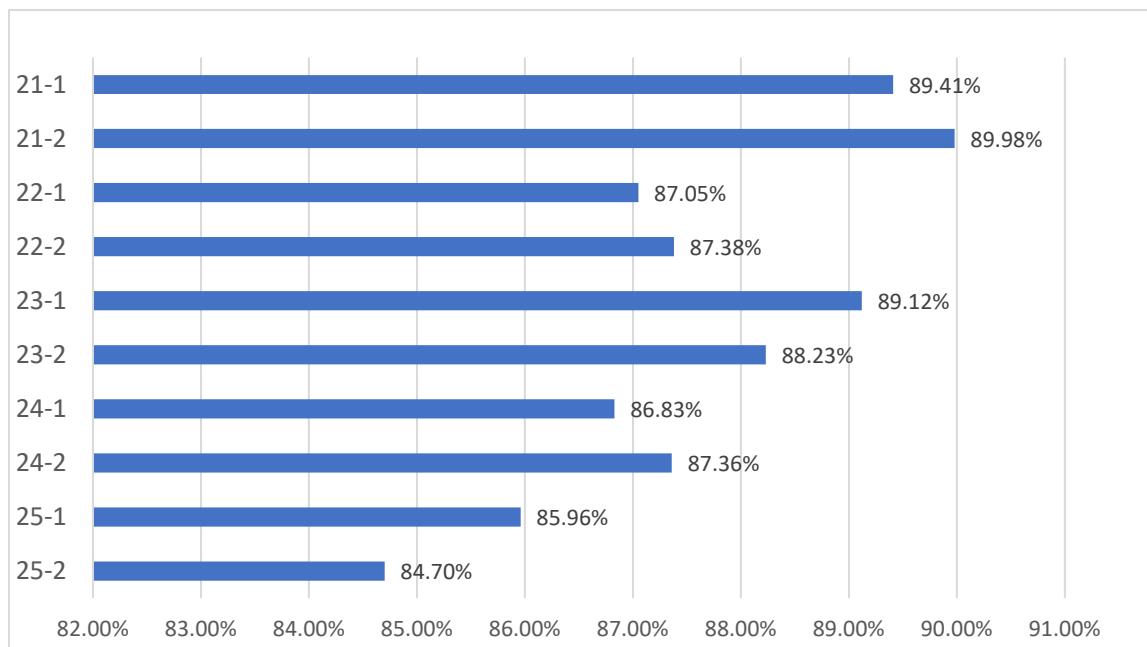
Graded Events		Points	% of Course
27-Jan	In Class Quiz 1	30.00	3%
29-Jan	Problem Set 1	75.00	8%
5-Feb	WPR #1	150.00	15%
18-Feb	Proficiency Project Part 1	30.00	3%
3-Mar	Proficiency Project Part 2	70.00	7%
7-Mar	In Class Quiz 2	20.00	2%
12-Mar	Proficiency Project Part 3	50.00	5%
31-Mar	Graded Lab	150.00	15%
2-Apr	Course Project Part 1	50.00	5%
9-Apr	In Class Quiz 3	25.00	3%
14-Apr	Course Project Part 2	75.00	8%
30-Apr	Course Project Part 3	75.00	8%
7-May	WPR #2	200.00	20%
Total		1000.00	100%

Section II – Course Assessment

1. Achievement of Course Objectives.

- Course Objectives/Syllabus. Enclosure 2.
- Course Objective Assessment Table. Enclosure 3.
- Results of Course End Feedback (web-based). Enclosure 5.
- Historical Course Final Grade Averages.

AT	Average
25-2	84.70%
25-1	85.96%
24-2	87.36%
24-1	86.83%
23-2	88.23%
23-1	89.12%
22-2	87.38%
22-1	87.05%
21-2	89.98%
21-1	89.41%
20-2	86.42%
20-1	86.40%
19-2	88.07%
19-1	89.61%



2. Course Objectives.

a. **Body of Knowledge.** The course objectives adequately cover the body of knowledge. This course teaches the tools and methods necessary to be proficient in discrete-event simulation theory and application. The course is split evenly between theory and application. The course is informally broken down into three primary blocks of material that follow the sequence of conducting a simulation study. The first block of the course focuses on the basics of discrete-event simulation, queuing theory, random number generation, data collection, and input data analysis. The second block introduces the students to ProModel and develops their proficiency in the software and in modeling the theoretical concepts introduced in the first block. The third and final block is focused on analyzing the simulation output and comparing alternatives. The WPRs and quizzes are designed to evaluate the theoretical material, whereas the graded lab and projects evaluate proficiency in ProModel. The projects, conducted in three parts, requires the students to work in teams to conduct a simulation study of a real system from start to finish, thereby requiring synthesis and application of the course material. This structure allows the course objectives to be met at a level appropriate for undergraduate students while simultaneously meeting systems engineering and engineering management program outcomes and objectives.

b. **Effective Assessment.** The end of course survey and the department's major-graded-event content approval process provide an adequate assessment tool for an impartial and effective assessment of the course objectives.

3. **Course Processes.**

a. **Textbook.** The Harrell, et al., textbook sufficiently covers the material outlined in the course objectives. The course supplements some material not covered in the text concerning queuing theory (the textbook only provides equations for M/M/1 queues) to provide additional tools to validate simulation output.

b. **Software.** ProModel is an exceptional tool that is critical to the course. In addition to having access to the software in each lab, students can download the software on their personal laptops. The user interface of the software is user friendly and the visual output that ProModel provides can be especially compelling to stakeholders and greatly aid in their decision making. Furthermore, the output analysis and statistical analysis packages that are associated with ProModel make data analysis and output analysis easier for the user.

c. **Graded Requirements.** In general, the graded requirements in EM481 are appropriate for gauging student comprehension, extending fundamental ideas, and demonstrating the real-world relevance of the material.

4. **Course Pedagogy.**

a. **Course Project.** The course is allotted 3.5 credit hours due to the significant course project. The basic objectives for the course project required the cadets to go through the entire simulation study process including raw data analysis, simulation synthesis, and sensitivity analysis.

b. **Instructor Points.** Instructor points were eliminated based off of AT25-1 Course End Feedback.

5. **EM Program Student Outcomes Crosswalk.** Enclosure 4.

Section III – Assessment of Recent Changes and Proposed Future Changes.

1. Changes to Current Academic Term.

- a. **Course Objectives.** None.
- b. **Textbook.** None
- c. **Curriculum.** Eliminating Instructor points and adding two quizzes was a change for good, as was adding ADV Output Viewer and Logic lessons.
- d. **Graded Requirements.** We conducted more grading via Gradescope in Canvas and would recommend using that again.
- e. **Other.** None.

2. Proposed Change for Next Academic Term.

- a. **Course Objectives.** None.
- b. **Textbook.** None.
- c. **Curriculum.** See recommended changes at the beginning of the memo.
- d. **Graded Requirements.** Recommend creating mandatory Homeworks to force repetitions in ProModel. WPR 2 can be reduced by up to 50 points to enable that change.
- e. **Other.** None.

Enclosure 2: EM481 Lesson Objectives/Syllabus

<u>LSN</u>	<u>LSN OBJs</u>
1	1. Define Simulation 2. Understand the purpose of simulation 3. Know how to conduct a simulation study
2	1. Define System and Processing System 2. Understand what drives complexity 3. Understand what a discrete event simulation is 4. Define system elements and the types of simulation
3	1. Know basic descriptive statistics and how to calculate them from a sample 2. Understand how we use distributions to model random behavior 3. Understand the difference between discrete and continuous random variables 4. Understand the characteristics of a Poisson Process 5. Understand key properties of the Poisson and exponential distributions
4	1. Know how to draw an effective and useful entity flow diagram 2. Discuss how to model a system 3. Understand guidelines for gathering data and what to do in the absence of data 4. Know steps for gathering data, the types of data, and the source of data
5	1. Define queue and utilization 2. Understand the purpose and limitations of analytical queuing models 3. Understand Kendall's notation 4. Know how to calculate evaluation measures of M/M/1 queues 5. Know Little's Law and general queueing relationships
6	1. Know how to calculate evaluation measures of G/G/1 and G/G/s queuing models
7	1. Know how to find critical values for statistical tests 2. Know how to interpret a p-value 3. Define independence and homogeneity (identical distribution) 4. Understand the importance of testing for independence and homogeneity 5. Know how to use scatter plots, runs tests, and autocorrelation to test for independence 6. Know how to conduct visual test for homogeneity
8	1. Understand the importance of fitting a distribution to data 2. Know how to use the Chi-Square goodness of fit test 3. Understand how the Komogorov-Smirnov (KS) and Anderson Darling (AD) goodness of fit tests work.
9	1. Understand how boundless distributions affect simulation 2. Understand how to model discrete distributions using continuous distributions 3. Know how to use Stat::Fit to test for independence and homogeneity
11	1. Understand the basic ProModel environment and how to build a discrete event simulation in ProModel 2. Know how to use the LEAP model building process 3. Know how to use the Logic Builder 4. Know how to implement the ProModel command WAIT
12	

13	1. Define Random Number Generator (RNG) and random variate 2. Know the desirable properties of a RNG 3. Know how to generate RNs using a linear congruential generator (LCG) and how to determine if the LCG will have a full cycle 4. Know how to apply the inverse transformation method to generate random numbers
14	1. Know the difference between simple average and time average statistics 2. Know how to assign local and global variables in ProModel 3. Know how to implement the ProModel commands REAL and INT 4. Interpret results in Output Viewer
15	1. Know the purpose of and how to create attributes and user distributions 2. Know how to assign attributes a value from a user distribution 3. Implement routing using attributes and logic 4. Know how to implement the ProModel commands: CLOCK, IF THEN ELSE, ROUTE, and GRAPHIC 5. Know how to use attributes to gather simple average statistics
16	1. Understand the purpose of and how to build resources and path networks 2. Know how to implement the ProModel commands: MOVE FOR, MOVE WITH, GET, FREE, and USE
17	1. Understand how to use temporary and permanent attachments and batching 2. Know how to create package files and import graphics 3. Know how to implement the ProModel commands: GROUP/ UNGROUP, COMBINE, JOIN, LOAD/ UNLOAD, ACCUM
18	1. Understand how to use temporary and permanent attachments and batching 2. Know how to create package files and import graphics 3. Know how to implement the ProModel commands: GROUP/ UNGROUP, COMBINE, JOIN, LOAD/ UNLOAD, ACCUM
20	1. Apply Output Viewer-created tables, plots, & reports to troubleshoot logic. 2. Apply Output Viewer to answer situation-specific questions.
21	1. Create logic that allows multiple conditions to be met. 2. Create IF THEN statements specific to complex scenarios. 3. Execute the Load/Group commands across multiple locations.
24	1. Understand the purpose and use of priorities and preemption 2. Identify and know how to model pull systems 3. Know how to implement the ProModel commands: SEND, ORDER
27	1. Know how to end simulations according to defined criteria using logic 2. Know how to model balking and jockeying customer behavior 3. Know how to implement the ProModel commands: DO WHILE, DO UNTIL, WHILE DO, STOP
28	1. Define validation & verification 2. Understand how to use validation and verification techniques 3. Know how to implement ProModel commands: TRACE, PAUSE, DEBUG 4. Understand how to simulate multiple scenarios in ProModel using Macros and Scenario features. 5. Know how to evaluate the output from multiple scenarios from ProModel Output Viewer.
29	1. Know how to construct confidence intervals using simulation results 2. Understand how to determine the appropriate number of replications to achieve desired error 3. Understand the implications of the Central Limit Theorem and replications in simulation

31	1. Understand difference between terminating and non-terminating simulations 2. Understand warm-up periods and how they apply to non-terminating simulations 3. Know how to determine the length of a warm-up period using visual methods and Welch Moving Average Method 4. Understand how to create independent observations from a non-terminating simulation using replications and the batch method
32	1. Understand hypothesis testing in comparing two simulation alternatives 2. Know how to use Welch Confidence Interval to compare two alternatives 3. Know how to use Paired-t Confidence Interval to compare two alternatives
33	1. Analyze simulation output of multiple alternatives using ANOVA (analysis of variance) 2. Analyze simulation output of multiple alternatives using Fisher's Least Significant Difference method

EM481 Syllabus

EM481: Systems Simulation								
AT 25-2								
LSN		Date	Lesson Name	Reading	Problems	Lesson Objectives	Notes	
Intro	1	7-Jan	Introduction to Simulation	1.1-1.6 1.8-1.9 1.11	1.1 1.3	1. Define Simulation 2. Understand the purpose of simulation 3. Know how to conduct a simulation study		
	2	13-Jan	Simulation Basics	2.1-2.7 3.1-3.2 4.1-4.2	3.1	1. Define System and Processing System 2. Understand what drives complexity 3. Understand what a discrete event simulation is 4. Define system elements and the types of simulation		
Technical	3	15-Jan	Random Behavior and Statistics	3.3 5.8.1-5.8.2 App A	5.5 5.12	1. Know basic descriptive statistics and how to calculate them from a sample 2. Understand how we use distributions to model random behavior 3. Understand the difference between discrete and continuous random variables 4. Understand the characteristics of a Poisson Process 5. Understand key properties of the Poisson and exponential distributions	Assign PS 1	
	4	21-Jan	Modeling a System and Data Collection	5.1-5.6, 5.9 5.12 5.7.1 6.1-6.3 6.4.1-6.4.4		1. Know how to draw an effective and useful entity flow diagram 2. Discuss how to model a system 3. Understand guidelines for gathering data and what to do in the absence of data 4. Know steps for gathering data, the types of data, and the source of data		
	5	23-Jan	Queuing Theory I	2.9-2.10	2.16 2.17	1. Define queue and utilization 2. Understand the purpose and limitations of analytical queuing models 3. Understand Kendall's notation 4. Know how to calculate evaluation measures of M/M/1 queues 5. Know Little's Law and general queueing relationships		
	6	27-Jan	Queueing Theory II	Handout	Handout	1. Know how to calculate evaluation measures of G/G/1 and G/G/s queueing models		
	7	29-Jan	Testing Input Data I	5.7.1-5.7.2		1. Know how to find critical values for statistical tests 2. Know how to interpret a p-value 3. Define independence and homogeneity (identical distribution) 4. Understand the importance of testing for independence and homogeneity 5. Know how to use scatter plots, runs tests, and autocorrelation to test for independence 6. Know how to conduct visual test for homogeneity	In- Class Quiz 1	
	8	31-Jan	Testing Input Data II	5.8.3	5.19	1. Understand the importance of fitting a distribution to data 2. Know how to use the Chi-Square goodness of fit test 3. Understand how the Komogorov-Smirnov (KS) and Anderson Darling (AD) goodness of fit tests work.	Problem Set 1 Due	
	9	3-Feb	Data Analysis using Stat::Fit	5.10-5.11	L5 Appendix A	1. Understand how boundless distributions affect simulation 2. Understand how to model discrete distributions using continuous distributions 3. Know how to use Stat::Fit to test for independence and homogeneity	1. Assign Proficiency Project 2. Install Pro Model	
	10	5-Feb				WPR 1		
		11	10-Feb	Simulation Modeling using LEAP I	Watch video!	L1 L2.1	1. Understand the basic ProModel environment and how to build a discrete event simulation in ProModel 2. Know how to use the LEAP model building process 3. Know how to use the Logic Builder 4. Know how to implement the ProModel command WAIT	
		12	12-Feb	Simulation Modeling using LEAP II				

Software-Assisted	13	14-Feb	Random Number and Random Variate Generation	3.4-3.6 8.2.1	3.6 3.8	1. Define Random Number Generator (RNG) and random variate 2. Know the desirable properties of a RNG 3. Know how to generate RNs using a linear congruential generator (LCG) and how to determine if the LCG will have a full cycle 4. Know how to apply the inverse transformation method to generate random numbers	
	14	18-Feb	Variables and the Output Viewer			1. Know the difference between simple average and time average statistics 2. Know how to assign local and global variables in ProModel 3. Know how to implement the ProModel commands REAL and INT 4. Interpret results in Output Viewer	Pro. Project Part 1 Due
	15	20-Feb	Attributes and User Distributions - Logic and Routing		L4.4 L4.11.1 L6.1-6.2	1. Know the purpose of and how to create attributes and user distributions 2. Know how to assign attributes a value from a user distribution 3. Implement routing using attributes and logic 4. Know how to implement the ProModel commands: CLOCK, IF THEN ELSE, ROUTE, and GRAPHIC 5. Know how to use attributes to gather simple average statistics	
	16	24-Feb	Resource and Path Networks		L4.3 L12.2	1. Understand the purpose of and how to build resources and path networks 2. Know how to implement the ProModel commands: MOVE FOR, MOVE WITH, GET, FREE, and USE	
	17	26-Feb	Batching and Attaching Entities I		L4.7-4.9 L6.10 L6.12	1. Understand how to use temporary and permanent attachments and batching 2. Know how to create package files and import graphics 3. Know how to implement the ProModel commands: GROUP/ UNGROUP, COMBINE, JOIN, LOAD/ UNLOAD, ACCUM	
	18	3-Mar	Batching and Attaching Entities II		L4.7-4.9 L6.10 L6.12	1. Understand how to use temporary and permanent attachments and batching 2. Know how to create package files and import graphics 3. Know how to implement the ProModel commands: GROUP/ UNGROUP, COMBINE, JOIN, LOAD/ UNLOAD, ACCUM	Pro. Project Part 2 Due
	19	5-Mar	X HOUR DROP- MSN CMD CONF				
	20	7-Mar	ADV Output Viewer			1. Apply Output Viewer-created tables, plots, & reports to troubleshoot logic. 2. Apply Output Viewer to answer situation-specific questions.	In Class Quiz 2
	21	10-Mar	ADV Routing Logic			1. Create logic that allows multiple conditions to be met. 2. Create IF THEN statements specific to complex scenarios. 3. Execute the Load/Group commands across multiple locations.	
	22	12-Mar	X HOUR DROP				
	23	14-Mar	X HOUR DROP= SPRING BREAK				
	24	24-Mar	Priorities, Preemption, & Pull; Course Project Intro		L6.7-6.8	1. Understand the purpose and use of priorities and preemption 2. Identify and know how to model pull systems 3. Know how to implement the ProModel commands: SEND, ORDER	Assign Course Project
	25	26-Mar	Practice Lab				
	26	31-Mar	GRADED LAB				
	27	2-Apr	Loops and Termination Logic		L4.11.2 L8.2-8.2.1	1. Know how to end simulations according to defined criteria using logic 2. Know how to model balking and jockeying customer behavior 3. Know how to implement the ProModel commands: DO WHILE, DO UNTIL, WHILE DO, STOP	CP Part 1 Due
	28	7-Apr	Verification, Validation, Macros and Scenarios		L7.2-7.3 L11.1-11.2	1. Define validation & verification 2. Understand how to use validation and verification techniques 3. Know how to implement ProModel commands: TRACE, PAUSE, DEBUG 4. Understand how to simulate multiple scenarios in ProModel using Macros and Scenario features. 5. Know how to evaluate the output from multiple scenarios from ProModel Output Viewer.	

Analysis	29	9-Apr	Output Analysis of a Single System I	8.2.2-8.3		1. Know how to construct confidence intervals using simulation results 2. Understand how to determine the appropriate number of replications to achieve desired error 3. Understand the implications of the Central Limit Theorem and replications in simulation	In Class Quiz 3
	30	11-Apr				DROP	
	31	14-Apr	Output Analysis of a Single System II	8.4 8.6	L8.3	1. Understand difference between terminating and non-terminating simulations 2. Understand warm-up periods and how they apply to non-terminating simulations 3. Know how to determine the length of a warm-up period using visual methods and Welch Moving Average Method 4. Understand how to create independent observations from a non-terminating simulation using replications and the batch method	CP Part 2 Due
	32	16-Apr	Output Analysis of Multiple Systems I	9.1-9.3	9.1a	1. Understand hypothesis testing in comparing two simulation alternatives 2. Know how to use Welch Confidence Interval to compare two alternatives 3. Know how to use Paired-t Confidence Interval to compare two alternatives	
	33	22-Apr	Output Analysis of Multiple Systems II	9.4-9.5		1. Analyze simulation output of multiple alternatives using ANOVA (analysis of variance) 2. Analyze simulation output of multiple alternatives using Fisher's Least Significant Difference method	
	34	25-Apr				X HOUR DROP	
	35	28-Apr				DROP	
	36	30-Apr	Presentations				CP Part 3 Due
	37	5-May	Presentations/Course Review				
	38	7-May	WPR2				
	39	9-May	X HOUR DROP				

Enclosure 3: EM481 Course Objective Assessment

EM481 Course Objectives		Direct Assessment	Course Director's Assessment	Applicable Graded Events
1.	Understand the process of conducting a simulation study	3.60	3.50	Quiz PS 1 WPR 1 Graded Lab Pro. Project Course Project
2.	Understand the application of closed-form analytical queuing methods when conducting a simulation study.	3.86	3.60	Quiz PS 1 WPR 1
3.	Demonstrate the ability to generate and test sequences of pseudo-random numbers and selected classes of random variates.	3.32	2.00	WPR 1 Quiz 2
4.	Collect, analyze, and determine the appropriate probability distribution for data used in a simulation study (input modeling analysis).	3.38	3.50	Graded Lab Pro. Project Course Project WPR 2
5.	Demonstrate comprehension of the simulation modeling process and be able to construct a simulation using course software.	3.09	3.50	Graded Lab Pro. Project Course Project Quiz 2 & 3
6.	Verify and validate simulation models.	3.10	3.00	Course Project WPR 2 Quiz 3
7.	Evaluate and interpret the quantitative output of simulation experiments for both single system and multiple systems (output analysis).	3.38	3.30	Graded Lab Pro. Project Course Project WPR 2
8.	Demonstrate the ability to design, conduct, and analyze simulation experiments using a simulation package.	3.09	3.00	Graded Lab Pro Project Course Project Quiz 2 & 3
9.	Prepare and brief a written analysis of a design problem supported by simulation results	3.17	2.50	Pro Project Course Project
Overall Assessment		3.33	3.10	All Graded Events
NOTES: 1) Graded Events included: Quizzes 1-3, Problem Sets 1, WPRs 1-2, Project Stage Gates 1-3, Graded Lab 2) Student Assessment is based on the End-of-Course Survey 3) Course Director Assessment is: -- *Based on course average for "All graded events" -- All others based on individual questions on graded events to include TEE question results Scale: 1 = Unsatisfactory, 2 = Developing, 3 = Satisfactory, 4 = Exemplary				

Enclosure 4: EM Program Student Outcome Crosswalk.

		Program Student Outcomes						
		An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	An ability to communicate effectively with a range of audiences.	An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.
Course Objectives		1	2	3	4	5	6	7
1	Understand the process of conducting a simulation study	X	X		X		X	X
2	Understand the application of closed-form analytical queuing methods when conducting a simulation study.	X	X				X	X
3	Demonstrate the ability to generate and test sequences of pseudo-random numbers and selected classes of random variates.	X						X
4	Collect, analyze, and determine the appropriate probability distribution for data used in a simulation study (input modeling analysis).	X						X
5	Demonstrate comprehension of the simulation modeling process and be able to construct a simulation using course software.	X	X				X	X
6	Verify and validate simulation models.	X	X	X	X		X	X
7	Evaluate and interpret the quantitative output of simulation experiments for both single system and multiple systems (output analysis).	X				X	X	X
8	Demonstrate the ability to design, conduct, and analyze simulation experiments using a simulation package.	X	X		X	X	X	X
9	Prepare and brief a written analysis of a design problem supported by simulation results		X	X				X

Enclosure 5: Course End Feedback



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB C

AY25-1 COURSE ASSESSMENT



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE
20204

18 December

MEMORANDUM FOR COL Brandon Thompson, Engineering Management Program
Director, Department of Systems Engineering

SUBJECT: AT25-1 Course Assessment – EM481, Project Management

1. The attached report summarizes the EM481 assessment and curriculum change actions for Academic Term 2025-2.
2. EM481 Systems Simulation is an introductory, discrete-event simulation course that consists of a thoughtful blend of simulation theory, quantitative and analytical concepts necessary for simulation, and the development of simulation models using ProModel software.
3. I recommend the following primary changes for improvement to the course:
 - a. The Random Number Generator (RNG) lesson will be moved to the middle of the course and tailored to be more ProModel specific. Previously, this was a full lesson and barely tested upon nor brought up again. I recommend it being more integrated with the ProModel curriculum.
 - b. Move the Modeling a System lesson to closer to the beginning of the course. This lesson shows Entity Flow Diagrams (EFDs) to Cadets and is instrumental in connecting conceptual theory to actual ProModel Simulations. I think it is important they see it sooner.
 - c. Add an “advanced” output viewer and logic creation lesson. Cadets struggled utilizing Output viewer to troubleshoot their code as well as being creative when creating code. I believe it is important we spend more time instructing these topics and showing Cadets the trial & error process when creating projects.
 - d. Replace Instructor points with in-class quizzes. Cadets struggled with conceptual questions on WPR 2 and I recommend objectively testing them on that more frequently throughout the course.
4. POC for this action is the undersigned at james.macgibbon@westpoint.edu.

JAMES E. MACGIBBON
MAJ, EN/FA49
EM481 Course Director

Encls

1. Course Analysis
2. Lesson Objectives / Syllabus
3. EM481 Course Objective Assessment
4. EM Program Student Outcomes Crosswalk
5. Course End Feedback

Enclosure 1: Course Analysis – EM481 Systems Simulation

Section I – Course Description

1. Redbook Description.

Cadets learn and explore discrete event simulation techniques and tools used to analyze and improve complex systems. Applications include operations, transportation, manufacturing and logistics systems. Topics include functional modeling with functional flow diagrams, simulation theory, the modeling process, input data analysis, generation of random numbers, verification and validation of simulation models, experimental design, output analysis, and application using simulation software. The course concepts provide cadets the tools to evaluate military and civilian systems. Emphasis is placed on using simulation in the Systems Decision Process (SDP). Cadets demonstrate proficiency and develop communication skills through design projects and briefings. Cadets spend several lessons in a computer lab environment.

- a. **Lessons.** 35 @ 75 min (2.50 Att/wk) LABS: 0 @ 0 min
- b. **Special Requirements.** One (1) half-day Trip Section
- c. **Offered** – Every Semester

2. Enrollment.

Academic Term	Enrollment
25-1	61
24-2	48
24-1	56
23-2	42
23-1	55
22-2	27
22-1	53
21-2	38
21-1	62
20-2	26
20-1	50
19-2	26
19-1	62
18-2	48
18-1	30

3. Current Course Objectives.

- (1) Understand the process of conducting a simulation study.
- (2) Understand the application of closed-form analytical queuing methods when conducting a simulation study.
- (3) Demonstrate the ability to generate and test sequences of pseudo-random numbers and selected classes of random variates.
- (4) Collect, analyze, and determine the appropriate probability distribution for data used in a simulation study (input modeling analysis).
- (5) Demonstrate comprehension of the simulation modeling process and be able to construct a simulation using course software.
- (6) Verify and validate simulation models.
- (7) Evaluate and interpret the quantitative output of simulation experiments for both single system and multiple systems (output analysis).
- (8) Demonstrate the ability to design, conduct, and analyze simulation experiments using a simulation package.

Prepare and brief a written analysis of a design problem supported by simulation results.

4. Textbooks.

- 1) Harrell, C., Biman, G., Gladwin, B. & Bowden, R. (2022). *Simulation Using ProModel, 4th edition*. San Diego, CA: Cognella Academic Publishing.
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5. **Course Strategy.** Through this course, cadets will learn and explore discrete event simulation techniques used to analyze and improve complex systems. Applications for discrete event simulation include manufacturing, transportation, logistics, and service systems. Topics include the simulation study process, queuing theory, input data analysis, generation and testing of random numbers, simulation construction, verification and validation of simulation models, and output analysis. Cadets will demonstrate proficiency of lesson objectives while developing communication skills through graded events, a comprehensive course project, and an interactive classroom and lab environment. Upon completion of the course, cadets will be able to simulate and analyze moderately complex civilian and military systems.

6. Graded Requirements.

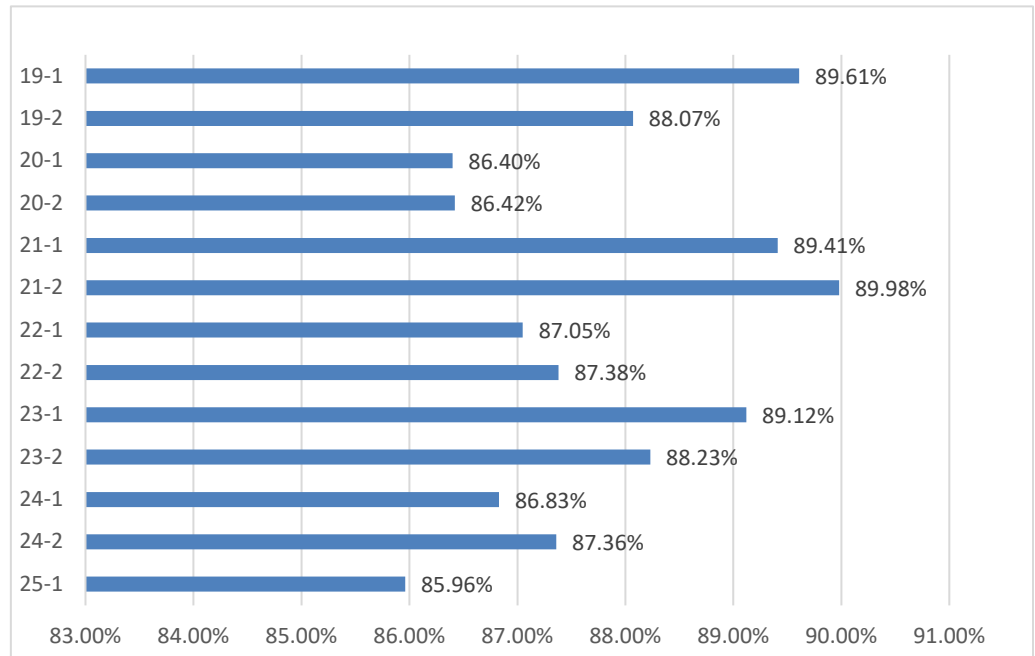
Graded Events		Points	% of Course
1	Quiz	50	5.0%
2	Problem Set 1	50	5.0%
3	Written Partial Review (WPR) #1	150	15.0%
4	Graded Lab	150	15.0%
5	Proficiency Project	150	15.0%
6	Course Project	200	20.0%
7	Written Partial Review (WPR) #2	200	20.0%
8	Instructor Points	50	5.0%
Total		1000	100.0%

Section II – Course Assessment

1. Achievement of Course Objectives.

- Course Objectives/Syllabus. Enclosure 2.
- Course Objective Assessment Table. Enclosure 3.
- Results of Course End Feedback (web-based). Enclosure 5.
- Historical Course Final Grade Averages.

AT	Average
25-1	85.96%
24-2	87.36%
24-1	86.83%
23-2	88.23%
23-1	89.12%
22-2	87.38%
22-1	87.05%
21-2	89.98%
21-1	89.41%
20-2	86.42%
20-1	86.40%
19-2	88.07%
19-1	89.61%



2. Course Objectives.

a. **Body of Knowledge.** The course objectives adequately cover the body of knowledge. This course teaches the tools and methods necessary to be proficient in discrete-event simulation theory and application. The course is split evenly between theory and application. The course is informally broken down into three primary blocks of material that follow the sequence of conducting a simulation study. The first block of the course focuses on the basics of discrete-event simulation, queuing theory, random number generation, data collection, and input data analysis. The second block introduces the students to ProModel and develops their proficiency in the software and in modeling the theoretical concepts introduced in the first block. The third and final block is focused on analyzing the simulation output and comparing alternatives. The WPR and the quiz are designed to evaluate the theoretical material, whereas the graded lab and proficiency project evaluate proficiency in ProModel. The project, conducted in four parts, requires the students to work in teams to conduct a simulation study of a real system from start to finish, thereby requiring synthesis and application of the course material. This structure allows the course objectives to be met at a level appropriate for undergraduate students while simultaneously meeting systems engineering and engineering management program outcomes and objectives.

b. **Effective Assessment.** The end of course survey and the department's major-graded-event content approval process provide an adequate assessment tool for an impartial and effective assessment of the course objectives.

3. **Course Processes.**

a. **Textbook.** The Harrell, et al., textbook sufficiently covers the material outlined in the course objectives. The course supplements some material not covered in the text concerning queuing theory (the textbook only provides equations for M/M/1 queues) to provide additional tools to validate simulation output.

b. **Software.** ProModel is an exceptional tool that is critical to the course. In addition to having access to the software in each lab, students can download the software on their personal laptops. The user interface of the software is user friendly and the visual output that ProModel provides can be especially compelling to stakeholders and greatly aid in their decision making. Furthermore, the output analysis and statistical analysis packages that are associated with ProModel make data analysis and output analysis easier for the user.

c. **Graded Requirements.** In general, the graded requirements in EM481 are appropriate for gauging student comprehension, extending fundamental ideas, and demonstrating the real-world relevance of the material.

4. **Course Pedagogy.**

a. **Course Project.** The course is allotted 3.5 credit hours due to the significant course project. The basic objectives for the course project required the cadets to go through the entire simulation study process including raw data analysis, simulation synthesis, and sensitivity analysis.

b. **Instructor Points.** Cadet performance was measured based on participation in class and preparation for class and awarded points accordingly. It is recommended to eliminate Instructor Points for next semester and replace them with In Class Quizzes.

5. **EM Program Student Outcomes Crosswalk.** Enclosure 5.

Section III – Assessment of Recent Changes and Proposed Future Changes.

1. Changes to Current Academic Term.

- a. **Course Objectives.** None.
- b. **Textbook.** None
- c. **Curriculum.** The Proficiency Project was based on a modified ACFT and the Course Project was based on the Expert Soldier Badge. All three Instructors this semester were new to the course so there were minimal changes this semester.
- d. **Graded Requirements.** None.
- e. **Other.** None.

2. Proposed Change for Next Academic Term.

- a. **Course Objectives.** None.
- b. **Textbook.** None.
- c. **Curriculum.** See recommended changes at beginning of the memo.
- d. **Graded Requirements.** Recommend replacing Instructor Points with In-Class Quizzes.
- e. **Other.** None.

Enclosure 2: EM481 Lesson Objectives/Syllabus

Lesson	1	1. Define Simulation 2. Understand the purpose of simulation 3. Know how to conduct a simulation study
Lesson	2	1. Define System and Processing System 2. Understand what drives complexity 3. Understand what a discrete event simulation is 4. Define system elements and the types of simulation
Lesson	3	1. Know basic descriptive statistics and how to calculate them from a sample 2. Understand how we use distributions to model random behavior 3. Understand the difference between discrete and continuous random variables 4. Understand the characteristics of a Poisson Process 5. Understand key properties of the Poisson and exponential distributions
Lesson	4	1. Define queue and utilization 2. Understand the purpose and limitations of analytical queuing models 3. Understand Kendall's notation 4. Know how to calculate evaluation measures of M/M/1 queues 5. Know Little's Law and general queueing relationships
Lesson	5	1. Know how to calculate evaluation measures of G/G/1 and G/G/s queueing models
Lesson	6	1. Know how to draw an effective and useful entity flow diagram 2. Discuss how to model a system 3. Understand guidelines for gathering data and what to do in the absence of data 4. Know steps for gathering data, the types of data, and the source of data
Lesson	7	1. Know how to find critical values for statistical tests 2. Know how to interpret a p-value 3. Define independence and homogeneity (identical distribution) 4. Understand the importance of testing for independence and homogeneity 5. Know how to use scatter plots, runs tests, and autocorrelation to test for independence 6. Know how to conduct visual test for homogeneity
Lesson	8	1. Understand the importance of fitting a distribution to data 2. Know how to use the Chi-Square goodness of fit test 3. Understand how the Komogorov-Smirnov (KS) and Anderson Darling (AD) goodness of fit tests work.

Lesson	10	1. Define Random Number Generator (RNG) and random variate 2. Know the desirable properties of a RNG 3. Know how to generate RNs using a linear congruential generator (LCG) and how to determine if the LCG will have a full cycle 4. Know how to apply the inverse transformation method to generate random numbers 5. Review Lessons 1-9
Lesson	12	1. Understand how boundless distributions affect simulation 2. Understand how to model discrete distributions using continuous distributions 3. Know how to use Stat::Fit to test for independence and homogeneity
Lesson	13	1. Understand the basic ProModel environment and how to build a discrete event simulation in ProModel 2. Know how to use the LEAP model building process 3. Know how to use the Logic Builder 4. Know how to implement the ProModel command WAIT
Lesson	14	
Lesson	15	1. Know the difference between simple average and time average statistics 2. Know how to assign local and global variables in ProModel 3. Know how to implement the ProModel commands REAL and INT 4. Interpret results in Output Viewer
Lesson	16	1. Know the purpose of and how to create attributes and user distributions 2. Know how to assign attributes a value from a user distribution 3. Implement routing using attributes and logic 4. Know how to implement the ProModel commands: CLOCK, IF THEN ELSE, ROUTE, and GRAPHIC 5. Know how to use attributes to gather simple average statistics
Lesson	18	1. Understand the purpose of and how to build resources and path networks 2. Know how to implement the ProModel commands: MOVE FOR, MOVE WITH, GET, FREE, and USE
Lesson	19	1. Understand how to use temporary and permanent attachments and batching 2. Know how to create package files and import graphics 3. Know how to implement the ProModel commands: GROUP/ UNGROUP, COMBINE, JOIN, LOAD/ UNLOAD, ACCUM
Lesson	20	1. Understand how to use temporary and permanent attachments and batching 2. Know how to create package files and import graphics 3. Know how to implement the ProModel commands: GROUP/ UNGROUP, COMBINE, JOIN, LOAD/ UNLOAD, ACCUM

Lesson	21	1. Understand the purpose and use of priorities and preemption 2. Identify and know how to model pull systems 3. Know how to implement the ProModel commands: SEND, ORDER
Lesson	23	1. Apply design methods to generate alternative solutions to address course project requirements. 2. Understand ethical considerations in solution design. 3. Develop Feasibility Screening Matrix (FSM) to identify candidate solutions.
Lesson	26	1. Know how to end simulations according to defined criteria using logic 2. Know how to model balking and jockeying customer behavior 3. Know how to implement the ProModel commands: DO WHILE, DO UNTIL, WHILE DO, STOP
Lesson	27	1. Define validation & verification 2. Understand how to use validation and verification techniques 3. Know how to implement ProModel commands: TRACE, PAUSE, DEBUG 4. Understand how to simulate multiple scenarios in ProModel using Macros and Scenario features. 5. Know how to evaluate the output from multiple scenarios from ProModel Output Viewer.
Lesson	28	1. Know how to construct confidence intervals using simulation results 2. Understand how to determine the appropriate number of replications to achieve desired error 3. Understand the implications of the Central Limit Theorem and replications in simulation
Lesson	30	1. Understand difference between terminating and non-terminating simulations 2. Understand warm-up periods and how they apply to non-terminating simulations 3. Know how to determine the length of a warm-up period using visual methods and Welch Moving Average Method 4. Understand how to create independent observations from a non-terminating simulation using replications and the batch method
Lesson	32	1. Understand hypothesis testing in comparing two simulation alternatives 2. Know how to use Welch Confidence Interval to compare two alternatives 3. Know how to use Paired-t Confidence Interval to compare two alternatives
Lesson	33	1. Analyze simulation output of multiple alternatives using ANOVA (analysis of variance) 2. Analyze simulation output of multiple alternatives using Fisher's Least Significant Difference method

EM481 Syllabus

EM481: Systems Simulation AY 25-1								
LSN		Date	Lesson Name	Reading	Problem	Lesson Objectives	Notes	
Intro	1	19-Aug	Introduction to Simulation	1.1-1.6 1.8-1.9 1.11	1.1 1.3	1. Define Simulation 2. Understand the purpose of simulation 3. Know how to conduct a simulation study		
	2	21-Aug	Simulation Basics	2.1-2.7 3.1-3.2 4.1-4.2	3.1	1. Define System and Processing System 2. Understand what drives complexity 3. Understand what a discrete event simulation is 4. Define system elements and the types of simulation		
Technical	3	23-Aug	Random Behavior and Statistics	3.3 5.8.1-5.8.2 App A	5.5 5.12	1. Know basic descriptive statistics and how to calculate them from a sample 2. Understand how we use distributions to model random behavior 3. Understand the difference between discrete and continuous random variables 4. Understand the characteristics of a Poisson Process 5. Understand key properties of the Poisson and exponential distributions		
	4	26-Aug	Queueing Theory I	2.9-2.10	2.16 2.17	1. Define queue and utilization 2. Understand the purpose and limitations of analytical queueing models 3. Understand Kendall's notation 4. Know how to calculate evaluation measures of M/M/1 queues 5. Know Little's Law and general queueing relationships	Assign Problem Set 1	
	5	28-Aug	Queueing Theory II	Handout	Handout	1. Know how to calculate evaluation measures of G/G/1 and G/G/s queueing models		
	6	3-Sep	Modeling a System and Data Collection	5.1-5.6, 5.9 5.12 5.7.1 6.1-6.3 6.4.1-6.4.4		1. Know how to draw an effective and useful entity flow diagram 2. Discuss how to model a system 3. Understand guidelines for gathering data and what to do in the absence of data 4. Know steps for gathering data, the types of data, and the source of data	Quiz Assign Proficiency Project	
	7	5-Sep	Testing Input Data I	5.7.1-5.7.2		1. Know how to find critical values for statistical tests 2. Know how to interpret a p-value 3. Define independence and homogeneity (identical distribution) 4. Understand the importance of testing for independence and homogeneity 5. Know how to use scatter plots, runs tests, and autocorrelation to test for independence 6. Know how to conduct visual test for homogeneity		
	8	9-Sep	Testing Input Data II	5.8.3	5.19	1. Understand the importance of fitting a distribution to data 2. Know how to use the Chi-Square goodness of fit test 3. Understand how the Komogorov-Smirnov (KS) and Anderson Darling (AD) goodness of fit tests work	Problem Set 1 Due	
	9	11-Sep	Drop - AORS					
	10	13-Sep	Random Number and Random Variate Generation	3.4-3.6 8.2.1	3.6 3.8	1. Define Random Number Generator (RNG) and random variate 2. Know the desirable properties of a RNG 3. Know how to generate RNs using a linear congruential generator (LCG) and how to determine if the LCG will have a full cycle 4. Know how to apply the inverse transformation method to generate random numbers 5. Review Lessons 1-9		
	11	16-Sep	VPR1					VPR1
		12	18-Sep	Data Analysis using Stat::Fit	5.10-5.11	L5 Appendix A	1. Understand how boundless distributions affect simulation 2. Understand how to model discrete distributions using continuous distributions 3. Know how to use Stat::Fit to test for independence and homogeneity	Install ProModel
13		23-Sep	Simulation Modeling using LEAP I		L1 L2.1	1. Understand the basic ProModel environment and how to build a discrete event simulation in ProModel 2. Know how to use the LEAP model building process 3. Know how to use the Logic Builder 4. Know how to implement the ProModel command WAIT	Proficiency Project P1 Due	
14		25-Sep	Simulation Modeling using LEAP II					
15		27-Sep	Variables and the Output Viewer			1. Know the difference between simple average and time average statistics 2. Know how to assign local and global variables in ProModel 3. Know how to implement the ProModel commands REAL and INT 4. Interpret results in Output Viewer		
16		30-Sep	Attributes and User Distributions - Logic and Routing		L4.4 L4.11.1 L6.1-6.2	1. Know the purpose of and how to create attributes and user distributions 2. Know how to assign attributes a value from a user distribution 3. Implement routing using attributes and logic 4. Know how to implement the ProModel commands: CLOCK, IF THEN ELSE, ROUTE, and GRAPHIC 5. Know how to use attributes to gather simple average statistics		
17		2-Oct	Compensatory Drop - P2 Prep					

Software-Assisted	18	7-Oct	Resource and Path Networks		L4.3 L12.2	1. Understand the purpose of and how to build resources and path networks 2. Know how to implement the ProModel commands: MOVE FOR, MOVE WITH, GET, FREE, and USE	Proficiency Project P2 Due
	19	9-Oct	Batching and Attaching Entities I		L4.7-4.9 L6.10 L6.12	1. Understand how to use temporary and permanent attachments and batching 2. Know how to create package files and import graphics 3. Know how to implement the ProModel commands: GROUP/ UNGROUP, COMBINE, JOIN, LOAD/ UNLOAD, ACCUM	
	20	11-Oct	Batching and Attaching Entities II		L4.7-4.9 L6.10 L6.12	1. Understand how to use temporary and permanent attachments and batching 2. Know how to create package files and import graphics 3. Know how to implement the ProModel commands: GROUP/ UNGROUP, COMBINE, JOIN, LOAD/ UNLOAD, ACCUM	
	21	15-Oct	Priorities, Preemption, and Pull Systems		L6.7-6.8	1. Understand the purpose and use of priorities and preemption 2. Identify and know how to model pull systems 3. Know how to implement the ProModel commands: SEND, ORDER	Proficiency Project P3 Due 18 OCT, 1600
	22	17-Oct	Drop				
	23	21-Oct	Course Project - Solution Design	Course Project Instructions Parnell (2e) Ch 11.1-11.2 (p.353-365), Ch 11.3.2 (p. 366-368) Parnell (3e) Ch 7.1-7.3		1. Apply design methods to generate alternative solutions to address course project requirements. 2. Understand ethical considerations in solution design. 3. Develop Feasibility Screening Matrix (FSM) to identify candidate solutions.	
	24	23-Oct	Compensatory Drop - Lab Prep				
	25	28-Oct	Graded Lab				Graded Lab
	26	30-Oct	Loops and Termination Logic		L4.11.2 L8.2-8.2.1	1. Know how to end simulations according to defined criteria using logic 2. Know how to model balking and jockeying customer behavior 3. Know how to implement the ProModel commands: DO WHILE, DO UNTIL, WHILE DO, STOP	
	27	1-Nov	Verification, Validation, Macros and Scenarios		L7.2-7.3 L11.1-11.2	1. Define validation & verification 2. Understand how to use validation and verification techniques 3. Know how to implement ProModel commands: TRACE, PAUSE, DEBUG 4. Understand how to simulate multiple scenarios in ProModel using Macros and Scenario features. 5. Know how to evaluate the output from multiple scenarios from ProModel Output Viewer.	Course Project P1 Due
Analysis	28	4-Nov	Output Analysis of a Single System I	8.2.2-8.3		1. Know how to construct confidence intervals using simulation results 2. Understand how to determine the appropriate number of replications to achieve desired error 3. Understand the implications of the Central Limit Theorem and replications in simulation	
	29	6-Nov	Drop ASEM				
	30	12-Nov	Output Analysis of a Single System II	8.4 8.6	L8.3	1. Understand difference between terminating and non-terminating simulations 2. Understand warm-up periods and how they apply to non-terminating simulations 3. Know how to determine the length of a warm-up period using visual methods and Welch Moving Average Method 4. Understand how to create independent observations from a non-terminating simulation using replications and the batch method	
	31	14-Nov	Compensatory Drop - P2 Prep				
	32	18-Nov	Output Analysis of Multiple Systems I	9.1-9.3	9.1a	1. Understand hypothesis testing in comparing two simulation alternatives 2. Know how to use Welch Confidence Interval to compare two alternatives 3. Know how to use Paired-t Confidence Interval to compare two alternatives	Course Project P2 Due
	33	20-Nov	Output Analysis of Multiple Systems II	9.4-9.5		1. Analyze simulation output of multiple alternatives using ANOVA (analysis of variance) 2. Analyze simulation output of multiple alternatives using Fisher's Least Significant Difference method	Class in MH 117
	34	22-Nov	Drop				
	35	25-Nov	Compensatory Drop				
	36	3-Dec	Compensatory Drop - Presentation Prep				
	37	5-Dec	Presentations				Course Project P3
	38	9-Dec	Presentations/Course Review				
	39	11-Dec	VPR2				VPR2 in MH 122
	40	13-Dec	Drop				

Enclosure 3: EM481 Course Objective Assessment

EM481 Course Objectives		Direct Assessment	Course Director's Assessment	Applicable Graded Events
1.	Understand the process of conducting a simulation study	2.95	3.5	Quiz PS 1 WPR 1 Graded Lab Pro. Project Course Project
2.	Understand the application of closed-form analytical queuing methods when conducting a simulation study.	3.13	4.0	Quiz PS 1 WPR 1
3.	Demonstrate the ability to generate and test sequences of pseudo-random numbers and selected classes of random variates.	3.11	2	WPR 1
4.	Collect, analyze, and determine the appropriate probability distribution for data used in a simulation study (input modeling analysis).	2.75	3.5	Graded Lab Pro. Project Course Project WPR 2
5.	Demonstrate comprehension of the simulation modeling process and be able to construct a simulation using course software.	2.78	3.0	Graded Lab Pro. Project Course Project
6.	Verify and validate simulation models.	2.72	3.0	Course Project WPR 2
7.	Evaluate and interpret the quantitative output of simulation experiments for both single system and multiple systems (output analysis).	2.75	3.0	Graded Lab Pro. Project Course Project WPR 2
8.	Demonstrate the ability to design, conduct, and analyze simulation experiments using a simulation package.	2.78	3.5	Graded Lab Pro Project Course Project
Overall Assessment		2.87	3.19	All Graded Events
NOTES: 1) Graded Events included: Quizzes 1-2, Problem Sets 1-4, WPRs 1-2, Project Stage Gates 1-3, Case Study, Participation 2) Student Assessment is based on the End-of-Course Survey 3) Course Director Assessment is: -- *Based on course average for "All graded events" -- All others based on individual questions on graded events to include TEE question results Scale: 1 = Unsatisfactory, 2 = Developing, 3 = Satisfactory, 4 = Exemplary				

Enclosure 4: EM Program Student Outcome Crosswalk.

		Program Student Outcomes						
		An ability to identify, formulate, and solve complex engineering problems by applying principles of	An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	An ability to communicate effectively with a range of audiences.	An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.
Course Objectives		1	2	3	4	5	6	7
1	Understand the process of conducting a simulation study	x	x		x		x	x
2	Understand the application of closed-form analytical queuing methods when conducting a simulation study.	x	x				x	x
3	Demonstrate the ability to generate and test sequences of pseudo-random numbers and selected classes of random variates.	x						x
4	Collect, analyze, and determine the appropriate probability distribution for data used in a simulation study (input modeling analysis).	x						x
5	Demonstrate comprehension of the simulation modeling process and be able to construct a simulation using course software.	x	x				x	x
6	Verify and validate simulation models.	x	x	x	x		x	x
7	Evaluate and interpret the quantitative output of simulation experiments for both single system and multiple systems (output analysis).	x					x	x
8	Demonstrate the ability to design, conduct, and analyze simulation experiments using a simulation package.	x	x			x	x	x

Enclosure 5: Course End Feedback



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TAB D

AT24-2 COURSE ASSESSMENT

EM481 Systems Simulation Course End Summary



Course Changes from AT24-1

- Directive course project
 - VA Health Clinic case study with simulated data
- Removed Multiple Choice questions on Graded Lab (improved output analysis performance by 20% from AT24-1)
- Made Proficiency Project “lead in” to Course Project (VA Clinic)

Course Recommendations for AT25-1

- Complete the integration of the course project with the proficiency project
 - Allows for improved model baseline for the analysis portion of the course project
- Change WPR 2 to a TEE to allow for sufficient time between closing course project requirements and the last exam

Programmatic Recommendations

- Introduce SimPy before starting ProModel with a simple “lead in” assignment, pending lesson build
 - Remove presentation requirement from group project to build two SimPy lessons
 - Aligns with other Python introduction material (distribution fit tests)

Instructional Recommendations

- Post pre-recorded videos of ProModel tutorials aligned to ProModel lessons
- Maintain practice Graded Lab as rehearsal for technological component
- Continue the use of the lab for future courses



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TAB E AY24-1 COURSE ASSESSMENT

EM481 Systems Simulation Course End Summary



Course Changes from AT23-2

- “Choose your own” course project
- Removed ProModel Problem Set
- Added Multiple Choice questions to Graded Lab
- Made Graded Lab in Canvas

Course Recommendations for AT24-2

- Directive course project
 - Named systems with simulated data for analysis
- Remove Multiple Choice questions on Graded Lab
- Make Proficiency Project “lead in” to Course Project

Programmatic Recommendations

- Cadets who have not taken SE375 struggle with the concept of hypothesis testing
- Introduce ANOVA and Fisher’s LSD as lesson objectives for proper analysis of simulations for the course project

Instructional Recommendations

- Continue to have cadets turn in a ProModel assignment for formatting standardization after the LEAP class (lesson 11)
- Practice Graded Lab as proof-of-concept/rehearsal for technological component
- Move to lab and use Graded Lab framework



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TAB F AT23-2 COURSE ASSESSMENT



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT NY 10996

MADN-SE

16 May 2023

MEMORANDUM FOR Record

SUBJECT: AT23-2 Course Assessment for EM481 Systems Simulation

1. Purpose. To assess EM481 and describe curriculum changes for Academic Term 23-2.
2. Description of Course. EM481 is an introductory, discrete-event simulation course that consists of a thoughtful blend of simulation theory, quantitative and analytical concepts necessary for simulation, and development of simulation models using ProModel software.
3. Summary of Performance. Performance this semester was 1.74% higher than last semester (89.12% compared to 87.38%).
4. The point of contact for this memorandum is the undersigned at

Encls

1. Course Summary
2. Course IM and Syllabus
3. WPR 1 Grade Approval
4. WPR 2 Grade Approval
5. Final Grades and OML

EDMUND J. YOON
Major, EN/49
EM481 Course Director

Enclosure 1: AT23-2 Course Summary – EM481 Systems Simulation

1. Redbook Description.

Cadets learn and explore discrete event simulation techniques and tools used to analyze and improve complex systems. Applications include operations, transportation, manufacturing, and logistics systems. Topics include functional modeling with functional flow diagrams, simulation theory, the modeling process, input data analysis, generation of random numbers, verification and validation of simulation models, experimental design, output analysis, and application using simulation software. The course concepts provide cadets with the tools to evaluate military and civilian systems. Emphasis is placed on using simulation in the Systems Decision Process (SDP). Cadets demonstrate proficiency and develop communication skills through design projects and briefings. Cadets spend several lessons in a computer lab environment.

a. Lessons. 30 @ 75 min

b. Special Requirements. In-progress reviews and course project; compensatory time provided.

c. Offered:

AY18		AY19		AY20		AY21		AY22		AY23	
18-1	18-2	19-1	19-2	20-1	20-2	21-1	21-2	22-1	22-2	23-1	23-2
X	X	X	X	X	X	X	X	X	X	X	X

2. Enrollment.

Academic Term	Enrollment
23-2	42
23-1	55
22-2	27
22-1	53
21-2	38
21-1	62
20-2	26
20-1	50
19-2	26
19-1	62
18-2	48
18-1	30

3. Current Course Objectives.

a. Understand the process of conducting a simulation study.

b. Understand the application of closed form analytical queuing methods when conducting a simulation study.

- c. Demonstrate the ability to generate and test sequences of pseudo-random numbers and selected classes of random variates.
- d. Collect, analyze, and determine the appropriate probability distribution for data used in a simulation study (input modeling analysis).
- e. Demonstrate comprehension of the simulation modeling process and be able to construct a simulation using course software.
- f. Verify and validate simulation models.
- g. Evaluate and interpret the quantitative output of simulation experiments for both single system and multiple systems (output analysis).
- h. Demonstrate the ability to design, conduct, and analyze simulation experiments using a simulation package.
- i. Prepare and brief a written analysis of a design problem supported by simulation results.

4. Textbooks.

- a. Harrell, C., Biman, G., Gladwin, B. & Bowden, R. (2022). Simulation Using ProModel, 4th edition. San Diego, CA: Cognella Academic Publishing.
- b. Parnell, G., Driscoll, P. & Henderson, D. (Eds.). (2011). Decision Making in Systems Engineering and Management, 2nd edition. Hoboken, NJ: John Wiley and Sons, Inc.

5. **Course Content.** Cadets learn and explore discrete event simulation techniques and tools used to analyze and improve complex systems. Applications include operations, transportation, manufacturing, and logistics systems. Topics include functional modeling with functional flow diagrams, simulation theory, the modeling process, input data analysis, generation of random numbers, verification and validation of simulation models, experimental design, output analysis, and application using simulation software. The course concepts provide cadets with the tools to evaluate military and civilian systems. Emphasis is placed on using simulation in the Systems Decision Process (SDP). Cadets demonstrate proficiency and develop communication skills through design projects and briefings. Cadets spend several lessons in a computer lab environment.

6. **Graded Requirements.** The point allocation for graded events changed slightly from AT23-1 to AT23-2. A new problem set was added to provide students with a measure of their ability to use ProModel before the proficiency project and course projects were due. The points for the problem set came from the ProModel Homework assignments.

Graded Events		Points	% of Course
1	Quiz	25	2.5%
2	Problem Set 1	25	2.5%
3	Written Partial Review (WPR) #1	150	15.0%
4	Graded Lab	100	10.0%
5	Problem Set 2	25	2.5%
6	Proficiency Project	150	15.0%
7	Course Project	275	27.5%
8	Written Partial Review (WPR) #2	200	20.0%
9	Instructor Points	50	5.0%
Total		1000	100.0%

7. **Special Emphasis.** EM481 builds upon the systems design / decision skills introduced in SE375 (Statistics for Engineers); SE387 (Deterministic Models); EM381 (Engineering Economy); SE388 (Stochastic Models); SE301 (Introduction to Systems Design); and / or EM384 (Analytical Methods for Engineering Management). Additionally, EM481 develops new analytical skills that aid in future Systems Engineering courses with the introduction of the concepts involved with Discrete Event Simulation, expanding the cadet's ability to model a large, complex, multi-disciplinary system. EM481 increases the cadet's ability to interface with a computer and produce a representation of a real-world system, to include statistical and graphical analysis of alternative systems. With the introduction of input analysis, random number / variate generation and testing, verification and validation, and output analysis, cadets are introduced to the power of simulation and its usefulness as a tool to assist in making sound decisions based upon a realistic simulation of real-world systems. Each of these topics is designed around real-world problems that the cadets must completely define, design alternative solutions for, create a simulation to represent the system, and analyze the results. EM481 uses state-of-the-art computer classroom settings, continually reinforcing the cadet's ability to interface with a computer, as well as observing the application of software products in teaching. Each succeeding exercise builds on previous learning objectives and increases the cadet's confidence and skill level in systems engineering analysis and design.

8. Historical Performance Averages.

a. Historical Course Final Grade Averages

Academic Term	Average
23-2	88.23%
23-1	89.12%
22-2	87.38%
22-1	87.05%
21-2	89.98%
21-1	89.41%
20-2	86.42%
20-1	86.40%
19-2	88.07%
19-1	89.60%
18-2	86.08%
18-1	83.86%

17-2	83.74%
17-1	85.59%
16-2	85.91%
16-1	83.80%
15-1	85.40%
14-2	85.56%
14-1	83.44%
13-2	86.08%
13-1	83.94%
12-1	84.78%
11-2	85.67%

9. Course Objectives.

a. **Body of Knowledge.** The course objectives adequately cover the body of knowledge. This course teaches the tools and methods necessary to be proficient in discrete-event simulation theory and application. The course is split evenly between theory and application. The course is informally broken down into three primary blocks of material that follow the sequence of conducting a simulation study. The first block of the course focuses on the basics of discrete-event simulation, queuing theory, random number generation, data collection, and input data analysis. The second block introduces the students to ProModel and develops their proficiency in the software and in modeling the theoretical concepts introduced in the first block. The third and final block is focused on analyzing the simulation output and comparing alternatives. The WPR and the quiz are designed to evaluate the theoretical material, whereas the graded lab and proficiency project evaluate proficiency in ProModel. The project, conducted in four parts, requires the students to work in teams to conduct a simulation study of a real system from start to finish, thereby requiring synthesis and application of the course material. This structure allows the course objectives to be met at a level appropriate for undergraduate students while simultaneously meeting systems engineering and engineering management program outcomes and objectives.

b. Effective Assessment.

(1) The WPRs and the quiz are designed to evaluate the theoretical material, whereas the graded lab and proficiency project evaluate proficiency in ProModel. The project, conducted in four parts, requires the students to work in teams to conduct a simulation study of a real system from start to finish, thereby requiring synthesis and application of the course material. This structure allows the course objectives to be met at a level appropriate for undergraduate students while simultaneously meeting systems engineering and engineering management program outcomes and objectives.

(2) The course design for AT23-2 included the addition of an additional Problem Set in lieu of five separate ProModel assignments. This assignment was only worth 25 points but allowed the cadets to have their skills tested prior to their Proficiency Project submission.

(3) The objective measures of cadet proficiency in ProModel are the Proficiency Project, the course project, and the final WPR. Average performance on the Proficiency Project increased slightly compared to 23-1. However, course project models and performance on the

WPR#2 were comparable.

(4) Although the evaluation metrics are inconclusive in stating that cadets are more proficient in ProModel and the simulation process, the course is now more properly weighted in analytical hand methods vs. simulation tools. Given that this is a simulation course with most lesson objectives focusing on simulation methods, it was determined that the graded events should follow the same distribution (35% vs. 65%). Furthermore, because the course average remained relatively the same as in previous semesters, we can conclude that general simulation proficiency increased from previous semesters.

10. **Course Processes.**

a. **Textbook.** The Harrell, et al., textbook sufficiently covers the material outlined in the course objectives. The course supplements some material not covered in the text concerning queuing theory (the textbook only provides equations for M/M/1 queues) to provide additional tools to validate simulation output.

b. **Software.** ProModel is an exceptional tool that is critical to the course. In addition to having access to the software in each lab, students can download the software on their personal laptops. The user interface of the software is user friendly and the visual output that ProModel provides can be especially compelling to stakeholders and greatly aid in their decision making. Furthermore, the output analysis and statistical analysis packages that are associated with ProModel make data analysis and output analysis easier for the user.

c. **Graded Requirements.** In general, the graded requirements in EM481 are appropriate for gauging student comprehension, extending fundamental ideas, and demonstrating the real-world relevance of the material.

11. **Changes.**

a. **Changes to Current Academic Term**

(1) **Course Objectives.** None.

(2) **Textbook.** Updated from 3rd edition to 4th edition.

(3) **Curriculum.** None.

(4) **Graded Requirements.**

(a) Problem Set 2. Removed the five ProModel assignments and combined them into an inclusive problem set that required the use of termination logic, loop logic, and balking logic.

(b) Course Project. Dictated a system (Grant Hall) with randomly generated data for analysis/implementation (student feedback indicated a major flaw in the original project design).

(5) **Other.** None.

b. **Proposed Changes for Next Academic Term**

MADN-SE

SUBJECT: AT23-2 Course Assessment for EM481 Systems Simulation

(1) **Course Objectives.** None.

(2) **Textbook.** None.

(3) **Curriculum.** None.

(4) **Graded Requirements.**

(a) Course Project. Change the system (or add for variability) into a more manufacturing-based system or

(b) WPR 2. Remove table for Chi-Square Goodness of Fit test question. Implement a “find the error” ProModel question.

(5) **Other.** None.



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT NY 10996



MADN-SE

6 January 2022

MEMORANDUM THRU Colonel James H. Schreiner, Engineering Management
Program Director, Department of Systems Engineering, West Point, NY 10996

FOR AY23-2 EM481 Sections, Department of Systems Engineering, West Point, NY
10996

SUBJECT: Instructional Memorandum for EM481 Systems Simulation

1. **PURPOSE.** The purpose of this memorandum is to provide each cadet with the administrative information required to successfully complete EM481 Systems Simulation.

2. **REFERENCES.**

a. Department of Systems Engineering (2022). *Administrative Memo 1-1: General Administrative Instructions*. West Point, NY: Department of Systems Engineering.

b. Department of Systems Engineering (2022). *Administrative Memo 1-2: Standards for Written Requirements*. West Point, NY: Department of Systems Engineering.

c. Department of Systems Engineering (2022). *Administrative Memo 1-3: Academic Documentation Policy*. West Point, NY: Department of Systems Engineering

d. Office of the Dean (2017). *Documentation of Academic Work*. West Point, NY: United States Military Academy.

e. Department of Systems Engineering (2022). *Communication Skills Handbook: A Basic Guide to Oral and Written Communications*. West Point, NY: Department of Systems Engineering.

f. Fowler, H. R. and Aaron, J. (2010). *The Little, Brown Handbook*. 11th ed. New York: Pearson Education, Inc.

3. **GENERAL.** This memorandum outlines the standards for EM481: Systems Simulation and should be used in conjunction with the course syllabus. It includes the course description, course objectives, course material, lesson assignments, grading plan, and other information relative to the execution of this course. This memorandum supersedes all previous editions. Deviation from this memorandum must be explicitly

communicated by the course instructor.

4. **SPECIFIC.**

a. **Course Scope.** Through this course, cadets will learn and explore discrete event simulation techniques used to analyze and improve complex systems. Applications for discrete event simulation include manufacturing, transportation, logistics, and service systems. Topics include the simulation study process, queuing theory, input data analysis, generation and testing of random numbers, simulation construction, verification and validation of simulation models, and output analysis. Cadets will demonstrate proficiency of lesson objectives while developing communication skills through graded events, a comprehensive course project, and an interactive classroom and lab environment. Upon completion of the course, cadets will be able to simulate and analyze moderately complex civilian and military systems.

b. **Course Strategy.** Upon completion of EM481, cadets will be able to:

- (1) Understand the process of conducting a simulation study.
- (2) Understand the application of closed form analytical queuing methods when conducting a simulation study.
- (3) Demonstrate the ability to generate and test sequences of pseudo-random numbers and selected classes of random variates.
- (4) Collect, analyze, and determine the appropriate probability distribution for data used in a simulation study (input modeling analysis).
- (5) Demonstrate comprehension of the simulation modeling process and be able to construct a simulation using course software.
- (6) Verify and validate simulation models.
- (7) Evaluate and interpret the quantitative output of simulation experiments for both single system and multiple systems (output analysis).
- (8) Demonstrate the ability to design, conduct, and analyze simulation experiments using a simulation package.
- (9) Prepare and brief a written analysis of a design problem supported by simulation results.

c. Course Material.

(1) Harrell, C., Biman, G., Gladwin, B. & Bowden, R. (2022). *Simulation Using ProModel, 4th edition*. San Diego, CA: Cognella Academic Publishing.

(2) Parnell, G., Driscoll, P. & Henderson, D. (Eds.). (2011). *Decision Making in Systems Engineering and Management, 2nd edition*. Hoboken, NJ: John Wiley and Sons, Inc.

Instructions to download ProModel will be posted in the Teams folder after WPR 1.

d. Lesson Assignments. All lesson preparation requirements including readings and practice problems are listed on the course syllabus by lesson. Additional lesson material, including class exercises and supplemental readings will be posted to the Teams folder.

e. Performance Assessment. This course consists of 1000 points and will follow the point scale below. The lesson and associated date for each graded event is listed in the syllabus.

Graded Events		Points	% of Course
1	Quiz	25	2.5%
2	Problem Set 1	25	2.5%
3	Written Partial Review (WPR) #1	150	15.0%
4	Graded Lab	100	10.0%
5	Problem Set 2	25	2.5%
6	Proficiency Project	150	15.0%
7	Course Project	275	27.5%
8	Written Partial Review (WPR) #2	200	20.0%
9	Instructor Points	50	5.0%
Total		1000	100.0%

5. ADMINISTRATIVE GUIDANCE.

a. Minimum Requirements.

Score at least 50% on WPR #2 and complete the course with at least 65%.

b. Graded Events.

Unless there are exceptional circumstances, cadets will take all WPRs, complete all exercises and projects, and participate in all briefings. Cadets who will miss an event must contact the instructor as soon they know about the conflict. It is the cadet's responsibility to coordinate with the instructor to make up any material/requirements missed.

c. Late Submissions.

Failure to submit a graded event on time for other than legitimate, compelling reasons will result in a reduced grade (5% per 24-hour period after the assignment was due). Any assignment not submitted before lesson 40 shall receive zero points. A cadet is still required to turn in a late assignment after point deductions reach zero.

d. Final Grades.

Final grades will be awarded IAW Department of Systems Engineering Administrative Memo 1-1.

e. Documentation.

Cadets may use any available reference material in preparing course work, unless otherwise noted. Use of any references for graded assignments must be properly documented IAW Reference 2c and 2d.

f. Written Requirements.

All assigned work will be submitted hard copy; digital copies will be submitted *by exception only*. Cadets will prepare project reports using the Memorandum For Record format IAW the MFR template Word document that will be posted to the Teams folder. Writing and printing will be on one side only on white, 8.5" x 11" paper. Cadets will use 12-point, Arial font for all typed or printed documents. Any digital submissions will be in .pdf format.

g. Additional Instruction (AI).

AI is available at mutually agreed upon times between the student and the instructor. You will have specific questions for an AI session. The instructor will not give general review sessions for groups of cadets prior to graded events outside of normal class periods. AI for specific assignments will not be administered 24 hours before an assignment is due. Be prepared to show your practice work and simulation models at any AI sessions. IAW Reference 2a, cadets must initiate all appointments for additional instruction.

MADN-SE

SUBJECT: Instructional Memorandum for EM481 Systems Simulation

h. Practice Problems.

Practice Problems are assigned along with reading assignments. These problems are intended to enhance understanding of text material. These problems will not be graded but solving them will enhance your understanding of the lesson objectives.

i. Class.

You will bring your textbook, charged laptop, EM481 Systems Simulation Reference Sheet, and calculator to each class. If you are not going to be in class, let your instructor know prior to the class in person or via email or Teams. It is emphasized that cadets are responsible for all lesson material *whether they attend a lesson or not*.

j. Availability and Use of Computer Laboratories.

The DSE labs are available daily from 0715 until 2300. Unless a class is being conducted in the labs, the facilities are available for use by any cadet enrolled in Systems Engineering programs on a first-come, first-served basis.

FOR THE HEAD OF THE DEPARTMENT:

EDMUND J. YOON
Major, EN/49
Course Director, EM481

EM481: Systems Simulation AY23-2							
LSN	Date	Lesson Name	Reading	Problems	Lesson Objectives	Notes	
Intro	1	11-Jan	Introduction to Simulation	1.1-1.6 1.8-1.9 1.11	1.1 1.3	1. Define Simulation 2. Understand the purpose of simulation 3. Know how to conduct a simulation study	
		13-Jan	No Planned Class #1				
	2	18-Jan	Simulation Basics	2.1-2.7 3.1-3.2 4.1-4.2	3.1	1. Define System and Processing System 2. Understand what drives complexity 3. Understand what a discrete event simulation is 4. Define system elements and the types of simulation	
Technical	3	20-Jan	Random Behavior and Statistics	3.3 5.8.2 App A	5.5 5.12	1. Know basic descriptive statistics and how to calculate them from a sample 2. Understand how we use distributions to model random behavior 3. Understand the difference between discrete and continuous random variables 4. Understand the characteristics of a Poisson Process 5. Understand key properties of the Poisson and exponential distributions 6. Know how to find critical values for statistical tests 7. Know how to interpret a p-value	
	4	23-Jan	Queuing Theory I	2.9-2.10	2.16 2.17	1. Define queue and utilization 2. Understand the purpose and limitations of analytical queuing models 3. Understand Kendall's notation 4. Know how to calculate evaluation measures of M/M/1 queues 5. Know Little's Law and general queueing relationships	Assign Proficiency Project
	5	26-Jan	Queueing Theory II	Handout	Handout	1. Know how to calculate evaluation measures of G/G/1 and G/G/s queueing models	
		30-Jan	No Planned Class #2				
	6	1-Feb	Modeling a System and Data Collection	5.1-5.6, 5.9 5.12 5.7.1 6.1-6.3 6.4.1-6.4.4		1. Know how to draw an effective and useful entity flow diagram 2. Discuss how to model a system 3. Understand guidelines for gathering data and what to do in the absence of data 4. Know steps for gathering data, the types of data, and the source of data 5. Form project groups and discuss project ideas	Quiz Assign Course Project
	7	3-Feb	Testing Input Data I	5.7.1-5.7.2		1. Define independence and homogeneity (identical distribution)	
	8	6-Feb	Testing Input Data II	5.8.3	5.19	2. Understand the importance of testing for independence and homogeneity	Problem Set 1
	9	9-Feb	Random Number and Random Variate Generation	3.4-3.6 8.2.1	3.6 3.8	1. Define Random Number Generator (RNG) and random variate 2. Know the desirable properties of a RNG 3. Know how to generate RNs using a linear congruential generator (LCG) and how to determine if the LCG will have a full cycle 4. Know how to apply the inverse transformation method to generate random numbers 5. Review Lessons 1-9	
	10	13-Feb	WPR1				WPR1
		16-Feb	No Planned Class #3				
Software-Assisted	11	22-Feb	Data Analysis using Stat::Fit	5.10-5.11	L5 Appendix A	1. Understand how boundless distributions affect simulation 2. Understand how to model discrete distributions using continuous distributions 3. Know how to use Stat::Fit to test for independence and homogeneity	Install ProModel
	12	24-Feb	Simulation Modeling using LEAP I		L1 L2.1	1. Understand the basic ProModel environment and how to build a discrete event simulation in ProModel 2. Know how to use the LEAP model building process 3. Know how to use the Logic Builder 4. Know how to implement the ProModel command WAIT	Proficiency Project P1 Due
	13	27-Feb	Simulation Modeling using LEAP II				
	14	1-Mar	Variables and the Output Viewer			1. Receive approval and feedback on course project 1. Know the difference between simple average and time average statistics 2. Know how to assign local and global variables in ProModel 3. Know how to implement the ProModel commands REAL and INT 4. Interpret results in Output Viewer	
		3-Mar	No Planned Class #4				Project P1 Due
	15	13-Mar	Attributes and User Distributions - Logic and Routing		L4.4 L4.11.1 L6.1-6.2	1. Know the purpose of and how to create attributes and user distributions 2. Know how to assign attributes a value from a user distribution 3. Implement routing using attributes and logic 4. Know how to implement the ProModel commands: CLOCK, IF THEN ELSE, ROUTE, and GRAPHIC 6. Know how to use attributes to gather simple average statistics	
	16	15-Mar	Resource and Path Networks		L4.3 L12.2	1. Understand the purpose of and how to build resources and path networks 2. Know how to implement the ProModel commands: MOVE FOR, MOVE WITH, GET, FREE, and USE	
		17-Mar	No Planned Class #5				
	17	20-Mar	Batching and Attaching Entities I		L4.7-4.9 L6.10	1. Understand how to use temporary and permanent attachments and batching 2. Know how to create package files and import graphics	
	18	23-Mar	Batching and Attaching Entities II		L6.12	3. Know how to implement the ProModel commands: GROUP/ UNGROUP, COMBINE, JOIN, LOAD/ UNLOAD, ACCUM	Project Part 2 Due 24MAR
	19	27-Mar	Priorities, Preemption, and Pull Systems		L6.7-6.8	1. Understand the purpose and use of priorities and preemption 2. Identify and know how to model pull systems 3. Know how to implement the ProModel commands: SEND, ORDER	
	20	29-Mar	Verification and Validation	7	L7.2-7.3 L11.1-11.2	1. Define validation 2. Define verification 3. Understand how to use validation and verification techniques 4. Know how to implement the ProModel commands: TRACE, PAUSE, DEBUG 5. Learn how to use ProModel Macro and Scenario features and apply them to a simulation	
		31-Mar	*Practice Graded Lab*				
	21	3-Apr	Graded Lab				Graded Lab
	22	6-Apr	Loops and Termination Logic		L4.11.2 L8.2-8.2.1	1. Know how to implement the ProModel commands: DO WHILE, DO UNTIL, WHILE DO, STOP 2. Know how to end simulations according to defined criteria using logic 3. Know how to model balking and jockeying customer behavior	
Analysis	23	10-Apr	Macros and Scenarios	L11.1-11.2		1. Understand how to simulate multiple scenarios in ProModel using macros. 2. Know how to evaluate the output from multiple scenarios from ProModel Output Viewer.	Problem Set 2
	24	12-Apr	Output Analysis of a Single System I	8.2.2-8.3		1. Know how to construct confidence intervals using simulation results 2. Understand how to determine the appropriate number of replications to achieve desired error 3. Understand the implications of the Central Limit Theorem and replications in simulation	
		14-Apr	No Planned Class #7				Proficiency Project Due
	25	17-Apr	Output Analysis of a Single System II	8.4 8.6	L8.3	1. Understand difference between terminating and non-terminating simulations 2. Understand warm-up periods and how they apply to non-terminating simulations 3. Know how to determine the length of a warm-up period using visual methods and Welch Moving Average Method 4. Understand how to create independent observations from a non-terminating simulation using replications and the batch method	
	26	20-Apr	Output Analysis of Multiple Systems	9.1-9.3 9.4.2 9.4.3	9.1a	1. Understand hypothesis testing in comparing two simulation alternatives 2. Know how to use Welch Confidence Interval to compare two alternatives 3. Know how to use Paired-t Confidence Interval to compare two alternatives 4. Analyze simulation output of multiple alternatives using ANOVA (analysis of variance) 5. Analyze simulation output of multiple alternatives using Fisher's Least Significant Difference method	
	27	24-Apr	Course Review				Project P3 Due
		26-Apr	No Planned Class #8				
		1-May	No Planned Class #9				
	28	5-May	Presentations				
	29	8-May	Presentations				Project P4 Due
		10-May	No Planned Class #10				
	30	11-May	WPR2				WPR2



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT NY 10996

MADN-SE-EM-481

16 February 2022

MEMORANDUM FOR COL James Schreiner, Engineering Management Program Director

SUBJECT: EM481 Written Partial Review (WPR) Grade Approval for Academic Term 23-2

1. EM481 WPR1 took place on lesson 12 on 13 February 2023. There are 42 students in the course. 41 students took the exam in MH102 on 13 February 2023. One student took the exam in MH414 with instructor oversight on 15 February 2023 due to being unable to take it with the class.

2. **Course Objective Mapping.** The following table identifies the course and lesson objectives that were tested according to each question on the WPR.

Question	Course Objectives	
1-5	Understand the process of conducting a simulation study.	
2, 4, 5	Understand the application of closed form analytical queuing methods when conducting a simulation study.	
3	Demonstrate the ability to generate and test sequences of pseudo-random numbers and selected classes of random variates.	
1, 2	Collect, analyze, and determine the appropriate probability distribution for data used in a simulation study (input modeling analysis).	
Question	Lesson Objectives	Obj #
1-5	Know how to conduct a simulation study	1.3
1	Understand what drives complexity	2.2
5	Understand key properties of the Poisson and exponential distributions	3.5
2, 4	Know how to find critical values for statistical tests	3.6
1, 2, 4	Know how to interpret a p-value	3.7
1, 5	Define queue and utilization	4.1
5	Understand Kendall's notation	4.3
5	Know Little's Law and general queueing relationships	4.5
5	Know how to calculate evaluation measures of G/G/1 and G/G/s queueing models	5.1
1,4	Understand the importance of testing for independence and homogeneity	7.2
1,4	Know how to use scatter plots, runs tests, and autocorrelation to test for independence	7.3
1,4	Know how to conduct visual test for homogeneity	7.4
2	Understand the importance of fitting a distribution to data	8.1
2	Know how to use the Chi-Square goodness of fit test	8.2
1, 3	Know the desirable properties of a RNG	9.2
3	Know how to generate RNs using a linear congruential generator	9.3
3	Know how to apply the inverse transformation method to generate random numbers	9.4

Table 1. EM481 Course Objective Mapping

3. **Scores and Historical Data.** The weight of this exam is 15 percent of the course. There are 42 students enrolled in the course. The overall course average on the WPR was 80.11%. This is a 6.94% decrease from AT23-1 and 6.47% below the 5-term moving average. There were no major changes to AT23-2 EM481 WPR1 from the AT23-1 EM481 WPR1.

Academic Term	Average	5-term Moving Avg
AT 23-1	87.05%	86.58%
AT 22-2	88.85%	84.97%
AT 21-2	87.00%	83.40%
AT 21-1	84.00%	80.80%
AT 20-2	86.00%	81.40%
AT 20-1	79.00%	80.80%

Table 2. EM481 Historical WPR Averages

4. **WPR Additional Statistics.** Additional descriptive statistics and the histogram of grades for WPR1 are depicted in Table 3 and Figure 1, respectively.

AY-T: 2023-2

Course Grading Scale	Cadets		Points:	Percent:
A+ >= 97.50	2	Highest:	150.00	100.00%
A >= 92.50	5	Average:	120.17	80.11%
A- >= 90.00	1	Lowest:	82.00	54.67%
B+ >= 87.50	4	Standard Deviation:		11.06
B >= 82.50	4	Average Letter Grade:		B-
B- >= 80.00	8	Section GPA:		2.49
C+ >= 77.50	2	Total Number of Cadets:		42
C >= 72.50	7			
C- >= 70.00	3			
D >= 65.00	2			
F >= 0.00	4			
NC	0			

Table 3. AT23-2 EM481 WPR1 Descriptive Statistics.

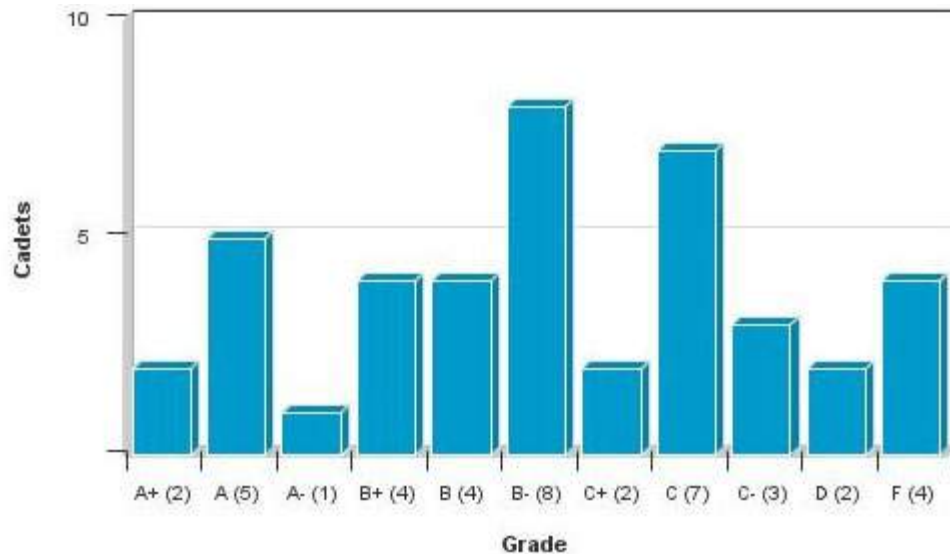


Figure 1. AT23-2 EM481 WPR1 Histogram of Grades

5. Comments/Recommendations.

a. Problem 1: Concept Matching.

The course average for problem 1 was 25.62 of 30 points (85.4%). This was mitigated by highlighting important terms in published lesson slides. In the future, it is recommended to tell the students to create an EM481 glossary by pulling out the key terms from each lesson into a separate document to better prepare for this specific style of question.

b. Problem 2: Testing Goodness of Fit.

The course average for problem 2 was 23 out of 30 points (76.67%). This question solely revolved around whether or not students understand how to state the null hypothesis properly and whether students knew how to use the chi square test or not. A trend emerged where students used the number of observations to calculate the critical value instead of using infinite degrees of freedom.

c. Problem 3: Random Number and Variate Generation.

The course average for problem 3 was 24.29 out of 30 points (80.95%). The students who did not receive maximum points for this question struggled to understand the concept of how to use “mod” or simply made a transcription error (“m” value written down as 13 instead of 31). The part of the question that tested the student’s ability to apply the inverse transformation method was missed by many of the students. The PDF and CDF were given (despite being on the reference sheet) and could possibly have misled the cadets since some tried to use the PDF instead of the CDF. Recommend no change as this differentiates those cadets who understand the material versus those who just know how to solve the math for it without understanding why.

d. Testing for Independence and Homogeneity.

The course average for problem 4 was 25.06 out of 30 points (83.53%). This section of the exam was meant to be easy points but the students showed a tendency to not understand what

the three tests for independence were. Additionally, there seems to be a trend of students not understanding the difference between independence and homogeneity. A slide was added to lesson 9 that succinctly states the definition of independence and how to test for it and contrast that with the definition of homogeneity (including the words “identically distributed”) and how to test for it. Although the question explicitly stated to use the Runs Above/Below the Median test, some students opted to use the Runs Up/Down Test, which caused them to lose a significant number of points on this question.

e. Problem 5. Queuing and Data Collection.

The course average for problem 5 was 22.2 out of 30 points (74.01%). The largest disconnect for this portion of the exam seems to have been from failing to understand when to use which equation for queuing questions. A “queuing theory review” slide was added at the end of lesson 7 before the quiz review between AT23-1 and AT23-2. Students largely knew when to use which equation but stopped calculating after only doing half of the last problem (they would stop after calculating W_q and the question asks for number of calls waiting, or L_q).

f. The slides were published after every lesson to the Teams folder in PDF format for download/review. All in class exercise solutions were posted with the slides for studying, as well as the solutions for problem set 1 and quiz 1.

6. The point of contact for this memorandum is the undersigned at edmund.yoon@westpoint.edu.

EDMUND J. YOON
Major, EN/49
EM481 Course Director



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT NY 10996

MADN-SE-EM

15 May 2023

MEMORANDUM FOR COL James Schreiner, Engineering Management Program Director

SUBJECT: EM481 Written Partial Review (WPR) 2 Grade Approval for Academic Term 23-2

1. EM481 WPR 2 took place on lesson 40 on 11 May 2023. There are 42 students in the course. 40 students took the exam in MH102 on 11 May 2023. Two students took the exam in MH414 with instructor oversight on 10 and 13 May 2023 due to being unable to take it with the class.

2. **Course Objective Mapping.** The following table identifies the course and lesson objectives that were tested according to each question on the WPR.

Question	Course Objectives	
1-6	Understand the process of conducting a simulation study.	
4,6	Understand the application of closed form analytical queuing methods when conducting a simulation study.	
1, 3, 4, 5	Demonstrate the ability to generate and test sequences of pseudo-random numbers and selected classes of random variates.	
1, 3, 5	Collect, analyze, and determine the appropriate probability distribution for data used in a simulation study (input modeling analysis).	
Question	Lesson Objectives	Obj #
6.1	Understand what drives complexity	2.2
6.2, 4.1	Understand Kendall's Notation	4.3
6.3	Know steps for gathering data, the types of data, and the source of data	6.4
6.4	Know how to use scatter plots, runs tests, and autocorrelation to test for independence	7.3
6.5	Understand the importance of fitting a distribution to data	7.2
6.6	Know how to use the LEAP model building process	12.2
6.7	Know the purpose of and how to create attributes and user distributions	15.1
6.8	Understand how to use temporary and permanent attachments and batching	17.1
6.9	Understand the purpose and use of priorities and preemption	19.1
6.10	Define verification	20.2
4.2-4.4	Know how to calculate evaluation measures of M/M/1 queues	4.4
1.1-1.3	Know how to use the Chi-Square goodness of fit test	10.2
2.1	Know how to implement the ProModel commands: IF/THEN/ELSE	15.4
2.2	Know how to implement the ProModel commands: DO WHILE	22.1
3.1	Know how to construct confidence intervals using simulation results	24.1

3.2	Understand how to determine the appropriate number of replications to achieve desired error	24.2
3	Know how to use Paired-t Confidence Interval to compare two alternatives	26.3

Table 1. EM481 Course Objective Mapping

3. **Scores and Historical Data.** The weight of this exam is 20 percent of the course. There are 42 students enrolled in the course. The overall course average on the WPR was 90.46%. This is a 3.41% increase from AT23-1. There were no major changes to AT23-2 EM481 WPR 2 from the AT23-1 EM481 WPR 2.

Academic Term	Average	5-term Moving Avg
AT23-2	90.46%	87.47%
AT23-1	87.05%	86.58%
AT 22-2	88.85%	84.97%
AT 21-2	87.00%	83.40%
AT 21-1	84.00%	80.80%

Table 2. EM481 Historical WPR 2 Averages

4. **WPR Additional Statistics.** Additional descriptive statistics and the histogram of grades for WPR 2 are depicted in Table 3 and Figure 1, respectively.

AY-T: 2023-2

Course Grading Scale	Cadets		Points:	Percent:
A+ >= 97.50	7	Highest:	200.00	100.00%
A >= 92.50	11	Average:	180.93	90.46%
A- >= 90.00	6	Lowest:	152.00	76.00%
B+ >= 87.50	5	Standard Deviation:		6.31
B >= 82.50	8	Average Letter Grade:		A-
B- >= 80.00	3	Section GPA:		3.56
C+ >= 77.50	1	Total Number of Cadets:		42
C >= 72.50	1			
C- >= 70.00	0			
D >= 65.00	0			
F >= 0.00	0			
NC	0			

Table 3. AT23-2 EM481 WPR2 Descriptive Statistics.

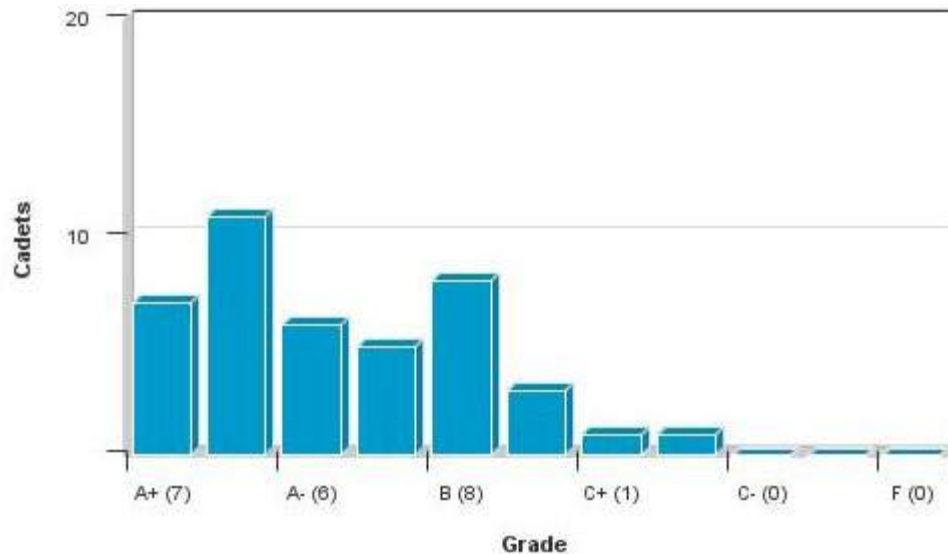


Figure 1. AT23-2 EM481 WPR1 Histogram of Grades

5. Comments/Recommendations.

a. Problem 1: Distribution Fitting.

The course average for problem 1 was 35.98 of 40 points (89.94%). Recommend removing the table to fill in and forcing Cadets to generate the table themselves. This will be a better gauge of their mastery of the Chi-Square Goodness of Fit Test.

b. Problem 2: Output Analysis of a Single System.

The course average for problem 2 was 28.74 out of 30 points (95.79%). Recommend adding a relative error question and changing the total points values for each sub question to 10 points each.

c. Problem 3: Output Analysis of Multiple Systems.

The course average for problem 3 was 38.83 out of 40 points (97.08%). Recommend removing the "Difference of Means" column as well as the Sample Mean and Sample Standard Deviation; Cadet calculations to solve for these will better fit the weight of this problem in future iterations of this exam.

d. Problem 4: Queueing Theory.

The course average for problem 4 was 27.21 out of 30 points (90.71%). Recommend increasing the difficulty of this problem by making it a Kingman's Multi-Server Problem (similar to WPR 1).

e. Problem 5: ProModel.

The course average for problem 5 was 25.88 out of 30 points (86.27%). Recommend restructuring the problem to include error-identification versus code-generation. This will better assess Cadets' understanding of ProModel without penalizing them for not having the coding language memorized.

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f. Problem 6: Multiple Choice.

The course average for problem 6 was 24.29 out of 30 points (80.95%). No change recommended.

g. The slides were published after every lesson to the Teams folder in PDF format for download/review. All in class exercise solutions were posted with the slides for studying, as well as the solutions for problem set 1 and quiz 1.

6. The point of contact for this memorandum is the undersigned at edmund.yoon@westpoint.edu.

EDMUND J. YOON
Major, EN/49
EM481 Course Director

Appendix A. Course OML with Cut Lines

OM	Name	Max	Points	Percent	Letter	Q1		PSet1		WPR1		GLab		Pset2		PProj		Proj		WPR2		Partic		Attitu		Standard Cut Line	Recommended Cut Line
		Points	Earned		Grade	25		25		150		100		25		150		275		200		30		20			
1	BARREDA, BROOKE G	1000	957.0	95.70%	A	22	88.00%	23.5	94.00%	141.5	94.33%	100	100.00%	22	88.00%	150	100.00%	272	98.91%	178	89.00%	28	93.33%	20	100.00%		
2	DOUTHWAITE, TRENTON R	1000	954.0	95.40%	A	25	100.00%	24	96.00%	134	89.33%	96	96.00%	25	100.00%	144	96.00%	272	98.91%	184	92.00%	30	100.00%	20	100.00%		95.40%
3	MOON, JACOB L	1000	947.5	94.75%	A	25	100.00%	22.5	90.00%	142	94.67%	73	73.00%	23	92.00%	147	98.00%	272	98.91%	197	98.50%	27	90.00%	19	95.00%		
4	BELLAMY, ELJAH J	1000	946.5	94.65%	A	24.5	98.00%	24	96.00%	148	98.67%	94	94.00%	22	88.00%	147	98.00%	239	86.91%	198	99.00%	30	100.00%	20	100.00%		
5	HOPKINS, EDA J	1000	944.5	94.45%	A	23.5	94.00%	25	100.00%	144	96.00%	79	79.00%	23	92.00%	142	94.67%	263	95.64%	195	97.50%	30	100.00%	20	100.00%		
6	GRABHER, STEPHEN M	1000	942.5	94.25%	A	24.5	98.00%	25	100.00%	139	92.67%	83	83.00%	22	88.00%	143	95.33%	264	96.00%	194	97.00%	30	100.00%	18	90.00%		
7	HWANG, MAX S	1000	939.5	93.95%	A	22.5	90.00%	25	100.00%	150	100.00%	90	90.00%	23	92.00%	141	94.00%	264	96.00%	178	89.00%	28	93.33%	18	90.00%		
8	KIM, MOSES J	1000	939.0	93.90%	A	25	100.00%	25	100.00%	134	89.33%	97	97.00%	24	96.00%	138	92.00%	248	90.18%	200	100.00%	29	96.67%	19	95.00%		
9	HOFFMANN, DANIEL D	1000	937.0	93.70%	A	16	64.00%	23	92.00%	133	88.67%	97	97.00%	22	88.00%	144	96.00%	264	96.00%	188	94.00%	30	100.00%	20	100.00%		
10	MARCHI, MAXIMUS A	1000	929.0	92.90%	A	23	92.00%	23.5	94.00%	140.5	93.67%	93	93.00%	25	100.00%	115	76.67%	272	98.91%	189	94.50%	28	93.33%	20	100.00%		92.50%
11	DU TOIT, SCHARL M	1000	921.5	92.15%	A-	21	84.00%	25	100.00%	118.5	79.00%	100	100.00%	25	100.00%	141	94.00%	247	89.82%	194	97.00%	30	100.00%	20	100.00%		
12	KNUDSEN, SPENCER B	1000	914.5	91.45%	A-	21	84.00%	24	96.00%	129.5	86.33%	100	100.00%	24	96.00%	137	91.33%	242	88.00%	189	94.50%	28	93.33%	20	100.00%		
13	WESLEY, SAMUEL J	1000	914.0	91.40%	A-	23.5	94.00%	24.5	98.00%	125	83.33%	82	82.00%	24	96.00%	145	96.67%	248	90.18%	193	96.50%	29	96.67%	20	100.00%		
14	JONES, JEMEL A	1000	913.5	91.35%	A-	24	96.00%	22.5	90.00%	110	73.33%	97	97.00%	25	100.00%	143	95.33%	247	89.82%	195	97.50%	30	100.00%	20	100.00%		
15	MICHEL, NOLAN T	1000	909.5	90.95%	A-	24	96.00%	24.5	98.00%	137	91.33%	58	58.00%	24	96.00%	145	96.67%	264	96.00%	190	95.00%	25	83.33%	18	90.00%		
16	BUDOFF, PAUL F	1000	907.0	90.70%	A-	21.5	86.00%	25	100.00%	112.5	75.00%	98	98.00%	24	96.00%	135	90.00%	246	89.45%	197	98.50%	28	93.33%	20	100.00%		
17	SARLITTO, CHRISTIAN J	1000	904.5	90.45%	A-	20.5	82.00%	25	100.00%	134	89.33%	98	98.00%	25	100.00%	147	98.00%	219	79.64%	188	94.00%	28	93.33%	20	100.00%		
18	WOODMAN, ANDREW J	1000	903.5	90.35%	A-	24	96.00%	22	88.00%	123.5	82.33%	77	77.00%	23	92.00%	138	92.00%	263	95.64%	190	95.00%	25	83.33%	18	90.00%		90.00%
19	ABBOTT, JASON	1000	898.0	89.80%	B+	21	84.00%	22.5	90.00%	122.5	81.67%	100	100.00%	23	92.00%	135	90.00%	248	90.18%	179	89.50%	28	93.33%	19	95.00%		89.80%
20	HARRISON, JACK M	1000	892.5	89.25%	B+	22	88.00%	24.5	98.00%	113	75.33%	89	89.00%	24	96.00%	139	92.67%	263	95.64%	168	84.00%	30	100.00%	20	100.00%		
21	PITHAN, JOHN N	1000	889.0	88.90%	B+	20	80.00%	21.5	86.00%	125.5	83.67%	85	85.00%	22	88.00%	138	92.00%	247	89.82%	180	90.00%	30	100.00%	20	100.00%		
22	SANGVIC, VICTOREA C	1000	881.0	88.10%	B+	22	88.00%	20.5	82.00%	109.5	73.00%	91	91.00%	23	92.00%	140	93.33%	246	89.45%	181	90.50%	28	93.33%	20	100.00%		
23	MURACO, JOHN F	1000	875.5	87.55%	B+	19	76.00%	25	100.00%	98.5	65.67%	97	97.00%	24	96.00%	148	98.67%	219	79.64%	197	98.50%	28	93.33%	20	100.00%		
24	MITCHELL, BROOK A	1000	875.0	87.50%	B+	20	80.00%	22.5	90.00%	120.5	80.33%	87	87.00%	23	92.00%	146	97.33%	239	86.91%	169	84.50%	28	93.33%	20	100.00%		87.50%
25	KONOPA, BRIDGET A	1000	866.5	86.65%	B	17.5	70.00%	21.5	86.00%	123.5	82.33%	83	83.00%	22	88.00%	137	91.33%	242	88.00%	174	87.00%	27	90.00%	19	95.00%		
26	BAZAN, ADAM L	1000	865.5	86.55%	B	19.5	78.00%	23	92.00%	108	72.00%	84	84.00%	25	100.00%	140	93.33%	247	89.82%	171	85.50%	28	93.33%	20	100.00%		
26	FENN, JAMES E	1000	865.5	86.55%	B	20.5	82.00%	17.5	70.00%	116.5	77.67%	77	77.00%	24	96.00%	138	92.00%	246	89.45%	181	90.50%	27	90.00%	18	90.00%		
28	KOCH, BENJAMIN S	1000	860.5	86.05%	B	23	92.00%	21	84.00%	113.5	75.67%	89	89.00%	24	96.00%	140	93.33%	238	86.55%	167	83.50%	27	90.00%	18	90.00%		
29	COSENKE, TREVOR H	1000	860.0	86.00%	B	17.5	70.00%	20.5	82.00%	109	72.67%	88	88.00%	23	92.00%	131	87.33%	238	86.55%	186	93.00%	28	93.33%	19	95.00%		
30	AL SHISHANY, AHMAD	1000	856.5	85.65%	B	22	88.00%	22	88.00%	124.5	83.00%	82	82.00%	22	88.00%	139	92.67%	246	89.45%	152	76.00%	28	93.33%	19	95.00%		
31	LOPEZ, BENJAMIN T	1000	848.5	84.85%	B	22.5	90.00%	25	100.00%	123	82.00%	80	80.00%	18	72.00%	134	89.33%	219	79.64%	184	92.00%	25	83.33%	18	90.00%		
32	BAKRE, ADAMA	1000	845.5	84.55%	B	22	88.00%	20.5	82.00%	122	81.33%	57	57.00%	23	92.00%	132	88.00%	248	90.18%	176	88.00%	27	90.00%	18	90.00%		
33	RUBIN, JOEL A	1000	843.5	84.35%	B	23	92.00%	17	68.00%	106.5	71.00%	71	71.00%	24	96.00%	138	92.00%	239	86.91%	180	90.00%	25	83.33%	20	100.00%		
34	MCMICKENS, TAVIN E	1000	831.0	83.10%	B	19.5	78.00%	21.5	86.00%	98	65.33%	75	75.00%	23	92.00%	143	95.33%	242	88.00%	163	81.50%	27	90.00%	19	95.00%		
34	SAYERS, JACOB H	1000	831.0	83.10%	B	10	40.00%	20	80.00%	122	81.33%	79	79.00%	23	92.00%	132	88.00%	238	86.55%	161	80.50%	27	90.00%	19	95.00%		
36	AGUILAR, RYAN R	1000	827.0	82.70%	B	18	72.00%	22	88.00%	91	60.67%	49	49.00%	25	100.00%	140	93.33%	263	95.64%	176	88.00%	25	83.33%	18	90.00%		82.50%
37	EBERLE, LINDEN J	1000	815.0	81.50%	B-	18.5	74.00%	20	80.00%	92.5	61.67%	69	69.00%	24	96.00%	127	84.67%	247	89.82%	169	84.50%	28	93.33%	20	100.00%		
38	KELLY, JOSEPH L	1000	814.5	81.45%	B-	18	72.00%	24	96.00%	110.5	73.67%	67	67.00%	22	88.00%	120	80.00%	239	86.91%	171	85.50%	25	83.33%	18	90.00%		
39	LEGGETT, CONNER R	1000	810.5	81.05%	B-	20	80.00%	18.5	74.00%	108	72.00%	66	66.00%	22	88.00%	130	86.67%	246	89.45%	155	77.50%	27	90.00%	18	90.00%		
40	BAUGH, DAYTON M	1000	804.0	80.40%	B-	16	64.00%	19.5	78.00%	88.5	59.00%	73	73.00%	24	96.00%	124	82.67%	242	88.00%	167	83.50%	30	100.00%	20	100.00%		80.00%
41	WEST, BENJAMIN S	1000	790.0	79.00%	C+	20.5	82.00%	21.5	86.00%	82	54.67%	67	67.00%	22	88.00%	121	80.67%	247	89.82%	164	82.00%	28	93.33%	17	85.00%		
42	MCGOVERN, EMMA L	1000	786.5	78.65%	C+	24	96.00%	25	100.00%	122.5	81.67%	75	75.00%	22	88.00%	83	55.33%	219	79.64%	191	95.50%	10	33.33%	15	75.00%		



UNITED STATES MILITARY ACADEMY
WEST POINT.

SECTION II

EXAMINATIONS



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB G

WPR #1

Instructor Solution

**DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996**

EM481 WRITTEN PARTIAL REVIEW #1

1. WPR 1 is a 75-minute exam worth 160 points. It consists of 24 problems on 12 pages, including the cover page. Prior to beginning, ensure that your exam is complete.
2. Authorized References: Pencil/pen, straight edge, calculator, instructor provided EM481 reference sheet.
3. Show all your work. Unsupported work earns little or no partial credit. Neatness will also help your cause.
4. Extra paper is available at the front of the classroom. Attach any additional pages to your exam when you turn them in. Label additional pages with your last name, section, and problem number.
5. Ensure you write in your name and section legibly in the boxes below.
6. Grading Summary:

Section	Possible Points	Score
1. Simulation Basics & Modeling	39	
2. Queuing	54	
3. Testing for Independence and Homogeneity	40	
4. Testing for Goodness of Fit	27	
Total	160	

Last, First Name:
(write large & legible)

SOLUTION

Section:

Section 1: Simulation Basics & Modeling (39 points)

- 1) **(2.5 Points, Each)** Match the DEFINITION to the TERM, using the table of terms below.

Definition	Term
A collection of elements that function together to achieve a desired goal.	B
Autocorrelation tests for this type of data characteristic.	A
In addition to variability, this is what makes a system complex.	I
A systems element that is a means by which activities are performed	P
A figure that visually demonstrates the route of entities in a system. Also contains useful quantitative information to describe a simulation system.	H
A system element that are the items processing through the system	N
This is helps classify queuing systems. It uses three different parameters to describe the queuing system.	C
Imitation of a dynamic system using a computer model in order to evaluate and improve system performance	L
A statistical assumption which implies that data come from the same underlying population (same distribution and parameter values)	G
A system element that dictates how, where, and when activities are performed	K
Poisson and exponential distributions fall under this category of distributions.	F
Technique that compares the empirical data histogram to the hypothesized probability mass or density function and requires the specification of intervals or grouping of data (bins)	Q

A Independence	K Controls
B System	L Simulation
C Kendall's Notation	M p -value
D Autocorrelation	N Entities
E Variability	O Structural
F Markovian	P Resources
G Homogeneity	Q Chi-Square Test
H Entity Flow Diagram	R Operational
I Interdependencies	S Activities
J Numerical	T Anderson-Darling Test

- 2) **(3 Points)** (TRUE | FALSE) In Static Simulations, state variables are updated as time advances.
- 3) **(3 Points)** (TRUE | FALSE) In Discrete Simulations, variables only change at discrete periods (triggered by events) and cannot incorporate any continuous distributions.
- 4) **(3 Points)** (TRUE | FALSE) In Deterministic Simulations, all future variable states are determined once the input data and initial state are defined.

Section #2: Queuing (54 points)

Vehicles arrive at the Grand Canyon South Rim Entrance following a Poisson process, with an average arrival rate of 220 vehicles per hour. Vehicles can use one of three ticket booths to purchase entry passes. It takes each vehicle an average of 1.33 minutes (exponentially distributed) to purchase a pass from a booth.

5) **(3 points)** What type of queuing model is this (in Kendall's notation)?

M / M / 3

ANS: _____

6) **(4 points)** Clearly state the arrival rate and the service rate, in vehicles per minute.

$$\lambda = \frac{220 \text{ vehicles}}{60 \text{ minutes}} = 3.667 \text{ vehicles/minute}$$

$$u = \frac{1}{(1.33 \text{ min})} = 0.752 \text{ vehicles/minute}$$

Arrival Rate: _____

Service Rate: _____

7) **(5 points)** Will the system reach steady-state conditions? Justify your answer mathematically.

$$p = \frac{\lambda}{s * \mu} = \frac{3.667}{3 * 0.752} = 1.625$$

No. It will not reach steady-state, because the utilization is greater than 1.

- 8) **(6 points)** On weekends, the arrival rate increases to an average of 260 vehicles per hour, according to a Poisson process. What is the probability that exactly 40 vehicles will arrive in the next 10 minutes?

$$\lambda = \frac{260 \text{ vehicles}}{60 \text{ minutes}} = 4.333 \text{ vehicles/minute}$$

$$P(X = x) = \frac{e^{-\lambda t} (\lambda t)^x}{x!}$$

$$P(X = 40) = \frac{e^{-4.333 \cdot 10} (4.333 \cdot 10)^{40}}{40!} = 0.0552 = 5.52\%$$

Probability: _____

- 9) **(12 points)** During peak times, the system operates as an M/G/2 model. The arrival rate is 5.4 vehicles per minute. The service time for each of the four booths follows a gamma distribution with a mean of 1.42 vehicles per minute and standard deviation of 0.1 minutes. The entrance is busy 95% of the time. How long do vehicles wait in the line on average? What is the expected number of vehicles in line?

$$p = 0.95; \quad \mu = 1.42; \quad \sigma_s = 0.1$$

$$W_q = \left(\frac{c_a^2 + c_s^2}{2} \right) \left(\frac{\rho^{\sqrt{2(s+1)}-1}}{s(1-\rho)} \right) \left(\frac{1}{\mu} \right)$$

$$c_a = 1 \text{ (Markovian)}$$

$$c_s = \mu \sigma_s = (1.42)(0.1) = 0.142$$

$$W_q = \left(\frac{1^2 + 0.142^2}{2} \right) \left(\frac{0.95^{\sqrt{2(4+1)}-1}}{4(1-0.95)} \right) \left(\frac{1}{0.714} \right) = (0.51)(4.475)(1.40) = 3.195 \text{ min}$$

$$L_q = \lambda W_q = 5.4 * 3.195 = 17.25 \text{ vehicles}$$

Average Wait in Line: _____

Expected # of Vehicles in Line: _____

The mean service time for the single server at the concession stand follows a Poisson process with a mean of 2.5 minutes. Customers arrive every 3 minutes, exponentially distributed. – **Round to the nearest hundredth for questions 10-15.**

10)(3 points) What is the utilization?

$$u = \frac{1 \text{ customer}}{2.5 \text{ minutes}} = 0.4 \text{ customers/minute}$$
$$\lambda = \frac{1 \text{ customer}}{3 \text{ minutes}} = 0.333 \text{ customer/minute}$$
$$\rho = \frac{\lambda}{\mu} = \frac{0.333}{0.4} = 0.833$$

ANS: _____

11)(3 points) How long will customers wait in line?

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)} = \frac{0.333}{0.4 * (0.4 - 0.333)} = 12.43 \text{ minutes}$$

ANS: _____

12)(3 points) How many customers would you expect in the system?

$$L = \frac{\rho}{1 - \rho} = \frac{0.833}{1 - 0.833} = 4.988 \text{ customers}$$

ANS: _____

13)(3 points) What is the expected average time in the system?

$$W = \frac{1}{\mu - \lambda} = \frac{1}{0.4 - 0.333} = 14.93 \text{ min}$$

ANS: _____

14)(4 points) What is the probability that 6 customers are in the system?

$$P_n = (1 - \rho)\rho^n = (1 - 0.833)0.833^6 = 0.0558 = 5.58\%$$

ANS: _____

15)(8 points) What is the probability that there will be more than 2 customers at the concession stand?

$$P_n = (1 - \rho)\rho^n = (1 - 0.833)0.833^n$$

$$P_n = (1 - \rho)\rho^0 = 0.167$$

$$P_n = (1 - \rho)\rho^1 = 0.139$$

$$P_n = (1 - \rho)\rho^2 = 0.116$$

$$P_{n>3} = 1 - \sum_0^3 \rho^n = 1 - 0.422 = 0.578 = 57.8\%$$

ANS: _____

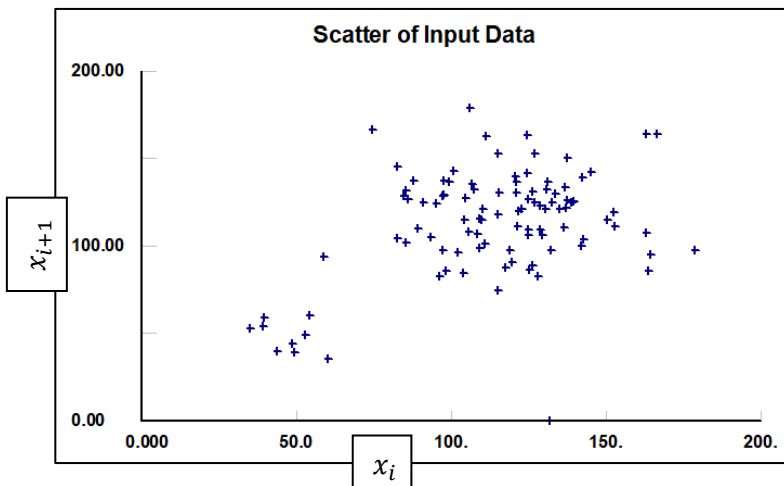
Section #3: Testing for Independence and Homogeneity (40 points)

You recently collected data on customer interarrival times, and now wish to analyze the data.

16)(6 Points) Name the three major methods to test for independence taught in EM481?

Checks if an observation is influenced by its preceding observation.	Scatterplot
Compares the observed number of sequenced observations to the expected number of sequenced observations	Runs Test
Checks if observations in a single dataset are correlated over different time lags.	Autocorrelation

17)(8 Points) You generated the following charts from historical data. What can you conclude about the data in terms of independency and homogeneity?

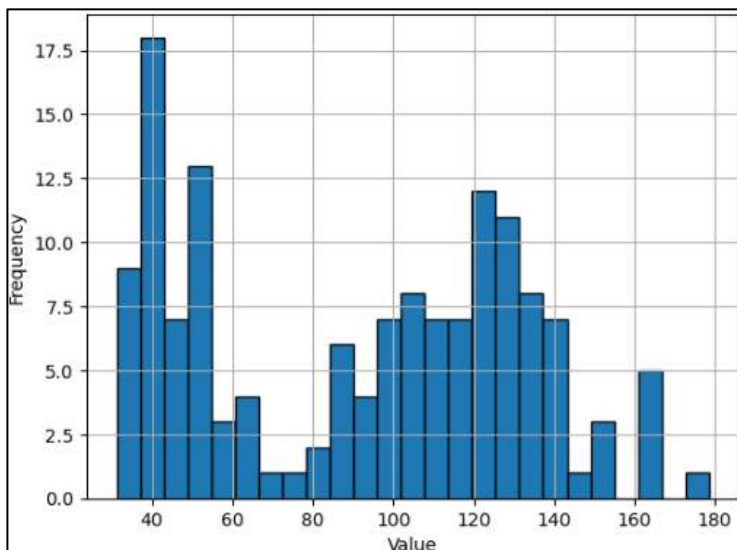


What type of chart is this?

Scatterplot

Conclusion from the graph:

Clustering indicates that the data is not / may not be independent



What type of chart is this?

Histogram

Conclusion from the graph:

Appears to be multi-modal / non-homogeneous

- 18) **(16 Points)** You decide to conduct a ***Runs Above/Below the Median Test*** on the 30 data points below to determine whether the data is independent (random) given a level of significance (α) of 0.05. The median for this data is 9.665. Assume the critical value follows a standard normal distribution. **Show all your work** by completing the chart on the left and then answering the following questions.

Observation	Value	Above/Below
1	10.6	
2	6.8	
3	19.0	
4	8.5	
5	14.2	
6	10.9	
7	8.2	
8	10.1	
9	3.5	
10	13.5	
11	7.9	
12	18.1	
13	3.8	
14	9.2	
15	10.6	
16	6.8	
17	19.0	
18	8.5	
19	14.2	
20	5.3	
21	6.8	
22	19.0	
23	8.5	
24	14.2	
25	10.9	
26	8.2	
27	10.1	
28	3.5	
29	13.5	
30	7.9	

State the null and alternative hypotheses for a runs test.

Null: Data is Independent

Alternative: Data is not Independent

$$r = 26$$

$$n_1 = 15$$

$$n_2 = 15$$

$$n = 30$$

$$\mu = \frac{2n_1n_2}{n} + 1 = \frac{2(15)(15)}{30} + 1 = 16$$

$$\sigma^2 = \frac{(\mu-1)(\mu-2)}{n-1} = \frac{(16-1)(16-2)}{29} = 7.241$$

$$Z_0 = \frac{r-\mu}{\sqrt{\sigma^2}} = \frac{26-16}{\sqrt{7.241}} = 3.716$$

$$Z_{df, \frac{\alpha}{2}} = Z_{\infty, 0.025} = 1.96$$

$$(n_1 \text{ or } n_2 > 10, df = \infty)$$

$$|Z_0| < Z_{crit}$$

$$|3.716| \not\leq 1.96$$

Reject H_0

Above/Below	
Median	9.665
Count	30
Runs Above Mean	13
Runs Below Mean	13
n1 (# Above Mean)	15
n2 (# Below Mean)	15
μ	16
σ^2	7.241
Z_0	3.716
$Z_{\alpha/2}, \alpha=0.05$	1.96
	REJECT

Test Statistic: _____

Critical Value: _____

Conclusion: _____

- 19) **(10 Points)** Analyze a Runs Up/Down Test for a dataset with 30 observations and 26 runs. Use a confidence level of 99%. Assume the critical value follows a standard normal distribution. **Show all your work** and answer the following questions. You are not required to write out your hypothesis.

$$n = 30; r = 26$$

$$\mu = \frac{2n-1}{3} = \frac{2(30)-1}{3} = 19.67$$

$$\sigma^2 = \frac{16n-29}{90} = \frac{16(30)-29}{90} = 5.011$$

$$Z_0 = \frac{r - \mu}{\sqrt{\sigma^2}} = \frac{26 - 19.67}{\sqrt{5.011}} = 2.827$$

$$\alpha = .01$$

$$Z_{\frac{\alpha}{2}} = 2.576 \text{ (reference sheet)}$$

$$|Z_0| = 2.827$$

$$|Z_0| < Z_{crit}$$

$$2.827 \not\leq 2.576$$

Reject H_0

Test Statistic: _____

Critical Value: _____

Conclusion: _____

Section #4: Testing for Goodness of Fit (27 points)

20)(3 Points) (TRUE | FALSE) The Chi-Squared Goodness of Fit test is only used for continuous distributions.

21)(3 Points) (TRUE | FALSE) The Chi-Squared Test has a two-tail rejection region.

Concessions Stand

You suspect that service times at the new South Rim Concessions Stand follow an exponential distribution with a mean of 2 minutes. Data from 22 customers is analyzed.

Use the chi-square goodness of fit test to determine if the time taken to get food is exponentially distributed with a mean of 2 minutes.

The three bins below are equiprobable; use them to conduct your test.

Bin	0 – 1.5 min	1.6 – 4.5 min	> 4.5 min
Observed Frequency	6	11	5

22)(3 points) Write a null hypothesis specific to this scenario?

H_0 = The wait times are exponentially distributed with a mean of 2 minutes.

23)(12 points) What is your observed chi-squared test statistic?

$$e = n\rho_i = 22 * 0.333 = 7.326$$

$$\chi^2_{calc} = \sum_{i=1}^k \frac{(o_i - e_i)^2}{e_i}$$

$$\chi^2_{calc} = \frac{(o_i - e_i)^2}{e_i} = \frac{(6 - 7.326)^2}{7.326} + \frac{(11 - 7.326)^2}{7.326} + \frac{(5 - 7.326)^2}{7.326}$$

$$\chi^2_{calc} = \frac{(o_i - e_i)^2}{e_i} = 0.240 + 1.8425 + 0.7385 = 2.821$$

Observed Test Statistic: _____

24)(6 Points) Using a 95% confidence level, determine whether to reject the null hypothesis or not. Write the critical value and your decision.

$$df = k - 1 = 2$$

$$\alpha = .05$$

$$\chi^2_{df,\alpha} = \chi^2_{2,.05} = 5.991$$

$$\chi^2_{calc} < \chi^2_{df,\alpha}$$

$$2.821 < 5.991 \quad \therefore \text{Fail to Reject } H_0$$

Critical Value: _____

Decision: _____

Outstanding Student Work

WPR #1

● Graded

Student

Ethan Ellis

Total Points

158 / 160 pts

Question 1

Match

30 / 30 pts

1.1	1a	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.2	1b	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.3	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.4	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.5	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.6	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.7	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.8	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.9	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.10	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.11	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.12	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	

Question 2

T/F - DES

3 / 3 pts

✓ - 0 pts Correct

Question 3

T/F - Discrete

3 / 3 pts

✓ - 0 pts Correct

Question 4

T/F - Deterministic

3 / 3 pts

✓ - 0 pts Correct

Question 5

Kendalls

3 / 3 pts

✓ - 0 pts Correct

Question 6

Arrival and Service Rates

4 / 4 pts

✓ - 0 pts Correct

Question 7

Steady State

5 / 5 pts

✓ - 0 pts Correct

Question 8

Probability

6 / 6 pts

✓ - 0 pts Correct

Question 9

Wq and L

12 / 12 pts

✓ - 0 pts Correct

Question 10

Utilization

3 / 3 pts

✓ - 0 pts Correct

Question 11

Customers in Line

3 / 3 pts

✓ - 0 pts Correct

Question 12

Customers in System

3 / 3 pts

✓ - 0 pts Correct

Question 13

Wait time in System

3 / 3 pts

✓ - 0 pts Correct

Question 14

Probability

4 / 4 pts

✓ - 0 pts Correct

Question 15

Probability 2

6 / 8 pts

✓ - 2 pts You found $P(X \leq x)$. Need to take complement to get $P(X > x)$

$$P_n = (1 - \rho)\rho^n = (1 - 0.833)0.833^n$$

$$P_n = (1 - \rho)\rho^0 = 0.167$$

$$P_n = (1 - \rho)\rho^1 = 0.139$$

$$P_n = (1 - \rho)\rho^2 = 0.116$$

$$P_{n>3} = 1 - \sum_0^3 \rho^n = 1 - 0.422 = 0.578 = 57.8\%$$

OR

$$P_n = \rho^3 = 0.833^3 = 0.578$$

Question 16

(no title)

6 / 6 pts

16.1 Ind Test 1

2 / 2 pts

✓ - 0 pts Correct

16.2 Ind Test 2

2 / 2 pts

✓ - 0 pts Correct

16.3 Ind Test 3

2 / 2 pts

✓ - 0 pts Correct

Question 17

(no title)

8 / 8 pts

17.1 Scatter 1

2 / 2 pts

✓ - 0 pts Correct

17.2 Scatter 2

2 / 2 pts

✓ - 0 pts Correct

17.3 Hist 1

2 / 2 pts

✓ - 0 pts Correct

17.4 Hist 2

2 / 2 pts

✓ - 0 pts Correct

Question 18

Above Below Med

16 / 16 pts

18.1 Hypothesis

4 / 4 pts

✓ - 0 pts Correct

18.2 Tstat, CV, Conclusion

12 / 12 pts

✓ - 0 pts Correct

Question 19

Up Down

10 / 10 pts

✓ - 0 pts Correct

Question 20

T/F Chi

3 / 3 pts

✓ - 0 pts Correct

Question 21

T/F Chi2

3 / 3 pts

✓ - 0 pts Correct

Question 22

Null

3 / 3 pts

✓ - 0 pts Correct

Question 23

Chi Square Stat

12 / 12 pts

✓ - 0 pts Correct

Question 24

CV and Decision

6 / 6 pts

✓ - 0 pts Correct

**DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996**

EM481 WRITTEN PARTIAL REVIEW #1

1. WPR 1 is a 75-minute exam worth 160 points. It consists of 24 problems on 12 pages, including the cover page. Prior to beginning, ensure that your exam is complete.
2. Authorized References: Pencil/pen, straight edge, calculator, instructor provided EM481 reference sheet.
3. Show all your work. Unsupported work earns little or no partial credit. Neatness will also help your cause.
4. Extra paper is available at the front of the classroom. Attach any additional pages to your exam when you turn them in. Label additional pages with your last name, section, and problem number.
5. Ensure you write in your name and section legibly in the boxes below.
6. Grading Summary:

Section	Possible Points	Score
1. Simulation Basics & Modeling	39	
2. Queuing	54	
3. Testing for Independence and Homogeneity	40	
4. Testing for Goodness of Fit	27	
Total	160	

Last, First Name:
(write large & legible)

Ellis, Ethan

Section:

G2

Section 1: Simulation Basics & Modeling (39 points)

- 1) **(2.5 Points, Each)** Match the DEFINITION to the TERM, using the table of terms below.

Definition	Term
A collection of elements that function together to achieve a desired goal.	B
Autocorrelation tests for this type of data characteristic.	A
In addition to variability, this is what makes a system complex.	I
A systems element that is a means by which activities are performed	P
A figure that visually demonstrates the route of entities in a system. Also contains useful quantitative information to describe a simulation system.	H
A system element that are the items processing through the system	N
This is helps classify queuing systems. It uses three different parameters to describe the queuing system.	C
Imitation of a dynamic system using a computer model in order to evaluate and improve system performance	L
A statistical assumption which implies that data come from the same underlying population (same distribution and parameter values)	G
A system element that dictates how, where, and when activities are performed	K
Poisson and exponential distributions fall under this category of distributions.	F
Technique that compares the empirical data histogram to the hypothesized probability mass or density function and requires the specification of intervals or grouping of data (bins)	Q

A Independence	K Controls
B System	L Simulation
G Kendall's Notation	M p-value
D Autocorrelation	N Entities
E Variability	O Structural
F Markovian	P Resources
G Homogeneity	Q Chi-Square Test
H Entity Flow Diagram	R Operational
I Interdependencies	S Activities
J Numerical	T Anderson-Darling Test

- 2) **(3 Points)** (TRUE | FALSE) In Static Simulations, state variables are updated as time advances.
- 3) **(3 Points)** (TRUE | FALSE) In Discrete Simulations, variables only change at discrete periods (triggered by events) and cannot incorporate any continuous distributions.
- 4) **(3 Points)** (TRUE | FALSE) In Deterministic Simulations, all future variable states are determined once the input data and initial state are defined.

Section #2: Queuing (54 points)

Vehicles arrive at the Grand Canyon South Rim Entrance following a Poisson process, with an average arrival rate of 220 vehicles per hour. Vehicles can use one of three ticket booths to purchase entry passes. It takes each vehicle an average of 1.33 minutes (exponentially distributed) to purchase a pass from a booth.

5) (3 points) What type of queuing model is this (in Kendall's notation)?

M/M/3

ANS: M/M/3

6) (4 points) Clearly state the arrival rate and the service rate, in vehicles per minute.

$$\lambda = \frac{220 \text{ v}}{1 \text{ hr}} = \frac{220}{60} = 3.667 \frac{\text{vehicles}}{\text{min}}$$

$$\mu = \frac{1 \text{ vehicle}}{1.33 \text{ min}} = .752 \frac{\text{vehicles}}{\text{min}}$$

Arrival Rate: 3.667 vehicles/min

Service Rate: .752 vehicles/min

7) (5 points) Will the system reach steady-state conditions? Justify your answer mathematically.

$$1) \text{ Find utilization } (\rho) \rightarrow \rho = \frac{3.667}{(3)(.752)} = 1.625 \rightarrow 1.625 > 1$$

No, the system will not reach steady-state conditions because utilization (1.625) is greater than 1.

- 8) (6 points) On weekends, the arrival rate increases to an average of 260 vehicles per hour, according to a Poisson process. What is the probability that exactly 40 vehicles will arrive in the next 10 minutes?

$$\lambda = \frac{260}{60} = 4.333 \frac{\text{vehicles}}{\text{min}}$$

$$P(X=40) = \frac{e^{(-4.333)(10)} \frac{(4.333(10))^{40}}{40!}}{1} = .0552$$

Probability: .0552 or 5.52%

- 9) (12 points) During peak times, the system operates as an ^{M/G/4}~~M/M/2~~ model. The arrival rate is 5.4 vehicles per minute. The service time for each of the four booths follows a gamma distribution with a mean of 1.42 vehicles per minute and standard deviation of 0.1 minutes. The entrance is busy 95% of the time. How long do vehicles wait in the line on average? What is the expected number of vehicles in line?

$$\lambda = 5.4 \frac{\text{veh}}{\text{min}} \quad \mu = 1.42 \frac{1}{\text{min}} \quad P = .95$$

$$C_a = 1$$

$$C_s = (1.42)(.1)$$

$$C_s = .142$$

$$W_q = \left(\frac{(1)^2 + (.142)^2}{2} \right) \left(\frac{.95^{\sqrt{2(4+1)} - 1}}{4(1 - .95)} \right) \left(\frac{1}{1.42} \right)$$

$$W_q = (.510082) \left(\frac{.895018989}{.2} \right) (.704225352)$$

$$W_q = 1.608 \text{ minutes}$$

$$L_q = (5.4)(1.608)$$

$$L_q = 8.683 \text{ vehicles}$$

Average Wait in Line: 1.608 minutes

Expected # of Vehicles in Line: 8.683 vehicles

The mean service time for the single server at the concession stand follows a Poisson process with a mean of 2.5 minutes. Customers arrive every 3 minutes, exponentially distributed. – Round to the nearest hundredth for questions 10-15.

10)(3 points) What is the utilization?

M/M/1

$$\lambda = \frac{1 \text{ cust}}{3 \text{ min}} \quad \mu = \frac{1 \text{ cust}}{2.5 \text{ min}} = .4 \text{ cust/min}$$

$$\rho = .33 \text{ cust/min}$$

$$\rho = \frac{.33}{(.4)} = .825 = .83$$

ANS: .83

$$W_q = \frac{\rho}{\mu(1-\rho)}$$

11)(3 points) How long will customers wait in line?

$$W_q = \frac{.33}{(.4)(.4-.33)} = \frac{.33}{.028} = 11.79$$

ANS: 11.79 minutes

$$L = \frac{\rho}{1-\rho}$$

12)(3 points) How many customers would you expect in the system?

$$L = \frac{.83}{(1-.83)} = 4.88$$

ANS: 4.88 customers

$$W = \frac{1}{\mu - \lambda}$$

13)(3 points) What is the expected average time in the system?

$$W = \frac{1}{.4 - .33} = 14.29$$

ANS: 14.29 minutes

14)(4 points) What is the probability that 6 customers are in the system?

$$P_n = (1-\rho)\rho^n$$

$$P_6 = (1-.83) \cdot .83^6$$

$$P_6 = .0556$$

$$P_6 = .06$$

ANS: .06 or 5.56%

15)(8 points) What is the probability that there will be more than 2 customers at the concession stand?

$$P > 2 = 1 - (P \leq 2) \rightarrow 1 - (P_0 + P_1 + P_2)$$

$$P_0 = (1 - .83) \cdot .83^0 = .17$$

$$P_1 = (1 - .83) \cdot .83^1 = .14$$

$$P_2 = (1 - .83) \cdot (.83)^2 = .12$$

$$\underline{\quad .43 \quad}$$

ANS: .43 or 43%

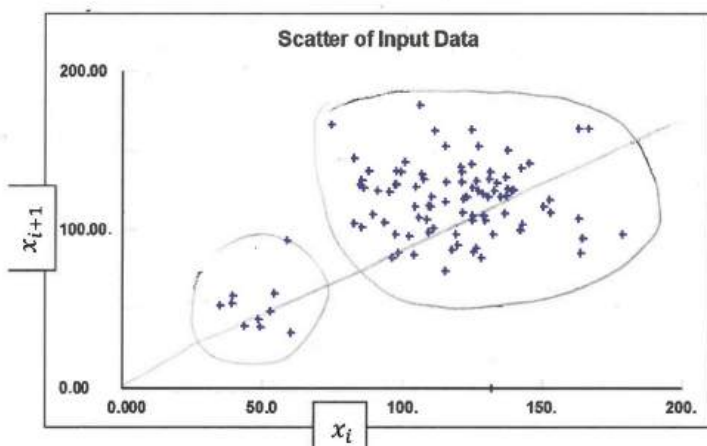
Section #3: Testing for Independence and Homogeneity (40 points)

You recently collected data on customer interarrival times, and now wish to analyze the data.

16)(6 Points) Name the three major methods to test for independence taught in EM481?

Checks if an observation is influenced by its preceding observation.	Scatter plot
Compares the observed number of sequenced observations to the expected number of sequenced observations	Runs Test
Checks if observations in a single dataset are correlated over different time lags.	Autocorrelation

17)(8 Points) You generated the following charts from historical data. What can you conclude about the data in terms of independency and homogeneity?

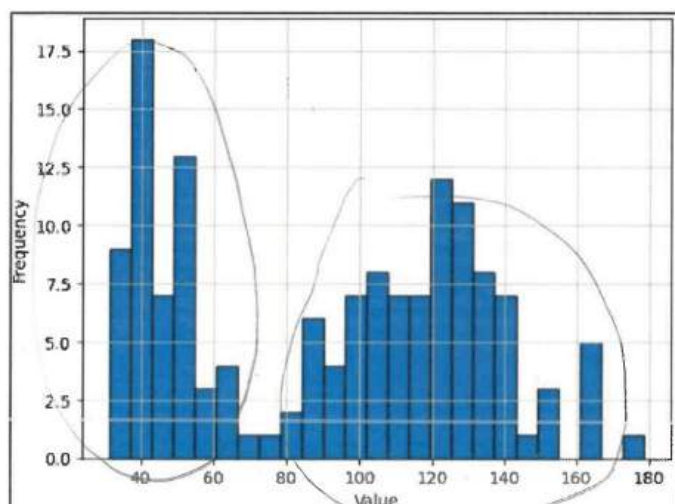


What type of chart is this?

Scatter plot

Conclusion from the graph:

Data is non-homogeneous.
There are two different clusters. Also does not appear independent as it generally follows a trend line.



What type of chart is this?

Histogram.

Conclusion from the graph:

Data is non-homogeneous. There are two distinct groups.

- 18) (16 Points) You decide to conduct a **Runs Above/Below the Median Test** on the 30 data points below to determine whether the data is independent (random) given a level of significance (α) of 0.05. The median for this data is 9.665. Assume the critical value follows a standard normal distribution. **Show all your work** by completing the chart on the left and then answering the following questions.

Observation	Value	Above/Below
1	10.6	+
2	6.8	-
3	19.0	+
4	8.5	-
5	14.2	+
6	10.9	+
7	8.2	-
8	10.1	+
9	3.5	-
10	13.5	+
11	7.9	-
12	18.1	+
13	3.8	-
14	9.2	-
15	10.6	+
16	6.8	-
17	19.0	+
18	8.5	-
19	14.2	+
20	5.3	-
21	6.8	-
22	19.0	+
23	8.5	-
24	14.2	+
25	10.9	+
26	8.2	-
27	10.1	+
28	3.5	-
29	13.5	+
30	7.9	-

State the null and alternative hypotheses for a runs test.

Null: H_0 : The data is independent

Alternative: H_a : The data is not independent

$$n_1 = 15$$

$$n_2 = 15$$

$$r = 26$$

$$\mu = \frac{2(15)(15)}{30} + 1$$

$$\mu = 16$$

$$\sigma^2 = \frac{(16-1)(16-2)}{30-1} = \frac{(15)(14)}{29}$$

$$\sigma^2 = 7.241$$

$$Z_0 = \frac{26 - 16}{\sqrt{7.241}} = 3.716$$

$$\alpha = .05$$

$$\alpha/2 = .025$$

$$Z_{crit} = 1.96$$

Test Statistic: 3.716

Critical Value: 1.96

Conclusion: Reject the null

- 19) (10 Points) Analyze a Runs Up/Down Test for a dataset with 30 observations and 26 runs. Use a confidence level of 99%. Assume the critical value follows a standard normal distribution. **Show all your work** and answer the following questions. You are not required to write out your hypothesis.

$$\mu = \frac{2(30) - 1}{3}$$

$$\mu = 19.667$$

$$\sigma^2 = \frac{16(30) - 29}{90}$$

$$\sigma^2 = 5.011$$

$$Z_o = \frac{26 - 19.667}{\sqrt{5.011}}$$

$$Z_o = 2.829$$

$$\alpha = .01$$

$$\alpha/2 = .005$$

$$Z_{crit} = 2.576$$

Test Statistic: 2.829

Critical Value: 2.576

Conclusion: Reject the null

Section #4: Testing for Goodness of Fit (27 points)

20)(3 Points) (TRUE | FALSE) The Chi-Squared Goodness of Fit test is only used for continuous distributions.

21)(3 Points) (TRUE | FALSE) The Chi-Squared Test has a two-tail rejection region.

Concessions Stand

You suspect that service times at the new South Rim Concessions Stand follow an exponential distribution with a mean of 2 minutes. Data from 22 customers is analyzed.

Use the chi-square goodness of fit test to determine if the time taken to get food is exponentially distributed with a mean of 2 minutes.

The three bins below are equiprobable; use them to conduct your test.

Bin	1 0 – 1.5 min	2 1.6 – 4.5 min	3 > 4.5 min
Observed Frequency	6	11	5

22)(3 points) Write a null hypothesis specific to this scenario?

H_0 : The data follows an exponential distribution with a mean of 2 minutes

23)(12 points) What is your observed chi-squared test statistic?

$$p_i = \frac{1}{3} \quad e_i = (22)\left(\frac{1}{3}\right) = 7.333 \quad df = 2$$

$$\chi_0^2 = \frac{(6 - 7.333)^2}{7.333} + \frac{(11 - 7.333)^2}{7.333} + \frac{(5 - 7.333)^2}{7.333}$$

$$\chi_0^2 = .2423 + 1.8338 + .7422$$

$$\chi_0^2 = 2.818$$

Observed Test Statistic: 2.818

24)(6 Points) Using a 95% confidence level, determine whether to reject the null hypothesis or not. Write the critical value and your decision.

$$\chi^2 = 2.818$$

$$\alpha = .05 \quad df = 2$$

$$crit = 5.991$$

$$2.818 < 5.991$$

Critical Value: 5.991

Decision: Do not Reject the null

Fair-to-Good Student Work

WPR #1

● Graded

Student

Kathryn Meyer

Total Points

137 / 160 pts

Question 1

Match

30 / 30 pts

1.1	1a	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.2	1b	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.3	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.4	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.5	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.6	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.7	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.8	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.9	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.10	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.11	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.12	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	

Question 2

T/F - DES

3 / 3 pts

✓ - 0 pts Correct

Question 3

T/F - Discrete

0 / 3 pts

✓ - 3 pts Correct ANS = FALSE

Variables only change at discrete periods.

Variables can incorporate continuous and discrete distributions

Question 4

T/F - Deterministic

3 / 3 pts

✓ - 0 pts Correct

Question 5

Kendalls

3 / 3 pts

✓ - 0 pts Correct

Question 6

Arrival and Service Rates

4 / 4 pts

✓ - 0 pts Correct

Question 7

Steady State

5 / 5 pts

✓ - 0 pts Correct

Question 8

Probability

6 / 6 pts

✓ - 0 pts Correct

Question 9

Wq and L

12 / 12 pts

✓ - 0 pts Correct

Question 10

Utilization

2 / 3 pts

✓ - 1 pt Service Rate Incorrect.

$$u = \frac{1 \text{ customer}}{2.5 \text{ minutes}} = 0.4 \text{ customers/minute}$$

$$\lambda = \frac{1 \text{ customer}}{3 \text{ minutes}} = 0.333 \text{ customer/minute}$$

$$p = \frac{\lambda}{\mu} = \frac{0.333}{0.4} = 0.833$$

1

Question 11

Customers in Line

3 / 3 pts

✓ - 0 pts ANS Follows with your values from Q10.

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)} = \frac{0.333}{0.4 * (0.4 - 0.333)} = 12.43 \text{ min}$$

2

Question 12

Customers in System

3 / 3 pts

✓ - 0 pts Follows with values from Q10.

$$L = \frac{\rho}{1 - \rho} = \frac{0.833}{1 - 0.833} = 4.988 \text{ customers}$$

3

Question 13

Wait time in System

3 / 3 pts

✓ - 0 pts Correct, based on your values from Q10

$$W = \frac{1}{\mu - \lambda} = \frac{1}{0.4 - 0.333} = 14.93 \text{ min}$$

Question 14

Probability

4 / 4 pts

✓ - 0 pts Correct, based on your value from Q11.

$$P_n = (1 - \rho)\rho^n = (1 - 0.833)0.833^6 = 0.0558 = 5.58\%$$

Question 15

Probability 2

5 / 8 pts

✓ - 3 pts You found probability of there being not 2 customers (any other number but 2). Need to find probability of more than 2 customers.

$$P_n = (1 - \rho)\rho^n = (1 - 0.833)0.833^n$$

$$P_n = (1 - \rho)\rho^0 = 0.167$$

$$P_n = (1 - \rho)\rho^1 = 0.139$$

$$P_n = (1 - \rho)\rho^2 = 0.116$$

$$P_{n>3} = 1 - \sum_0^3 \rho^n = 1 - 0.422 = 0.578 = 57.8\%$$

OR

$$P_n = \rho^3 = 0.833^3 = 0.578$$

4

Value Follows

Question 16

(no title)

2 / 6 pts

16.1 Ind Test 1

2 / 2 pts

✓ - 0 pts Correct

16.2 Ind Test 2

0 / 2 pts

✓ - 2 pts ANS = Runs Test

16.3 Ind Test 3

0 / 2 pts

✓ - 2 pts ANS = Autocorrelation

Question 17

(no title)

6 / 8 pts

17.1 Scatter 1

2 / 2 pts

✓ - 0 pts Correct

17.2 Scatter 2

2 / 2 pts

✓ - 0 pts Correct

17.3 Hist 1

0 / 2 pts

✓ - 2 pts ANS = Histogram

17.4 Hist 2

2 / 2 pts

✓ - 0 pts Correct

Question 18

Above Below Med

14 / 16 pts

18.1 Hypothesis

4 / 4 pts

✓ - 0 pts Correct

18.2 Tstat, CV, Conclusion

10 / 12 pts

✓ - 2 pts Correct calculations; wrong conclusion

Conclusion: Reject Null Hypothesis

Question 19

Up Down

7 / 10 pts

✓ - 1 pt Math Error

$$n = 30; r = 26$$

$$\mu = \frac{2n-1}{3} = \frac{2(30)-1}{3} = 19.67$$

$$\sigma^2 = \frac{16n-29}{90} = \frac{16(30)-29}{90} = 5.011$$

$$Z_0 = \frac{r - \mu}{\sqrt{\sigma^2}} = \frac{26 - 19.67}{\sqrt{5.011}} = 2.827$$

6

✓ - 2 pts Conclusion, not supported by Z_test & Z_crit values that you found.

7

5

Follows

Question 20

T/F Chi

0 / 3 pts

✓ - 3 pts Correct answer: FALSE

Question 21

T/F Chi2

3 / 3 pts

✓ - 0 pts Correct

Question 22

Null

3 / 3 pts

✓ - 0 pts Correct

Question 23

Chi Square Stat

10 / 12 pts

✓ - 2 pts wrong Ei value

$$E_i = nP_i = 22.333 = 7.326$$

Question 24

CV and Decision

6 / 6 pts

✓ - 0 pts Correct

**DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996**

EM481 WRITTEN PARTIAL REVIEW #1

1. WPR 1 is a 75-minute exam worth 160 points. It consists of 24 problems on 12 pages, including the cover page. Prior to beginning, ensure that your exam is complete.
2. Authorized References: Pencil/pen, straight edge, calculator, instructor provided EM481 reference sheet.
3. Show all your work. Unsupported work earns little or no partial credit. Neatness will also help your cause.
4. Extra paper is available at the front of the classroom. Attach any additional pages to your exam when you turn them in. Label additional pages with your last name, section, and problem number.
5. Ensure you write in your name and section legibly in the boxes below.
6. Grading Summary:

Section	Possible Points	Score
1. Simulation Basics & Modeling	39	
2. Queuing	54	
3. Testing for Independence and Homogeneity	40	
4. Testing for Goodness of Fit	27	
Total	160	

Last, First Name:
(write large & legible)

Meyer, Kathryn

Section:

G2-1

Section 1: Simulation Basics & Modeling (39 points)

- 1) (2.5 Points, Each) Match the DEFINITION to the TERM, using the table of terms below.

Definition	Term
A collection of elements that function together to achieve a desired goal.	B
Autocorrelation tests for this type of data characteristic.	A
In addition to variability, this is what makes a system complex.	I
A systems element that is a means by which activities are performed	P
A figure that visually demonstrates the route of entities in a system. Also contains useful quantitative information to describe a simulation system.	H
A system element that are the items processing through the system	N
This is helps classify queuing systems. It uses three different parameters to describe the queuing system.	C
Imitation of a dynamic system using a computer model in order to evaluate and improve system performance	L
A statistical assumption which implies that data come from the same underlying population (same distribution and parameter values)	G
A system element that dictates how, where, and when activities are performed	K
Poisson and exponential distributions fall under this category of distributions.	F
Technique that compares the empirical data histogram to the hypothesized probability mass or density function and requires the specification of intervals or grouping of data (bins)	Q

A Independence	K Controls
B System	L Simulation
C Kendall's Notation	M p-value
D Autocorrelation	N Entities
E Variability	O Structural
F Markovian	P Resources
G Homogeneity	Q Chi-Square Test
H Entity Flow Diagram	R Operational
I Interdependencies	S Activities
J Numerical	T Anderson-Darling Test

- 2) (3 Points) (TRUE | FALSE) In Static Simulations, state variables are updated as time advances.
- 3) (3 Points) (TRUE | FALSE) In Discrete Simulations, variables only change at discrete periods (triggered by events) and cannot incorporate any continuous distributions.
- 4) (3 Points) (TRUE | FALSE) In Deterministic Simulations, all future variable states are determined once the input data and initial state are defined.

Section #2: Queuing (54 points)

Vehicles arrive at the Grand Canyon South Rim Entrance following a Poisson process, with an average arrival rate of 220 vehicles per hour. Vehicles can use one of three ticket booths to purchase entry passes. It takes each vehicle an average of 1.33 minutes (exponentially distributed) to purchase a pass from a booth.

- 5) (3 points) What type of queuing model is this (in Kendall's notation)?

ANS: M|M|3

- 6) (4 points) Clearly state the arrival rate and the service rate, in vehicles per minute.

$$\lambda = \frac{220 \text{ vehicle}}{\text{hr}} \times \frac{1 \text{ hr}}{60 \text{ min}}$$

$$\mu = 1.33 \frac{\text{min}}{\text{vehicle}}$$
$$\frac{1}{1.33} \frac{\text{vehicle}}{\text{min}}$$

Arrival Rate: 3.6667 $\frac{\text{vehicles}}{\text{min}}$

Service Rate: 0.7519 $\frac{\text{vehicle}}{\text{min}}$

- 7) (5 points) Will the system reach steady-state conditions? Justify your answer mathematically.

$$\rho = \frac{\lambda}{s\mu}$$

$$= \frac{3.6667}{3 \cdot 0.7519}$$

$$= 1.6255$$

No, system will not reach steady-state conditions since $\rho > 1$.

- 8) (6 points) On weekends, the arrival rate increases to an average of 260 vehicles per hour, according to a Poisson process. What is the probability that exactly 40 vehicles will arrive in the next 10 minutes?

$$\lambda = \frac{260}{1 \text{ hr}} \times \frac{1 \text{ hr}}{60 \text{ min}} = 4.3333$$

$$P(X=x) = \frac{e^{-\lambda t} (\lambda t)^x}{x!}$$

$$P(X=40) = \frac{e^{-4.3333(10)} (4.3333 \cdot 10)^{40}}{40!}$$

$$= 0.055185$$

Probability: 5.5185 %

- 9) (12 points) During peak times, the system operates as an M/G/4 model. The arrival rate is 5.4 vehicles per minute. The service time for each of the four booths follows a gamma distribution with a mean of 1.42 vehicles per minute and standard deviation of 0.1 minutes. The entrance is busy 95% of the time. How long do vehicles wait in the line on average? What is the expected number of vehicles in line?

$$W_q + L_q$$

$$W_q = \left(\frac{C_a^2 + C_s^2}{2} \right) \left(\frac{\rho \sqrt{2(s+1)} - 1}{s(1-\rho)} \right) \left(\frac{1}{\mu} \right)$$

$$= \left(\frac{1^2 + 0.14^2}{2} \right) \left(\frac{0.9507 \sqrt{2(4+1)} - 1}{4(1-0.9507)} \right) \left(\frac{1}{1.42} \right)$$

$$= 1.632 \text{ min}$$

$$L_q = \lambda W_q$$

$$= (5.4)(1.632)$$

$$= 8.8128$$

$$\lambda = 5.4$$

$$\mu = 1.42$$

$$C_a = 1$$

$$C_s = 1.4(0.1) = 0.14$$

$$\rho = \frac{\lambda}{s \cdot \mu} = \frac{5.4}{4 \cdot 1.42} = 0.9507$$

Average Wait in Line: 1.632 min

Expected # of Vehicles in Line: 8.8128 vehicles

The mean service time for the single server at the concession stand follows a Poisson process with a mean of 2.5 minutes. Customers arrive every 3 minutes, exponentially distributed. – Round to the nearest hundredth for questions 10-15.

* M/M/1

10)(3 points) What is the utilization?

$$\rho = \frac{\lambda}{s \cdot \mu}$$

$$\lambda = 0.33 \text{ min}$$

$$\mu = 2.5 \text{ min}$$

$$= \frac{0.33}{1 \cdot 2.5}$$

$$= 0.132$$

ANS: 0.13

11)(3 points) How long will customers wait in line?

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)}$$

$$= \frac{0.33}{2.5(2.5 - 0.33)}$$

$$= 0.0608$$

ANS: 0.06 min

12)(3 points) How many customers would you expect in the system?

$$L = \frac{\lambda}{\mu - \lambda}$$

$$= \frac{0.33}{2.5 - 0.33}$$

ANS: 0.15 customers

13)(3 points) What is the expected average time in the system?

$$W = \frac{1}{\mu - \lambda}$$

$$= \frac{1}{2.5 - 0.33}$$

ANS: 0.46 min

14)(4 points) What is the probability that 6 customers are in the system?

$$P_n = (1 - \rho) \rho^n$$

$$= (1 - 0.13) (0.13)^6$$

ANS: 0.00042 %

15)(8 points) What is the probability that there will be more than 2 customers at the concession stand?

$$\begin{aligned} P_n &= 1 - ((1-p)p^n) \\ &= 1 - (1-0.13) 0.13^2 \\ &= 0.98529 \end{aligned}$$

ANS: 98.53%

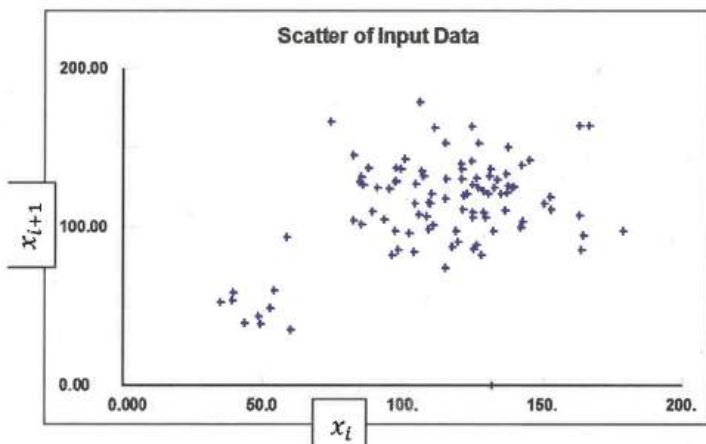
Section #3: Testing for Independence and Homogeneity (40 points)

You recently collected data on customer interarrival times, and now wish to analyze the data.

16)(6 Points) Name the three major methods to test for independence taught in EM481?

Checks if an observation is influenced by its preceding observation.	scatter plot
Compares the observed number of sequenced observations to the expected number of sequenced observations	autocorrelation
Checks if observations in a single dataset are correlated over different time lags.	runs tests

17)(8 Points) You generated the following charts from historical data. What can you conclude about the data in terms of independence and homogeneity?

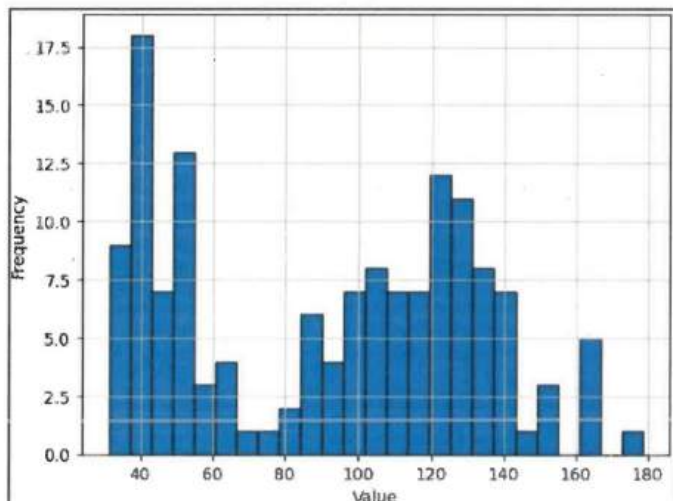


What type of chart is this?

scatter plot

Conclusion from the graph:

data is independent



What type of chart is this?

autocorrelation

Conclusion from the graph:

data is not
homogeneous

18) (16 Points) You decide to conduct a **Runs Above/Below the Median Test** on the 30 data points below to determine whether the data is independent (random) given a level of significance (α) of 0.05. The median for this data is 9.665. Assume the critical value follows a standard normal distribution. **Show all your work** by completing the chart on the left and then answering the following questions.

Observation	Value	Above/Below
1	10.6	+
2	6.8	-
3	19.0	+
4	8.5	-
5	14.2	+
6	10.9	+
7	8.2	-
8	10.1	+
9	3.5	-
10	13.5	+
11	7.9	-
12	18.1	+
13	3.8	-
14	9.2	-
15	10.6	+
16	6.8	-
17	19.0	+
18	8.5	-
19	14.2	+
20	5.3	-
21	6.8	-
22	19.0	+
23	8.5	-
24	14.2	+
25	10.9	+
26	8.2	-
27	10.1	+
28	3.5	-
29	13.5	+
30	7.9	-

State the null and alternative hypotheses for a runs test.

Null: H_0 : data is independent

Alternative: H_A : data is not independent

$$\mu = \frac{2(15)(15)}{30} + 1 = 16 \quad \left(\frac{2n_1n_2}{n} + 1 \right)$$

$$\sigma^2 = \frac{(16-1)(16-2)}{30-1} = \frac{(15)(14)}{29} = \frac{210}{29}$$

$$z_0 = \frac{r - \mu}{\sqrt{\sigma^2}} = \frac{26 - 16}{\sqrt{210/29}} = 3.716$$

$$z_{crit} \rightarrow z_{\alpha, \frac{0.05}{2}} = 1.96$$

Test Statistic: 3.716

Critical Value: 1.96

Conclusion: fail to reject null $\therefore$
data is independent

- 19) (10 Points) Analyze a Runs Up/Down Test for a dataset with 30 observations and 26 runs. Use a confidence level of 99%. Assume the critical value follows a standard normal distribution. **Show all your work** and answer the following questions. You are not required to write out your hypothesis.

$$\mu = \frac{2n-1}{3} = \frac{2(30)-1}{3} = 9.667$$

$$\sigma^2 = \frac{16n-29}{90} = \frac{16(30)-29}{90} = \frac{451}{90}$$

$$z_0 = \frac{r-\mu}{\sqrt{\sigma^2}} = \frac{26-9.667}{\sqrt{\frac{451}{90}}} = 7.296$$

$$z_{crit} \rightarrow z_{\alpha, \frac{0.01}{2}} = 2.576$$

Test Statistic: 7.296

Critical Value: 2.576

Conclusion: data is independent +

Section #4: Testing for Goodness of Fit (27 points)

20)(3 Points) (TRUE | FALSE) The Chi-Squared Goodness of Fit test is only used for continuous distributions.

21)(3 Points) (TRUE | FALSE) The Chi-Squared Test has a two-tail rejection region.

Concessions Stand

You suspect that service times at the new South Rim Concessions Stand follow an exponential distribution with a mean of 2 minutes. Data from 22 customers is analyzed.

Use the chi-square goodness of fit test to determine if the time taken to get food is exponentially distributed with a mean of 2 minutes.

The three bins below are equiprobable; use them to conduct your test.

Bin	0 - 1.5 min	1.6 - 4.5 min	> 4.5 min
Observed Frequency	6	11	5

22)(3 points) Write a null hypothesis specific to this scenario?

service times at new South Rim Concessions Stand
H₀: λ follows an exponential distribution with a mean of 2 min

23)(12 points) What is your observed chi-squared test statistic?

o_i	P_i	e_i
1	0.33	2
2	0.33	3.67
3	0.33	1.67

$$\chi^2 = \frac{(6-2)^2}{2} + \frac{(11-3.67)^2}{3.67} + \frac{(5-1.67)^2}{1.67}$$

$$= 29.328$$

Observed Test Statistic: 29.328

EM481 AY26-1
WPR#1 v1 (160 points)
26 August 2025

Initials: KM

24)(6 Points) Using a 95% confidence level, determine whether to reject the null hypothesis or not. Write the critical value and your decision.

$$z_{crit} \rightarrow z_{df, \alpha} \rightarrow z_{k-1, \alpha} \rightarrow \chi^2_{crit} = 5.991$$

Critical Value: 5.991

Decision: fail to reject null

Poor Student Work

WPR #1

● Graded

Student

Jacob Azizi

Total Points

122.5 / 160 pts

Question 1

Match

22.5 / 30 pts

1.1	1a	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.2	1b	0 / 2.5 pts
	✓ - 2.5 pts ANS = A	
1.3	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.4	(no title)	0 / 2.5 pts
	✓ - 2.5 pts ANS = P	
1.5	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.6	(no title)	0 / 2.5 pts
	✓ - 2.5 pts ANS = N	
1.7	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.8	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.9	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.10	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.11	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	
1.12	(no title)	2.5 / 2.5 pts
	✓ - 0 pts Correct	

Question 2

T/F - DES

0 / 3 pts

✓ - 3 pts Correct ANS = FALSE

Question 3

T/F - Discrete

0 / 3 pts

✓ - 3 pts Correct ANS = FALSE

Variables only change at discrete periods.

Variables can incorporate continuous and discrete distributions

Question 4

T/F - Deterministic

3 / 3 pts

✓ - 0 pts Correct

Question 5

Kendalls

3 / 3 pts

✓ - 0 pts Correct

Question 6

Arrival and Service Rates

3 / 4 pts

✓ - 1 pt Gave Service Time instead of Service Rate

$$\lambda = \frac{220 \text{ vehicles}}{60 \text{ minutes}} = 3.667 \text{ vehicles/minute}$$

$$u = \frac{1}{(1.33 \text{ min})} = 0.752 \text{ vehicles/minute}$$

Question 7

Steady State

5 / 5 pts

✓ - 0 pts Wrong value due to incorrect service rate, follows credit from Question 6

Question 8

Probability

6 / 6 pts

✓ - 0 pts Correct

Question 9

Wq and L

12 / 12 pts

✓ - 0 pts Correct

Question 10

Utilization

3 / 3 pts

✓ - 0 pts Correct

Question 11

Customers in Line

3 / 3 pts

✓ - 0 pts Correct

Question 12

Customers in System

3 / 3 pts

✓ - 0 pts Correct

Question 13

Wait time in System

3 / 3 pts

✓ - 0 pts Correct

Question 14

Probability

4 / 4 pts

✓ - 0 pts Correct

Question 15

Probability 2

6 / 8 pts

✓ - 2 pts You found $P(X \leq x)$. Need to take complement to get $P(X > x)$

$$P_n = (1 - \rho)\rho^n = (1 - 0.833)0.833^n$$

$$P_n = (1 - \rho)\rho^0 = 0.167$$

$$P_n = (1 - \rho)\rho^1 = 0.139$$

$$P_n = (1 - \rho)\rho^2 = 0.116$$

$$P_{n>3} = 1 - \sum_0^3 \rho^n = 1 - 0.422 = 0.578 = 57.8\%$$

OR

$$P_n = \rho^3 = 0.833^3 = 0.578$$

1

Question 16

(no title)

2 / 6 pts

16.1 Ind Test 1

0 / 2 pts

✓ - 2 pts ANS = Scatterplot

16.2 Ind Test 2

2 / 2 pts

✓ - 0 pts Correct

16.3 Ind Test 3

0 / 2 pts

✓ - 2 pts ANS = Autocorrelation

Question 17

(no title)

2 / 8 pts

17.1 Scatter 1

0 / 2 pts

✓ - 2 pts ANS = Scatterplot

17.2 Scatter 2

2 / 2 pts

✓ - 0 pts Correct

17.3 Hist 1

0 / 2 pts

✓ - 2 pts ANS = Histogram

17.4 Hist 2

0 / 2 pts

✓ - 2 pts ANS = Not Homogeneous.

Question 18

Above Below Med

14 / 16 pts

18.1 Hypothesis

4 / 4 pts

✓ - 0 pts Correct

18.2 Tstat, CV, Conclusion

10 / 12 pts

✓ - 2 pts Correct calculations; wrong conclusion

Conclusion: Reject Null Hypothesis

Question 19

Up Down

10 / 10 pts

✓ - 0 pts Correct

Question 20

T/F Chi

3 / 3 pts

✓ - 0 pts Correct

Question 21

T/F Chi2

0 / 3 pts

✓ - 3 pts Answer: False

Question 22

Null

3 / 3 pts

✓ - 0 pts Correct

Question 23

Chi Square Stat

9 / 12 pts

✓ - 3 pts Incorrect Chi Square Statistic Summation Implementation

$$e = n\rho_i = 22 * 0.333 = 7.326$$

$$\chi^2_{calc} = \sum_{i=1}^k \frac{(o_i - e_i)^2}{e_i}$$

$$\chi^2_{calc} = \frac{(o_i - e_i)^2}{e_i} = \frac{(6 - 7.326)^2}{7.326} + \frac{(11 - 7.326)^2}{7.326} + \frac{(5 - 7.326)^2}{7.326}$$

$$\chi^2_{calc} = \frac{(o_i - e_i)^2}{e_i} = 0.240 + 1.8425 + 0.7385 = 2.821$$

Question 24

CV and Decision

3 / 6 pts

✓ - 3 pts Wrong Critical Value

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EM481 WRITTEN PARTIAL REVIEW #1

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3. Testing for Independence and Homogeneity	40	
4. Testing for Goodness of Fit	27	
Total	160	

Last, First Name:
(write large & legible)

Azizi, Jacob

Section:

H1

Section 1: Simulation Basics & Modeling (39 points)

- 1) **(2.5 Points, Each)** Match the DEFINITION to the TERM, using the table of terms below.

Definition	Term
A collection of elements that function together to achieve a desired goal.	B
Autocorrelation tests for this type of data characteristic.	T
In addition to variability, this is what makes a system complex.	I
A systems element that is a means by which activities are performed	R
A figure that visually demonstrates the route of entities in a system. Also contains useful quantitative information to describe a simulation system.	H
A system element that are the items processing through the system	S
This is helps classify queuing systems. It uses three different parameters to describe the queuing system.	C
Imitation of a dynamic system using a computer model in order to evaluate and improve system performance	L
A statistical assumption which implies that data come from the same underlying population (same distribution and parameter values)	G
A system element that dictates how, where, and when activities are performed	K
Poisson and exponential distributions fall under this category of distributions.	F
Technique that compares the empirical data histogram to the hypothesized probability mass or density function and requires the specification of intervals or grouping of data (bins)	Q

A Independence	K Controls
B System	L Simulation
C Kendall's Notation	M p-value
D Autocorrelation	N Entities
E Variability	O Structural
F Markovian	P Resources
G Homogeneity	Q Chi-Square Test
H Entity Flow Diagram	R Operational
I Interdependencies	S Activities
J Numerical	T Anderson-Darling Test

- 2) (3 Points) (TRUE | FALSE) In Static Simulations, state variables are updated as time advances.
- 3) (3 Points) (TRUE | FALSE) In Discrete Simulations, variables only change at discrete periods (triggered by events) and cannot incorporate any continuous distributions.
- 4) (3 Points) (TRUE | FALSE) In Deterministic Simulations, all future variable states are determined once the input data and initial state are defined.

Section #2: Queuing (54 points)

Vehicles arrive at the Grand Canyon South Rim Entrance following a Poisson process, with an average arrival rate of 220 vehicles per hour. Vehicles can use one of three ticket booths to purchase entry passes. It takes each vehicle an average of 1.33 minutes (exponentially distributed) to purchase a pass from a booth.

5) (3 points) What type of queuing model is this (in Kendall's notation)?

ANS: M/M/3

6) (4 points) Clearly state the arrival rate and the service rate, in vehicles per minute.

$$\lambda = \frac{220 \text{ veh/hr}}{1 \text{ hr} / 60 \text{ min}} = 3.67 \text{ veh per min}$$

Arrival Rate: 3.67 veh/min

Service Rate: 1.33 pass/min

7) (5 points) Will the system reach steady-state conditions? Justify your answer mathematically.

$$\rho = \frac{\lambda}{s \cdot m} = \frac{3.67}{3 \cdot 1.33} = .919 < 1$$



Yes, this system will reach a steady state because we found the utilization and $.919 < 1$ is true.

- 8) (6 points) On weekends, the arrival rate increases to an average of 260 vehicles per hour, according to a Poisson process. What is the probability that exactly 40 vehicles will arrive in the next 10 minutes?

$$S = 3$$

$$\mu = 1.33 \frac{\text{pass}}{\text{min}}$$

$$\lambda = 4.33 \frac{\text{veh}}{\text{min}}$$

$$\lambda = \frac{260 \text{ veh}}{1 \text{ hour}} \times \frac{1 \text{ hour}}{60 \text{ min}} = 4.33 \frac{\text{veh}}{\text{min}}$$

$$P(X=40) = \frac{e^{-(4.33)(10)} (4.33 \cdot 10)^{40}}{40!}$$

$$\frac{e^{(-43.33)} (4.33 \cdot 10)^{40}}{40!} = 0.053$$

Probability: 5.3% chance

- 9) (12 points) During peak times, the system operates as an M/G/2 model. The arrival rate is 5.4 vehicles per minute. The service time for each of the four booths follows a gamma distribution with a mean of 1.42 vehicles per minute and standard deviation of 0.1 minutes. The entrance is busy 95% of the time. How long do vehicles wait in the line on average? What is the expected number of vehicles in line?

$$M/G/2$$

$$\lambda = 5.4 \frac{\text{veh}}{\text{min}}$$

$$S = 4$$

$$\mu = 1.42 \frac{\text{veh}}{\text{min}}$$

$$s.d. = 0.1 \text{ min}$$

$$C_a = 1$$

$$C_s = 0.142$$

$$W_q = \left(\frac{1^2 + 1.42^2}{2} \right) \left(\frac{.9507 \sqrt{2(4 \cdot 1)} - 1}{4(1 - .9507)} \right) \left(\frac{1}{1.42} \right)$$

$$W_q = (0.510082)(4.178632771)(0.704225352)$$

$$W_q = 1.5 \text{ minutes}$$

$$L_q = \lambda W_q$$

$$L_q = (5.4)(1.5) = 8.1 \approx 8$$

Average Wait in Line: 1.5 minutes

Expected # of Vehicles in Line: 8 vehicles

The mean service time for the single server at the concession stand follows a Poisson process with a mean of 2.5 minutes. Customers arrive every 3 minutes, exponentially distributed. – **Round to the nearest hundredth for questions 10-15.**

10)(3 points) What is the utilization?

$$\lambda = \frac{1 \text{ cust}}{3 \text{ min}} \quad \rho = \frac{\frac{1}{3}}{1 \cdot \frac{1}{2.5}} = .83\bar{3}$$

$$\mu = \frac{1 \text{ cust}}{2.5 \text{ mins}}$$

$$S = 1$$

ANS: $\rho = .83\bar{3}$

11)(3 points) How long will customers wait in line? queue

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)} \rightarrow \frac{\frac{1}{3}}{\frac{1}{2.5}(\frac{1}{2.5} - \frac{1}{3})} \rightarrow \frac{\frac{1}{3}}{.02\bar{6}} = 12.49$$

ANS: 12.5 minutes

12)(3 points) How many customers would you expect in the system? system

$$L = \frac{\rho}{(1 - \rho)} = \frac{.83\bar{3}}{1 - .83\bar{3}} = 4.99\bar{9} \approx 5$$

ANS: 5 customers

13)(3 points) What is the expected average time in the system?

$$W = \frac{1}{\mu - \lambda} = \frac{1}{\frac{1}{2.5} - \frac{1}{3}} = \frac{1}{0.06\bar{6}} = 14.9 \approx 15$$

ANS: 15 minutes

14)(4 points) What is the probability that 6 customers are in the system?

$$P_6 = (1 - .83\bar{3}) \cdot .83\bar{3}^6$$

$$(.16667)(.33489)$$

$$= 0.0558 \rightarrow 5.58\%$$

ANS: 5.58%

15)(8 points) What is the probability that there will be more than 2 customers at the concession stand?

$$P > 2 = (1 - .833) \cdot .833^0 = 0.166667$$

$$(1 - .833) \cdot .833^1 = 0.13889111$$

$$(1 - .833) \cdot .833^2 = 0.115962945$$

Add all

=

$$0.421520756$$

$$42.2\%$$

ANS: 42.2% ^{chance} ^{more than} _{2 customers}

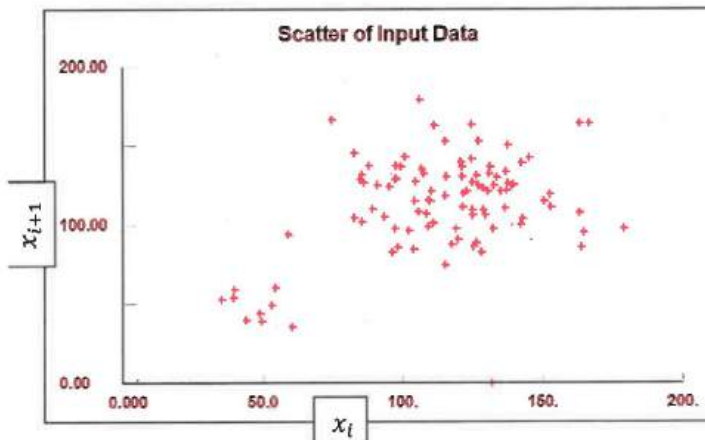
Section #3: Testing for Independence and Homogeneity (40 points)

You recently collected data on customer interarrival times, and now wish to analyze the data.

16)(6 Points) Name the three major methods to test for independence taught in EM481?

Checks if an observation is influenced by its preceding observation.	Up/Down
Compares the observed number of sequenced observations to the expected number of sequenced observations	Above/Below
Checks if observations in a single dataset are correlated over different time lags.	Chi squared goodness of fit

17)(8 Points) You generated the following charts from historical data. What can you conclude about the data in terms of independency and homogeneity?

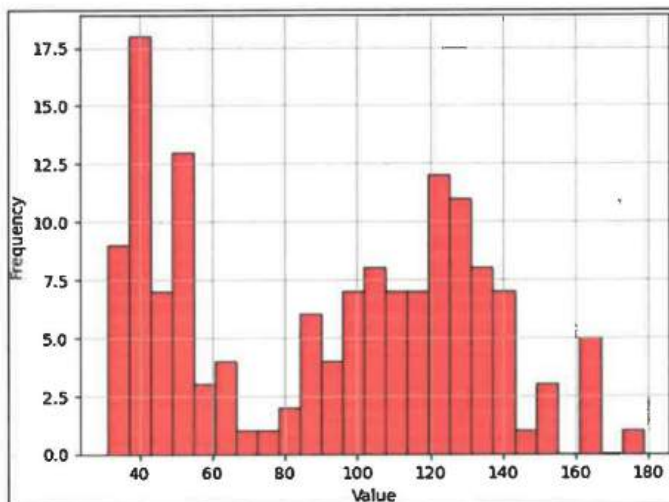


What type of chart is this?

Positive - random correlation

Conclusion from the graph:

The graph shows a positive correlation but random as K_i increases.



What type of chart is this?

Negative - random correlation

Conclusion from the graph:

The frequency goes down as the value increases.

18) (16 Points) You decide to conduct a **Runs Above/Below the Median Test** on the 30 data points below to determine whether the data is independent (random) given a level of significance (α) of 0.05. The median for this data is 9.665. Assume the critical value follows a standard normal distribution. **Show all your work** by completing the chart on the left and then answering the following questions.

Observation	Value	Above/Below
1	10.6	+
2	6.8	-
3	19.0	+
4	8.5	-
5	14.2	+
6	10.9	+
7	8.2	-
8	10.1	+
9	3.5	-
10	13.5	+
11	7.9	-
12	18.1	+
13	3.8	-
14	9.2	-
15	10.6	+
16	6.8	-
17	19.0	+
18	8.5	-
19	14.2	+
20	5.3	-
21	6.8	-
22	19.0	+
23	8.5	-
24	14.2	+
25	10.9	+
26	8.2	-
27	10.1	+
28	3.5	-
29	13.5	+
30	7.9	-

State the null and alternative hypotheses for a runs test.

Null: H_0 : Data is independent

Alternative: H_a : Data is not independent

$$M = \frac{2(15 \cdot 15)}{30} + 1 \rightarrow 16$$

$$n_1 = 15$$

$$n_2 = 15$$

$$r = 26$$

$$M = 16$$

$$G^2 = 7.24$$

$$G^2 = \frac{(16-1)(16-2)}{30-1}$$

$$G^2 = \frac{(15)(14)}{29} = \frac{210}{29} = 7.24$$

$$Z_0 = \frac{26-16}{\sqrt{7.24}} = \frac{10}{\sqrt{7.24}} = 3.716$$

$$3.716 < 1.96$$

$$\alpha = 0.05$$

↓

two tail

$$\alpha = 0.025$$

$$1.96$$

Test Statistic: 3.72

Critical Value: 1.96

Conclusion: Fail to reject the null

↖ (might've switched that up)

- 19) (10 Points) Analyze a Runs Up/Down Test for a dataset with 30 observations and 26 runs. Use a confidence level of 99%. Assume the critical value follows a standard normal distribution. **Show all your work** and answer the following questions. You are not required to write out your hypothesis.

$$\mu = \frac{2(30) - 1}{3} = 19.6\bar{6}$$

$$\sigma^2 = \frac{16(30) - 29}{90} = 5.011$$

$$Z_0 = \frac{26 - 19.6\bar{6}}{\sqrt{5.011}} = 2.829$$

$$2.829 < 2.579$$

$$\alpha = .01$$

two tailed
.005

Test Statistic: 2.829

Critical Value: 2.579

Conclusion: Fail to reject null

Section #4: Testing for Goodness of Fit (27 points)

20)(3 Points) (TRUE | **FALSE**) The Chi-Squared Goodness of Fit test is only used for continuous distributions.

21)(3 Points) (**TRUE** | FALSE) The Chi-Squared Test has a two-tail rejection region.

Concessions Stand

You suspect that service times at the new South Rim Concessions Stand follow an exponential distribution with a mean of 2 minutes. Data from 22 customers is analyzed.

Use the chi-square goodness of fit test to determine if the time taken to get food is exponentially distributed with a mean of 2 minutes.

The three bins below are equiprobable; use them to conduct your test.

Bin	0 – 1.5 min	1.6 – 4.5 min	> 4.5 min
Observed Frequency	6	11	5

22)(3 points) Write a null hypothesis specific to this scenario?

H₀: Time taken to get food is exponentially distributed with mean of 2 minutes

23)(12 points) What is your observed chi-squared test statistic?

$$\begin{aligned}
 \chi^2 &= \sum_{i=1}^k \frac{(22 - 7.33)^2}{7.33} \\
 &= -4.3346 \\
 &|-4.3346| <
 \end{aligned}$$

Used all formulas in reference manual

$$\begin{aligned}
 k &= 3 \\
 p_i &= \frac{1}{3} \\
 e_i &= (22)\left(\frac{1}{3}\right) = 7.33\bar{3} \\
 df &= 2
 \end{aligned}$$

Observed Test Statistic: |-4.3346|

24)(6 Points) Using a 95% confidence level, determine whether to reject the null hypothesis or not. Write the critical value and your decision.

$$95\% = 0.05 \rightarrow 0.025$$

$$\text{and } df = K - 1 = 4 - 1 = 3$$

$$C.V_{0.025, 3} = 9.348$$

$$|-4.3346| < 9.348$$

Critical Value: 9.348

Decision: Reject the null



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB H

WPR #2

Instructor Solution

**DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996**

EM481 WRITTEN PARTIAL REVIEW #2

(Show all work)

EM481 AT 26-1
WPR #2 (165 points)

Name: Solution

Hour/Section: _____

1. **Contents.** WPR #2 is an individual 75-minute exam worth 165 points.
2. **Show Your Work.** Unsupported work earns *little* or *no* partial credit. Neatness will also help your cause.
3. **References.** The authorized references are your calculator, pencil/pen, and the EM481 reference manual (provided).
4. **Grading Summary.**

Section	Possible Points	Score
A. Modeling Concepts	38	
B. Output Analysis of a Single System	23	
C. Output Analysis of Two Systems	40	
D. Output Analysis of Multiple Systems	51	
E. Output Analysis Concepts	13	
Total	165	

Show all work to receive full credit.

Section A Multiple Choice

11 questions (38 points)

Clearly Circle the correct answer.

1. (3 points) In ProModel, where do you dictate the frequency at which entities enter the system?
 - a. Locations
 - b. Entities
 - c. Arrivals**
 - d. Processing

2. (3 points) In ProModel, you've created a seed of 1 with a stream 4. Select the distribution that correctly applies the stream/seed.
 - a. Wait N(3, 2) min
 - b. Wait N(3, 2, 1, 4) min
 - c. Wait N(3, 2, 1) min
 - d. Wait N(3, 2, 4) min**

3. (3 points) What is the remainder for 17 mod (5)?
 - a. 1
 - b. 2**
 - c. 3.4
 - d. 12

4. (3 points) True or False: In ProModel, a global variable can be updated in a location's operation logic and then decremented in the move logic when the entity leaves the location.
 - a. True**
 - b. False

Show all work to receive full credit.

5. (3 points) **True or False:** Static resources generally do not appear graphically/visually in ProModel?
- a. **True**
 - b. False
6. (3 points) What is required for a dynamic resource?
- a. Attribute linked to User Distribution
 - b. Variables
 - c. Modules
 - d. **Path Network**
7. (3 points) **True or False:** After using the JOIN command to join entities, the entities can be separated later in the model.
- a. True
 - b. **False**
8. (3 points) **True or False:** The Group command requires an “If Group” request.
- a. True
 - b. **False**
9. (3 points) A _____ system starts and ends at a specified time or event; or at a time point specified by management.
- a. Continuous
 - b. Non-Continuous
 - c. **Terminating**
 - d. Non-Terminating

Show all work to receive full credit.

10. (3 points) Which of the following is the least useful when verifying a simulation?

- a. Output Viewer Results
- b. Trace Window
- c. Macro Scenario Parameters**
- d. Display Command

11. (8 points) Review the logic below. Select all that **HAVE** an error or create an infinite loop. (Circle all that apply).

a)

```
Int lv_syst_check = 5
Do
{
    Wait 10 min
    Dec a_system_check
}
Until lv_syst_check = 0
```

b)

```
a_rebake = 5
Do
{
    Wait 10 min
    Dec a_rebake
}
While a_rebake > 0
```

c)

```
a_class = 1
a_toppings = 3
If a_class = 1 Then
{
    Do
    {
        Wait 10 min
        Dec a_toppings
    }
    Until a_toppings < 0
}
```

d)

```
a_class = 1
a_CT_A = 2
Int lv_CT_Y = 5

If a_class = 2 Or a_CT_A > 1 Then
{
    Do
    {
        Wait 10 min
        Dec lv_CT_Y, 2
    }
    Until lv_CT_Y = 0
}
```


Show all work to receive full credit.

Section B Output Analysis of a Single System

2 Questions (23 points)

You are simulating a T-Shirt Production Factory. You build a DES model for the factory and would like to determine a confidence interval for the average number of T-shirts produced daily. You run your simulation for twenty (20) replications and calculate the following point estimates:

Descriptive Statistics	
1005.7	Mean
125	STD DEV (Sample)
6750.326	Variance (Sample)
20	n (# of Repetitions)

12. (12 points) Compute a 95% confidence interval for the true value of the average number of T-shirts produced.

$$\bar{x} \pm t_{df, \frac{\alpha}{2}} \left(\frac{s}{\sqrt{n}} \right)$$

$$df = n - 1 = 20 - 1 = 19$$

$$\frac{\alpha}{2} = \frac{0.05}{2} = 0.025$$

$$t_{19, 0.025} = 2.093$$

$$1005.7 \pm 2.093 \left(\frac{125}{\sqrt{20}} \right)$$

$$= 1005.7 \pm 58.5$$

$$[947.2 , 1064.2]$$

Confidence Interval: [_____ , _____]

(Upper & Lower confidence limit format)

13. (11 points) If you wanted to achieve a relative error of five percent (5%), how many replications would you have to run?

$$n' = \left(\frac{\frac{z_{\alpha/2} \cdot S}{2}}{\bar{x} \cdot \frac{re}{1+re}} \right)^2$$

$$df = inf$$

$$\frac{\alpha}{2} = \frac{0.05}{2} = 0.025$$

$$z_{0.025} = 1.96$$

$$n' = \left(\frac{1.96 \times 125}{1005.7 \cdot \frac{0.05}{1+0.05}} \right)^2 = 26.17 \approx 27$$

of Required Replications: _____

Show all work to receive full credit.

Section C Output Analysis of Two Systems

3 Questions (40 points)

You update your model and develop an alternative configuration for your T-Shift Factory. You run ten (10) replications of both the original configuration (Alternative 1) and your new alternative (Alternative 2). Using the data below, you will now determine if there is any difference in the average number of T-Shirts produced daily. A higher quantity is better.

14. (6 points) State your null and alternative hypotheses.

H_0 : Alternatives produce same AVG number of Tshirts ($\mu_1 = \mu_2$)

H_0 : _____

H_a : Alternatives produce different AVG number of Tshirts ($\mu_1 \neq \mu_2$)

H_a : _____

15. (12 points) Given the information below, calculate the degrees of freedom needed if you were to conduct a Welch CI Test.

Rep.	ALT 1	ALT 2
1	957.00	961.00
2	939.00	942.00
3	952.00	960.00
4	960.00	962.00
5	951.00	951.00
6	936.00	937.00
7	961.00	961.00
8	949.00	952.00
9	956.00	957.00
10	954.00	963.00

ALT 1	ALT 2	
951.5	954.6	Mean
8.317	9.009	STD DEV (Sample)
69.167	81.156	Variance (Sample)
10	10	n (# of Repetitions)

$$df \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1-1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2-1}} = \frac{\left(\frac{69.167}{10} + \frac{81.156}{10}\right)^2}{\frac{\left(\frac{69.167}{10}\right)^2}{10-1} + \frac{\left(\frac{81.156}{10}\right)^2}{10-1}} = \frac{225.97}{5.316 + 7.318} = \frac{225.97}{12.634}$$

$$df = 17.89 \approx 18$$

Degrees of Freedom: _____

	0.05	Significance Level
	17.89	Degrees of Freedom
	2.101	t critical value

Show all work to receive full credit.

16. (22 points) You receive new information so you update your model and rerun both alternatives for 15 replications each. You now have the following information. Complete your Welch CI Test and provide your determination by circling the appropriate answers below.

ALT 1	ALT 2	
794.8	798.07	Mean
7.37	9.208	STD DEV (Sample)
54.314	84.781	Variance (Sample)
15	15	n (# of Repetitions)

0.05	Significance Level
27	Degrees of Freedom

$$(\bar{x}_1 - \bar{x}_2) \pm t_{df, \frac{\alpha}{2}} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$(794.8 - 798.07) \pm 2.052 \sqrt{\frac{54.314}{15} + \frac{84.781}{15}}$$

$$(-3.27) \pm 2.052 \sqrt{9.273} = (-3.27) \pm 2.052 * 3.045$$

$$(-3.27) \pm 6.249 \rightarrow [-9.519, 2.979]$$

Confidence Interval: [_____ , _____]

Determination: (Fail to Reject H_0) (Reject the H_0)
(Circle One)

Better Alternative: (ALT 1) (ALT 2) (Neither / Not Applicable)
(Circle One)

IF REJECT → THEN ALT 2 IS THE FOLLOW ANS.

Show all work to receive full credit.

Section D Output Analysis of Multiple Systems

3 Questions (51 points)

You developed three (3) alternatives for a widget assembly plant and ran each alternative for sixteen (16) replications each, recording the total widgets produced for each replication. You will now conduct an ANOVA to determine if there is any significant difference between the alternatives.

17. (6 points) State your null and alternative hypotheses. Use an $\alpha = 0.05$.

H₀: _____

H₀: Alternatives all produce the same number of widgets ... ($\mu = \mu_2 = \mu_3 = \mu_4$)

H_a: At least one alternatives produces a different number of widgets ... ($\mu \neq \mu_i$, for at least 1 Alternative)

H_a: _____

18. (25 points) Use the information below to complete your ANOVA test. Fill out the associated table on the right and state your test determination.

	Alt 1	Alt 2	Alt 3
$\bar{x}_i$	1129	1156.8	1247.4
Variance	1.92	3.74	4.18
$\bar{\bar{x}}$	70.56	72.3	77.96
$x_{..} = \sum_{i=1}^K x_i$	3533.20		
$\sum_{j=1}^n x_{ij}^2$	79693.88	83692.70	97313.08
SS_i	28.82	56.06	62.66

Source of Variation	Degrees of Freedom	Sum of Squares	Mean Square	Fcalc	Fcritical
Treatment					
Error					
Total (corrected)					

$$SST = \frac{1}{n} \left[\left(\sum_{i=1}^K x_i^2 \right) - \frac{x_{..}^2}{K} \right] = \frac{1}{16} \left[(1129^2 + 1156.8^2 + 1247.4^2) - \frac{3533.2^2}{3} \right] = 479.16$$

$$1129^2 + 1156.8^2 + 1247.4^2 = 4168834$$

$$SSE = \sum_{i=1}^K SS_i = 28.82 + 56.06 + 62.66 = 147.54$$

$$MS_{TC} = SST + SSE = 479.16 + 147.54 = 626.7$$

$$MST = \frac{SST}{K-1} = \frac{479.16}{3-1} = 239.58$$

$$MSE = \frac{SSE}{N-K} = \frac{147.54}{(48-3)} = 3.28$$

$$F_{calc} = \frac{MST}{MSE} = \frac{239.58}{3.28} = 73.04$$

Source of Variation	Degrees of Freedom	Sum of Squares	Mean Square	Fcalc	Fcritical	
Treatment	2	479.16	239.58	73.075	3.204	3.20 -- From Reference
Error	45	147.54	3.28			
Total (corrected)	47	626.70				

ANOVA Determination: Reject the Null. Significant Difference

Show all work to receive full credit.

19. (20 points) You complete an ANOVA test on a different set of alternatives for a different factory, one that produces bottles. You run each of 3 alternatives for 11 replications. The owner wants to minimize daily costs. Use the information below to complete a Fisher Least Significance Difference (LSD) Test on two of the possible alternative pairings. Use a $\alpha = 0.05$. Write your conclusions in the table below.

Factor	Daily Cost (in Dollars)		
	Alternatives		
Replications	Alt 1	Alt 2	Alt 3
$\bar{x}_i$	14053.73	12672.23	13965.77
Variance	1087.88	575.87	1279.09
$\bar{\bar{x}}$	1277.61	1152.02	1269.62

Source of Variation	Degrees of Freedom	Sum of Squares	Mean Square
Treatment	2	108773.5	54386.74
Error	30	29428.37	980.9457
Total (corrected)	32	138201.9	

$$LSD = t_{N-K, \alpha/2} \sqrt{\frac{2MS_E}{n}} = t_{30, 0.025} \sqrt{\frac{2 \cdot 33931.77}{11}}$$

$$LSD = 2.042 * 13.355 = 27.27$$

$$|\bar{x}_i - \bar{x}_{i'}| > LSD$$

$$|\bar{x}_i - \bar{x}_{i'}| > 27.27$$

LSD Value: _____

	<u>Alt 2</u>	<u>Alt 3</u>
<u>Alt 1</u>	$ \bar{x}_1 - \bar{x}_2 = 125.59$ $125.59 > 27.27 \Rightarrow \text{Significant Difference}$	$ \bar{x}_1 - \bar{x}_3 = 7.99$ $7.99 > 27.27 \Rightarrow \text{NO Significant Diff}$
<u>Alt 3</u>		

Show all work to receive full credit.

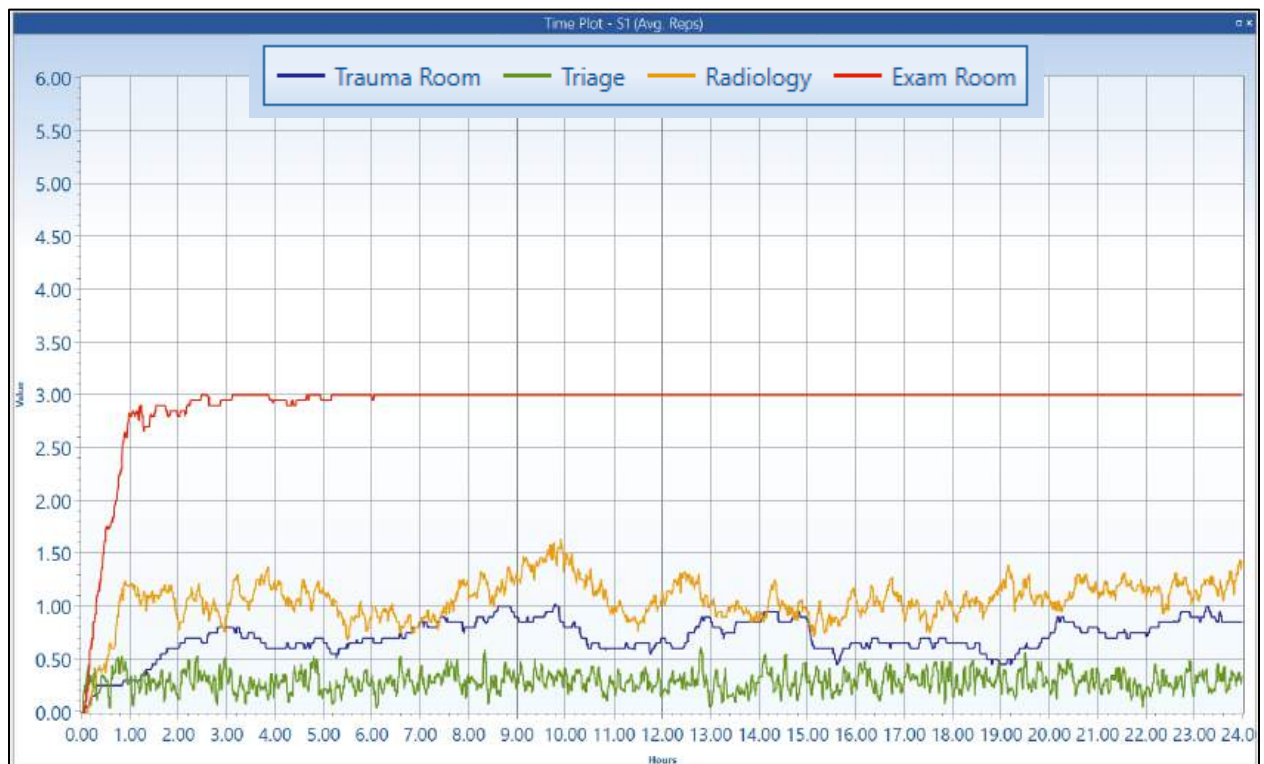
Section E Output Analysis Concepts

3 Questions (13 points)

You are modeling a Hospital Emergency Room. Answer the following questions. In the model, each location has its own respective queue. The location capacities are:

Service Location	Capacity
Trauma Room	3
Triage	1
Radiology	2
Exam Rooms	3

20. (5 points) The time plot below shows the average number of patients in each service location over 10 replications for 24 hours. Which service location(s) may have an unstable/runaway queue. (Circle all that apply).



- a. Trauma Room
- b. Triage
- c. Radiology
- d. Exam Rooms
- e. None of the locations

Show all work to receive full credit.

You update your model with new information and run it for 24 hours and 20 replications. The following chart and table show the average number of patients in the hospital ER at any point in time.



21. (4 points) If this is a non-terminating model, what needs to be done in order to calculate a more accurate average number of patients?

Response needs to mention:

Determine the warm-up time, so that it is not included in the collected data.

Optional:

Use of Welch Moving Average.

22. (4 points) You create two alternative scenarios for the hospital simulation. You run each scenario for 20 replications. You don't use a set seed, so each replication (within and across scenarios) is independent. Which testing method should you use to see if there is a significant difference between the two alternatives.

a) Welch CI Test

b) Paired-t Test

Outstanding Student Work

WPR #2

● Graded

Student

Ethan Ellis

Total Points

162 / 165 pts

Question 1

M- LEAP

3 / 3 pts

✓ - 0 pts Correct

Question 2

M- Stream/Seed

0 / 3 pts

✓ - 3 pts Incorrect

Question 3

M- RNG Mod

3 / 3 pts

✓ - 0 pts Correct

Question 4

T/F - G.Variable

3 / 3 pts

✓ - 0 pts Correct

Question 5

T/F - Static Res.

3 / 3 pts

✓ - 0 pts Correct

Question 6

M - Dyn. Res.

3 / 3 pts

✓ - 0 pts Correct

Question 7

T/F - JOIN

3 / 3 pts

✓ - 0 pts Correct

Question 8

T/F - GROUP

3 / 3 pts

✓ - 0 pts Correct

Question 9

M - Non-Terminating

3 / 3 pts

✓ - 0 pts Correct

Question 10

M - Verification

3 / 3 pts

✓ - 0 pts Correct

Question 11

M - Looping

8 / 8 pts

✓ - 0 pts Correct

Question 12

C - CI, Single

12 / 12 pts

✓ - 0 pts Correct

Question 13

C - Rel. Error

11 / 11 pts

✓ - 0 pts Correct

Question 14

S - Welch Ho & Ha

6 / 6 pts

✓ - 0 pts Correct

Question 15

C - Welch df

12 / 12 pts

✓ - 0 pts Correct

Question 16

C - Welch Test

22 / 22 pts

✓ - 0 pts Correct

Question 17

S - ANOVA Ho & Ha

6 / 6 pts

✓ - 0 pts Correct

Question 18

C - ANOVA

25 / 25 pts

✓ - 0 pts Correct

Question 19

C - LSD

20 / 20 pts

✓ - 0 pts Correct

Question 20

M - Unstable Queue

5 / 5 pts

✓ - 0 pts Correct

Question 21

S - Warm-up

4 / 4 pts

✓ - 0 pts Correct

Question 22

M - Welch vs. Paired-t

4 / 4 pts

✓ - 0 pts Correct

**DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996**

EM481 WRITTEN PARTIAL REVIEW #2

(Show all work)

EM481 AT 26-1
WPR #2 (165 points)

Name: Ethan Ellis

Hour/Section: 62

1. **Contents.** WPR #2 is an individual 75-minute exam worth 165 points.
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B. Output Analysis of a Single System	23	
C. Output Analysis of Two Systems	40	
D. Output Analysis of Multiple Systems	51	
E. Output Analysis Concepts	13	
Total	165	

Show all work to receive full credit.

Section A Modeling Concepts
11 questions (38 points)

Clearly Circle the correct answer.

1. (3 points) In ProModel, where do you dictate the frequency at which entities enter the system?
 - a. Locations
 - b. Entities
 - ☒ c. Arrivals
 - d. Processing

2. (3 points) In ProModel, you've created a seed of 1 with a stream 4. Select the distribution that correctly applies the stream/seed.
 - a. Wait N(3, 2) min
 - b. Wait N(3, 2, 1, 4) min
 - ☒ c. Wait N(3, 2, 1) min
 - d. Wait N(3, 2, 4) min

3. (3 points) What is the remainder for 17 mod (5)?
 - a. 1
 - ☒ b. 2
 - c. 3.4
 - d. 12

4. (3 points) **True or False:** In ProModel, a global variable can be updated in a location's operation logic and then decremented in the move logic when the entity leaves the location.
 - ☒ a. True
 - b. False

Show all work to receive full credit.

5. (3 points) **True or False:** Static resources generally do not appear graphically/visually in ProModel?
- ☒ a. True
 - b. False
6. (3 points) What is required for a dynamic resource?
- a. Attribute linked to User Distribution
 - b. Variables
 - c. Modules
 - ☒ d. Path Network
7. (3 points) **True or False:** After using the JOIN command to join entities, the entities can be separated later in the model.
- a. True
 - ☒ b. False
8. (3 points) **True or False:** The Group command requires an "If Group" request.
- a. True
 - ☒ b. False
9. (3 points) A _____ system starts and ends at a specified time or event; or at a time point specified by management.
- a. Continuous
 - b. Non-Continuous
 - ☒ c. Terminating
 - d. Non-Terminating

Show all work to receive full credit.

10. (3 points) Which of the following is the least useful when verifying a simulation?

- a. Output Viewer Results
- b. Trace Window
- ☒ c. Macro Scenario Parameters
- d. Display Command

11. (8 points) Review the logic below. Select all that HAVE an error or create an infinite loop. (Circle all that apply).

a)

```
Int lv_syst_check = 5
Do
{
    Wait 10 min
    Dec a_system_check
}
Until lv_syst_check = 0
```

b)

```
a_rebake = 5
Do
{
    Wait 10 min
    Dec a_rebake
}
While a_rebake > 0
```

c)

```
a_class = 1
a_toppings = 3
If a_class = 1 Then
{
    Do
    {
        Wait 10 min
        Dec a_toppings
    }
    Until a_toppings < 0
}
```

d)

```
a_class = 1
a_CT_A = 2
Int lv_CT_Y = 5

If a_class = 2 Or a_CT_A > 1 Then
{
    Do
    {
        Wait 10 min
        Dec lv_CT_Y, 2
    }
    Until lv_CT_Y = 0
}
```

Show all work to receive full credit.

Section B Output Analysis of a Single System
2 Questions (23 points)

You are simulating a T-Shirt Production Factory. You build a DES model for the factory and would like to determine a confidence interval for the average number of T-shirts produced daily. You run your simulation for twenty (20) replications and calculate the following point estimates:

Descriptive Statistics	
1005.7	Mean
125	STD DEV (Sample)
6750.326	Variance (Sample)
20	n (# of Repetitions)

use 15625 for variance

12. (12 points) Compute a 95% confidence interval for the true value of the average number of T-shirts produced. $\bar{x} \pm t_{n-1, \frac{\alpha}{2}} \left(\frac{s}{\sqrt{n}} \right)$

$$1005.7 \pm 2.093 \left(\frac{125}{\sqrt{20}} \right)$$

$$1005.7 \pm 58.501$$

$$(947.199, 1064.201)$$

Confidence Interval: [947.199 , 1064.201]
(Upper & Lower confidence limit format)

13. (11 points) If you wanted to achieve a relative error of five percent (5%), how many replications would you have to run?

$$n' = \left(\frac{(1.96)(125)}{\left(\frac{0.05}{1.05} \right)(1005.7)} \right)^2 = \left(\frac{245}{47.3905} \right)^2 = 26.17 \rightarrow 27$$

of Required Replications: 27 replications

Show all work to receive full credit.

Section C Output Analysis of Two Systems

3 Questions (40 points)

You update your model and develop an alternative configuration for your T-Shift Factory. You run ten (10) replications of both the original configuration (Alternative 1) and your new alternative (Alternative 2). Using the data below, you will now determine if there is any difference in the average number of T-Shirts produced daily. A higher quantity is better.

14. (6 points) State your null and alternative hypotheses.

$H_0: \mu_1 = \mu_2$; alternative 1 and alternative 2 produce the same average # of t-shirts daily

$H_a: \mu_1 \neq \mu_2$; alt 1 and alt 2 produce different average # of t-shirts daily

15. (12 points) Given the information below, calculate the degrees of freedom needed if you were to conduct a Welch CI Test.

Rep.	ALT 1	ALT 2
1	957.00	961.00
2	939.00	942.00
3	952.00	960.00
4	960.00	962.00
5	951.00	951.00
6	936.00	937.00
7	961.00	961.00
8	949.00	952.00
9	956.00	957.00
10	954.00	963.00

ALT 1	ALT 2	
951.5	954.6	Mean
8.317	9.009	STD DEV (Sample)
69.167	81.156	Variance (Sample)
10	10	n (# of Repetitions)

$$df = \frac{\left(\frac{69.167}{10} + \frac{81.156}{10} \right)^2}{\frac{\left(\frac{69.167}{10} \right)^2}{9} + \frac{\left(\frac{81.156}{10} \right)^2}{9}} = \frac{225.97}{5.3156 + 7.3181}$$

$$df = 17.886 \rightarrow 18$$

Degrees of Freedom: 18

Show all work to receive full credit.

16. (22 points) You receive new information so you update your model and rerun both alternatives for 15 replications each. You now have the following information. Complete your Welch CI Test and provide your determination by circling the appropriate answers below.

ALT 1	ALT 2	
794.8	798.07	Mean
7.37	9.208	STD DEV (Sample)
54.314	84.781	Variance (Sample)
15	15	n (# of Repetitions)

0.05	Significance Level
27	Degrees of Freedom

$$794.8 - 798.07 \pm 2.052 \sqrt{\frac{54.314}{15} + \frac{84.781}{15}}$$

$$-3.27 \pm 6.249$$

$$(-9.519, 2.979)$$

Confidence Interval: -9.519 , 2.979

Determination:
(Circle One)

(Fail to Reject H_0)

(Reject the H_0)

Better Alternative:
(Circle One)

(ALT 1)

(ALT 2)

(Neither / Not Applicable)

Show all work to receive full credit.

Section D Output Analysis of Multiple Systems

3 Questions (51 points)

You developed three (3) alternatives for a widget assembly plant and ran each alternative for sixteen (16) replications each, recording the total widgets produced for each replication. You will now conduct an ANOVA to determine if there is any significant difference between the alternatives.

17. (6 points) State your null and alternative hypotheses. Use an $\alpha = 0.05$.

$H_0: \mu_1 = \mu_2 = \mu_3$ There is no difference in widgets produced between the 3 plants

$H_a: \mu_1 \neq \mu_2 \neq \mu_3$ There is a difference in widgets produced between the 3 plants

18. (25 points) Use the information below to complete your ANOVA test. Fill out the associated table on the right and state your test determination.

	Alt 1	Alt 2	Alt 3
$\bar{x}_i$	1129	1156.8	1247.4
Variance	1.92	3.74	4.18
$\bar{\bar{x}}$	70.56	72.3	77.96
$x_{..} = \sum_{i=1}^K x_i$	3533.20		
$\sum_{j=1}^n x_{ij}^2$	79693.88	83692.70	97313.08
SS_i	28.82	56.06	62.66

Source of Variation	Degrees of Freedom	Sum of Squares	Mean Square	Fcalc	Fcritical
Treatment	2	479.16	239.58	73.04	3.20
Error	45	147.54	3.28		
Total (corrected)	47	626.7			

$$\begin{aligned} DOFT_r &= 3-1=2 \\ DOFE &= \\ DOFT_{total} &= 3 \times 16-1 \\ &= 47 \end{aligned}$$

$$SST = \frac{1}{16} \left[(1129)^2 + (1156.8)^2 + (1247.4)^2 - \frac{(3533.2)^2}{3} \right]$$

$$SST = \frac{1}{16} [4168834 - 4161167.413]$$

$$SST = \frac{1}{16} [766.58667]$$

$$SST = 479.16$$

$$SSE = 28.82 + 56.06 + 62.66$$

$$SSE = 147.54$$

$$MST = \frac{479.16}{2} = 239.58$$

$$\begin{aligned} SST_c &= 479.16 + 147.54 \\ SST_c &= 626.7 \end{aligned}$$

$$F_{calc} = \frac{239.58}{3.28}$$

$$F_{calc} = 73.04$$

$$MSE = \frac{147.54}{48-3} = 3.28$$

ANOVA Determination: Reject Null, there is a difference in widgets produced

between the 3 plants ($F_{calc} > F_{crit}$)
73.04 > 3.2

Show all work to receive full credit.

19. (20 points) You complete an ANOVA test on a different set of alternatives for a different factory, one that produces bottles. You run each of 3 alternatives for 11 replications. The owner wants to minimize daily costs. Use the information below to complete a Fisher Least Significance Difference (LSD) Test on two of the possible alternative pairings. Use a $\alpha = 0.05$. Write your conclusions in the table below.

Factor	Daily Cost (in Dollars)		
	Alternatives		
Replications	Alt 1	Alt 2	Alt 3
$\bar{x}_i$	14053.73	12672.23	13965.77
Variance	1087.88	575.87	1279.09
$\bar{x}_i$	1277.61	1152.02	1269.62

$$LSD(0.05) = 2.042 \sqrt{\frac{2(980.9457)}{11}}$$

$$LSD(0.05) = 27.27$$

Source of Variation	Degrees of Freedom	Sum of Squares	Mean Square
Treatment	2	108773.5	54386.74
Error	30	29428.37	980.9457
Total (corrected)	32	138201.9	

Alt 1 vs Alt 2

$$|1277.61 - 1152.02| = 125.59$$

$$125.59 > 27.27$$

Sig Diff

LSD Value: 27.27

	Alt 2	Alt 3
Alt 1	$ 1277.61 - 1152.02 = 125.59$ $125.59 > 27.27$ <u>Significant difference, Alt 2 is better than alt 1</u>	$ 1277.61 - 1269.62 = 7.99$ $7.99 < 27.27$ <u>No Significant Difference</u>
Alt 3		

Show all work to receive full credit.

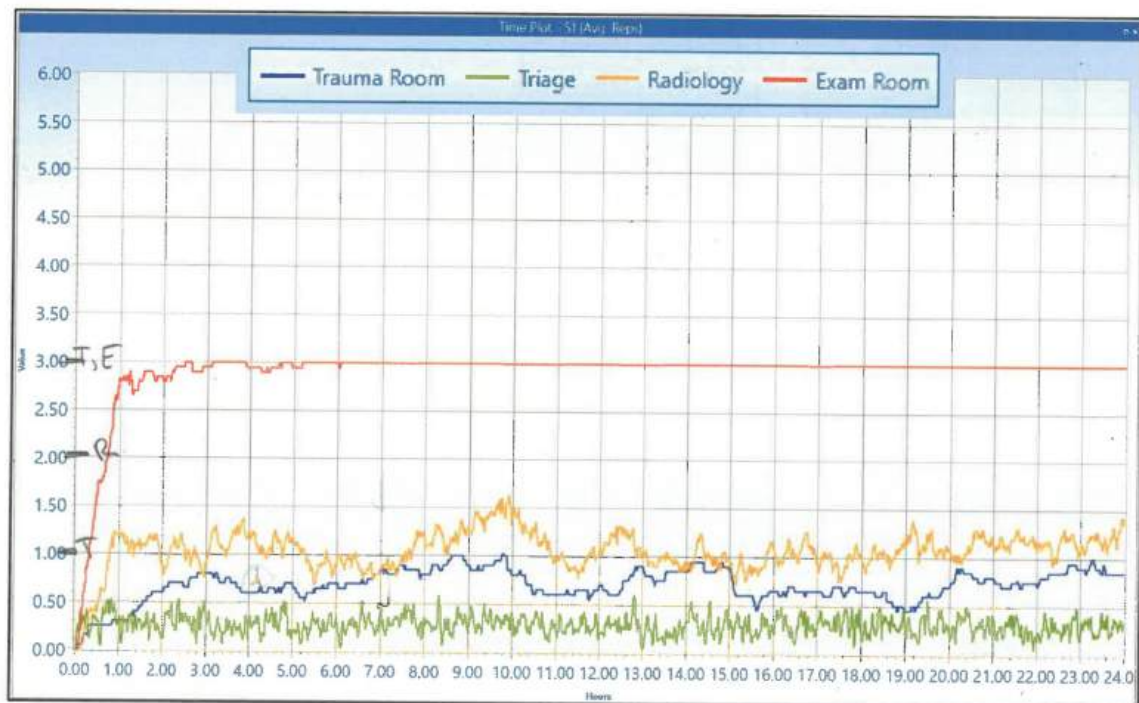
Section E Output Analysis Concepts

3 Questions (13 points)

You are modeling a Hospital Emergency Room. Answer the following questions. In the model, each location has its own respective queue. The location capacities are:

Service Location	Capacity
Trauma Room	3
Triage	1
Radiology	2
Exam Rooms	3

20. (5 points) The time plot below shows the average number of patients in each service location over 10 replications for 24 hours. Which service location(s) may have an unstable/runaway queue. (Circle all that apply).



- a. Trauma Room
- b. Triage
- c. Radiology
- ☒ d. Exam Rooms
- e. None of the locations

Show all work to receive full credit.

You update your model with new information and run it for 24 hours and 20 replications. The following chart and table show the average number of patients in the hospital ER at any point in time.



21. (4 points) If this is a non-terminating model, what needs to be done in order to calculate a more accurate average number of patients?

You should use Welch's Moving Average to eliminate the warm-up period from the true average number of patients. Using Batching versus more replications will also give a more accurate average since the model is non-terminating.

22. (4 points) You create two alternative scenarios for the hospital simulation. You run each scenario for 20 replications. You don't use a set seed, so each replication (within and across scenarios) is independent. Which testing method should you use to see if there is a significant difference between the two alternatives.

- a) ☒ Welch CI Test
- b) ☐ Paired-t Test

Fair-to-Good Student Work

Student
Caleb Uhorchuk

Total Points
146 / 165 pts

Question 1

M- LEAP

3 / 3 pts

✓ - 0 pts Correct

Question 2

M- Stream/Seed

0 / 3 pts

✓ - 3 pts Incorrect

Question 3

M- RNG Mod

3 / 3 pts

✓ - 0 pts Correct

Question 4

T/F - G.Variable

3 / 3 pts

✓ - 0 pts Correct

Question 5

T/F - Static Res.

3 / 3 pts

✓ - 0 pts Correct

Question 6

M - Dyn. Res.

3 / 3 pts

✓ - 0 pts Correct

Question 7

T/F - JOIN

3 / 3 pts

✓ - 0 pts Correct

Question 8

T/F - GROUP

3 / 3 pts

✓ - 0 pts Correct

Question 9

M - Non-Terminating

3 / 3 pts

✓ - 0 pts Correct

Question 10

M - Verification

0 / 3 pts

✓ - 3 pts Incorrect

Question 11

M - Looping

8 / 8 pts

✓ - 0 pts Correct

Question 12

C - CI, Single

12 / 12 pts

✓ - 0 pts Correct

Question 13

C - Rel. Error

11 / 11 pts

✓ - 0 pts Correct

Question 14

S - Welch Ho & Ha

3 / 6 pts

✓ - 3 pts Did not define what is being compared.

Question 15

C - Welch df

10 / 12 pts

✓ - 1 pt Equation Setup Error

✓ - 1 pt Math Error

1 Additional Math Error -- Your equation, as written, does not equal 27.69

2 Dropped Square

Question 16

C - Welch Test

22 / 22 pts

✓ - 0 pts Correct

Question 17

S - ANOVA Ho & Ha

6 / 6 pts

✓ - 0 pts Correct

Question 18

C - ANOVA

25 / 25 pts

✓ - 0 pts Correct

Question 19

C - LSD

20 / 20 pts

✓ - 0 pts Correct

Question 20

M - Unstable Queue

5 / 5 pts

✓ - 0 pts Correct

Question 21

S - Warm-up

0 / 4 pts

✓ - 4 pts Blank / No meaningful response

Question 22

M - Welch vs. Paired-t

0 / 4 pts

✓ - 4 pts Incorrect

**DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996**

EM481 WRITTEN PARTIAL REVIEW #2

(Show all work)

EM481 AT 26-1
WPR #2 (165 points)

Name: Caleb Uborchuk

Hour/Section: I1

1. **Contents.** WPR #2 is an individual 75-minute exam worth 165 points.
2. **Show Your Work.** Unsupported work earns *little* or *no* partial credit. Neatness will also help your cause.
3. **References.** The authorized references are your calculator, pencil/pen, and the EM481 reference manual (provided).
4. **Grading Summary.**

Section	Possible Points	Score
A. Modeling Concepts	38	
B. Output Analysis of a Single System	23	
C. Output Analysis of Two Systems	40	
D. Output Analysis of Multiple Systems	51	
E. Output Analysis Concepts	13	
Total	165	

Show all work to receive full credit.

Section A Modeling Concepts

11 questions (38 points)

Clearly Circle the correct answer.

1. (3 points) In ProModel, where do you dictate the frequency at which entities enter the system?
 - a. Locations
 - b. Entities
 - ☒ c. Arrivals
 - d. Processing

2. (3 points) In ProModel, you've created a seed of 1 with a stream 4. Select the distribution that correctly applies the stream/seed.
 - a. Wait N(3, 2) min
 - b. Wait N(3, 2, 1, 4) min
 - ☒ c. Wait N(3, 2, 1) min
 - d. Wait N(3, 2, 4) min

3. (3 points) What is the remainder for 17 mod (5)?
 - a. 1
 - ☒ b. 2
 - c. 3.4
 - d. 12

4. (3 points) **True or False:** In ProModel, a global variable can be updated in a location's operation logic and then decremented in the move logic when the entity leaves the location.
 - ☒ a. True
 - b. False

Show all work to receive full credit.

5. (3 points) **True or False:** Static resources generally do not appear graphically/visually in ProModel?
☒ a. True
b. False
6. (3 points) What is required for a dynamic resource?
a. Attribute linked to User Distribution
b. Variables
c. Modules
☒ d. Path Network
7. (3 points) **True or False:** After using the JOIN command to join entities, the entities can be separated later in the model.
a. True
☒ b. False
8. (3 points) **True or False:** The Group command requires an "If Group" request.
a. True
☒ b. False
9. (3 points) A _____ system starts and ends at a specified time or event; or at a time point specified by management.
a. Continuous
b. Non-Continuous
☒ c. Terminating
d. Non-Terminating

Show all work to receive full credit.

10. (3 points) Which of the following is the least useful when verifying a simulation?

- ☒ a. Output Viewer Results
- b. Trace Window
- c. Macro Scenario Parameters
- d. Display Command

11. (8 points) Review the logic below. Select all that HAVE an error or create an infinite loop. (Circle all that apply).

a)

```
Int lv_syst_check = 5
Do
{
    Wait 10 min
    Dec a_system_check
}
Until lv_syst_check = 0
```

b)

```
a_rebake = 5
Do
{
    Wait 10 min
    Dec a_rebake
}
While a_rebake > 0
```

c)

```
a_class = 1
a_toppings = 3
If a_class = 1 Then
{
    Do
    {
        Wait 10 min
        Dec a_toppings
    }
    Until a_toppings < 0
}
```

☒ d)

```
a_class = 1
a_CT_A = 2
Int lv_CT_Y = 5

If a_class = 2 Or a_CT_A > 1 Then
{
    Do
    {
        Wait 10 min
        Dec lv_CT_Y, 2
    }
    Until lv_CT_Y = 0
}
```


Show all work to receive full credit.

Section B Output Analysis of a Single System

2 Questions (23 points)

You are simulating a T-Shirt Production Factory. You build a DES model for the factory and would like to determine a confidence interval for the average number of T-shirts produced daily. You run your simulation for twenty (20) replications and calculate the following point estimates:

Descriptive Statistics	
1005.7	Mean
125	STD DEV (Sample)
6750.326	Variance (Sample)
20	n (# of Repetitions)

12. (12 points) Compute a 95% confidence interval for the true value of the average number of T-shirts produced.

$$t_{19, 0.025}$$

$$1005.7 \pm 2.093 \left(\frac{125}{\sqrt{20}} \right)$$

$$1005.7 \pm 58.501$$

$$= 947.2, 1064.2$$

Confidence Interval: 947.2, 1064.2
(Upper & Lower confidence limit format)

13. (11 points) If you wanted to achieve a relative error of five percent (5%), how many replications would you have to run?

$$n' = \left(\frac{1.96 (125)}{\frac{0.05}{1.05} (1005.7)} \right)^2$$

$$n' = 26.17$$

of Required Replications:

27

Show all work to receive full credit.

Section C Output Analysis of Two Systems

3 Questions (40 points)

You update your model and develop an alternative configuration for your T-Shift Factory. You run ten (10) replications of both the original configuration (Alternative 1) and your new alternative (Alternative 2). Using the data below, you will now determine if there is any difference in the average number of T-Shirts produced daily. A higher quantity is better.

14. (6 points) State your null and alternative hypotheses.

$H_0: \mu_1 = \mu_2$, there is no difference in the configuration

$H_a: \mu_1 \neq \mu_2$, there is a difference in the configuration

15. (12 points) Given the information below, calculate the degrees of freedom needed if you were to conduct a Welch CI Test.

Rep.	ALT 1	ALT 2
1	957.00	961.00
2	939.00	942.00
3	952.00	960.00
4	960.00	962.00
5	951.00	951.00
6	936.00	937.00
7	961.00	961.00
8	949.00	952.00
9	956.00	957.00
10	954.00	963.00

ALT 1	ALT 2	
951.5	954.6	Mean
8.317	9.009	STD DEV (Sample)
69.167	81.156	Variance (Sample)
10	10	n (# of Repetitions)

$$\frac{8.317^2}{10} + \frac{9.009^2}{10} = 6.92 + 8.12$$

$$df = \frac{(6.92 + 8.12)^2}{\frac{(6.92)^2}{9} + \frac{(8.12)^2}{9}}$$

$$df = 27.69$$

Degrees of Freedom: 28

EM481 AT26-1
WPR #2

Last Name (Printed): Whorchuk

Show all work to receive full credit.

16. (22 points) You receive new information so you update your model and rerun both alternatives for 15 replications each. You now have the following information. Complete your Welch CI Test and provide your determination by circling the appropriate answers below.

ALT 1	ALT 2	
794.8	798.07	Mean
7.37	9.208	STD DEV (Sample)
54.314	84.781	Variance (Sample)
15	15	n (# of Repetitions)

0.05	Significance Level
27	Degrees of Freedom

$$\frac{237^2}{15} + \frac{9.208^2}{15} = 3.62 + 5.65$$

$$t_{27, 0.025} = 2.052$$

$$\begin{aligned} &794.8 - 798.07 \pm 2.052 \sqrt{3.62 + 5.65} \\ &-3.27 \pm 2.025 \sqrt{9.27} \\ &-9.435, 2.895 \end{aligned}$$

Confidence Interval: -9.435, 2.895

Determination:
(Circle One)

(Fail to Reject H_0)

(Reject the H_0)

Better Alternative:
(Circle One)

(ALT 1)

(ALT 2)

(Neither / Not Applicable)

Show all work to receive full credit.

Section D Output Analysis of Multiple Systems

3 Questions (51 points)

You developed three (3) alternatives for a widget assembly plant and ran each alternative for sixteen (16)ⁿ replications each, recording the total widgets produced for each replication. You will now conduct an ANOVA to determine if there is any significant difference between the alternatives.

17. (6 points) State your null and alternative hypotheses. Use an $\alpha = 0.05$.

H₀: There is no statistical difference between μ_1 , μ_2 , and μ_3

H_a: There is a statistical difference between μ_1 , μ_2 or μ_3

18. (25 points) Use the information below to complete your ANOVA test. Fill out the associated table on the right and state your test determination.

	Alt 1	Alt 2	Alt 3
$\bar{x}_i$	1129	1156.8	1247.4
Variance	1.92	3.74	4.18
$\bar{x}_i$	70.56	72.3	77.96
$x_{..} = \sum_{i=1}^K x_i$	3533.20		
$\sum_{j=1}^n x_{ij}^2$	79693.88	83692.70	97313.08
SS_i	28.82	56.06	62.66

Source of Variation	Degrees of Freedom	Sum of Squares	Mean Square	Fcalc	Fcritical
Treatment	2	479.2	239.6	73.1	3.2
Error	45	147.5	3.28		
Total (corrected)	47	626.7			

$$\frac{1}{16} \left[(1129^2 + 1156.8^2 + 1247.4^2) - \frac{3533.20^2}{3} \right]$$

$$\frac{1}{16} [4168834 - 4161167.413]$$

$$= 479.2$$

$$73.1 > 3.2$$

ANOVA Determination: Reject the Null

Show all work to receive full credit.

19. (20 points) You complete an ANOVA test on a different set of alternatives for a different factory, one that produces bottles. You run each of 3 alternatives for 11 replications. The owner wants to minimize daily costs. Use the information below to complete a Fisher Least Significance Difference (LSD) Test on two of the possible alternative pairings. Use a $\alpha = 0.05$. Write your conclusions in the table below.

Factor	Daily Cost (in Dollars)		
	Alternatives		
Replications	Alt 1	Alt 2	Alt 3
$\bar{x}_i$	14053.73	12672.23	13965.77
Variance	1087.88	575.87	1279.09
$\bar{x}_i$	1277.61	1152.02	1269.62

Source of Variation	Degrees of Freedom	Sum of Squares	Mean Square
Treatment	2	108773.5	54386.74
Error	30	29428.37	980.9457
Total (corrected)	32	138201.9	

$$t_{30, 0.025} = 2.042$$

$$LSD = 2.042 \sqrt{\frac{2(980.9457)}{11}}$$

$$LSD = 27.3$$

LSD Value: 27.3

	Alt 2	Alt 3
Alt 1	$11277.61 - 1152.021$ $125.59 > 27.3$ Significant	$11277.61 - 1269.621$ $7.99 < 27.3$ Not Significant
Alt 3	$11269.62 - 1152.021$ $117.6 > 27.3$ Significant	

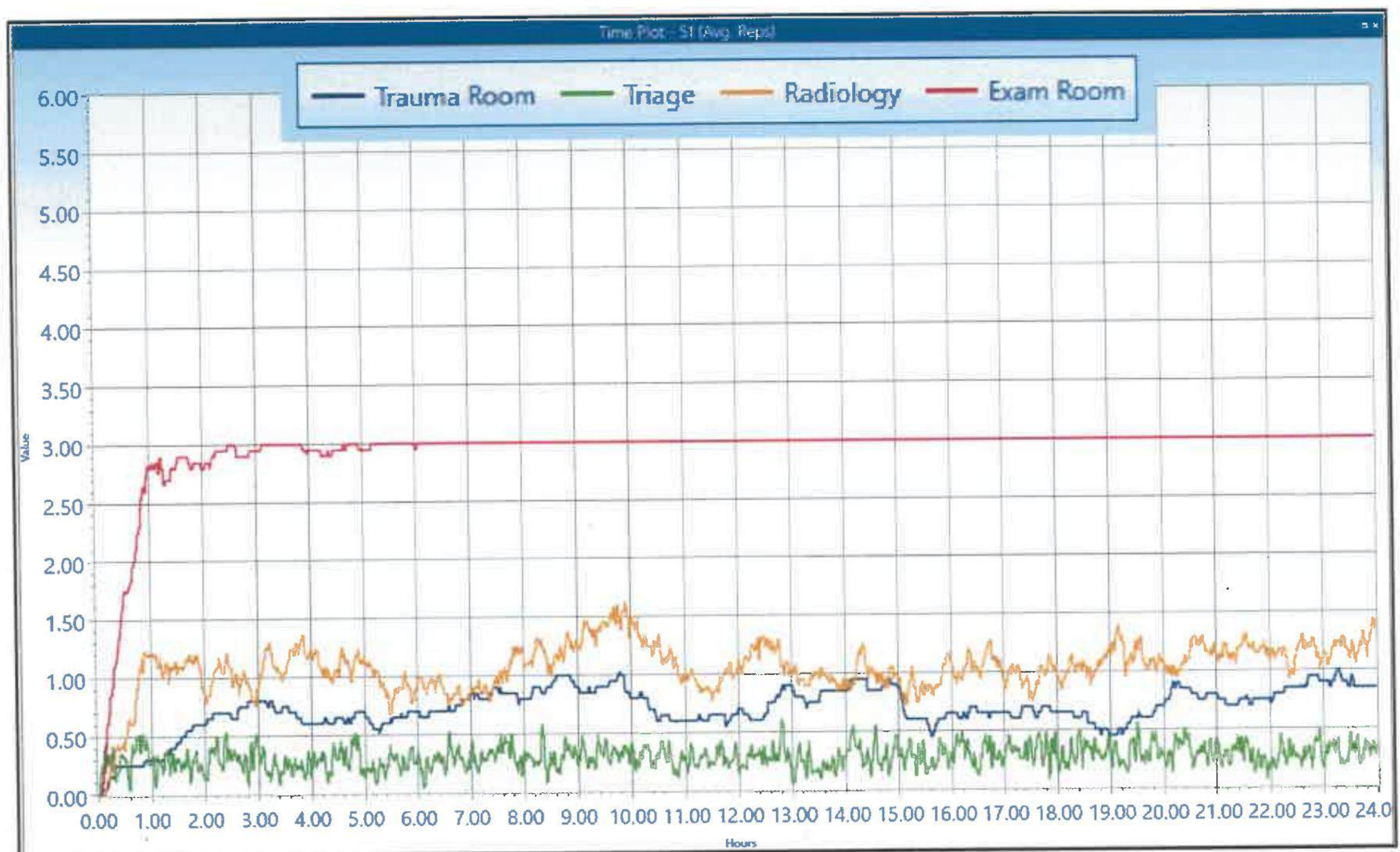
Show all work to receive full credit.

Section E Output Analysis Concepts
3 Questions (13 points)

You are modeling a Hospital Emergency Room. Answer the following questions. In the model, each location has its own respective queue. The location capacities are:

Service Location	Capacity
Trauma Room	3
Triage	1
Radiology	2
Exam Rooms	3

20. (5 points) The time plot below shows the average number of patients in each service location over 10 replications for 24 hours. Which service location(s) may have an unstable/runaway queue. (Circle all that apply).



- a. Trauma Room
- b. Triage
- c. Radiology
- ☒ d. Exam Rooms
- e. None of the locations

EM481 AT26-1
WPR #2

Last Name (Printed): Whorchuk

Show all work to receive full credit.

You update your model with new information and run it for 24 hours and 20 replications. The following chart and table show the average number of patients in the hospital ER at any point in time.



21. (4 points) If this is a non-terminating model, what needs to be done in order to calculate a more accurate average number of patients?

Create an average number of Patients variable
or make the model terminating

22. (4 points) You create two alternative scenarios for the hospital simulation. You run each scenario for 20 replications. You don't use a set seed, so each replication (within and across scenarios) is independent. Which testing method should you use to see if there is a significant difference between the two alternatives.

a) Welch CI Test

☒ b) Paired-t Test

Poor Student Work

Student	
Chloe Rambo	
Total Points	
120.5 / 165 pts	
Question 1	
M- LEAP	3 / 3 pts
✔ - 0 pts Correct	
Question 2	
M- Stream/Seed	0 / 3 pts
✔ - 3 pts Incorrect	
Question 3	
M- RNG Mod	3 / 3 pts
✔ - 0 pts Correct	
Question 4	
T/F - G.Variable	0 / 3 pts
✔ - 3 pts Incorrect	
Question 5	
T/F - Static Res.	3 / 3 pts
✔ - 0 pts Correct	
Question 6	
M - Dyn. Res.	3 / 3 pts
✔ - 0 pts Correct	
Question 7	
T/F - JOIN	3 / 3 pts
✔ - 0 pts Correct	
Question 8	
T/F - GROUP	3 / 3 pts
✔ - 0 pts Correct	
Question 9	
M - Non-Terminating	3 / 3 pts
✔ - 0 pts Correct	
Question 10	
M - Verification	0 / 3 pts
✔ - 3 pts Incorrect	
Question 11	
M - Looping	0 / 8 pts
✔ - 8 pts Incorrect x4	
Question 12	
C - CI, Single	10 / 12 pts
✔ - 2 pts df calculated incorrectly	
Question 13	
C - Rel. Error	11 / 11 pts
✔ - 0 pts Correct	
Question 14	
S - Welch Ho & Ha	3 / 6 pts
✔ - 3 pts Did not define what is being compared.	
Question 15	
C - Welch df	11.5 / 12 pts
✔ - 0.5 pts df Reporting error. Report as an integer.	
Question 16	
C - Welch Test	16 / 22 pts
✔ - 3 pts Incorrect Determination (based on reported ANS)	
✔ - 3 pts Incorrect Alternative Choice (based on reported ANS & Determination)	
Question 17	
S - ANOVA Ho & Ha	6 / 6 pts
✔ - 0 pts Correct	

Question 18

C - ANOVA

22 / 25 pts

- ✓ - 2 pts Sum X^2 Incorrect Setup
- ✓ - 1 pt MSE Incorrect

- 145
- 21129^2+1156.8^2+1247.4^2
- 345
- 4Follows
- 5Follows
- 6Follows

Question 19

C - LSD

14 / 20 pts

- ✓ - 2 pts t_crit look-up error
- ✓ - 2 pts (ALT 1, ALT2) Determination Incorrect
- ✓ - 2 pts (ALT 1, ALT3) Determination Incorrect

Question 20

M - Unstable Queue

2 / 5 pts

- ✓ - 3 pts Select "e" -- Incorrect. This is plot of service location, Exam room has CAP = 3.

Question 21

S - Warm-up

4 / 4 pts

- ✓ - 0 pts Correct

Question 22

M - Welch vs. Paired-t

0 / 4 pts

- ✓ - 4 pts Incorrect

**DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996**

EM481 WRITTEN PARTIAL REVIEW #2
(Show all work)

EM481 AT 26-1
WPR #2 (165 points)

Name: Chloe Bambo

Hour/Section: H1

1. **Contents.** WPR #2 is an individual 75-minute exam worth 165 points.
2. **Show Your Work.** Unsupported work earns *little* or *no* partial credit. Neatness will also help your cause.
3. **References.** The authorized references are your calculator, pencil/pen, and the EM481 reference manual (provided).
4. **Grading Summary.**

Section	Possible Points	Score
A. Modeling Concepts	38	
B. Output Analysis of a Single System	23	
C. Output Analysis of Two Systems	40	
D. Output Analysis of Multiple Systems	51	
E. Output Analysis Concepts	13	
Total	165	

Show all work to receive full credit.

Section A Modeling Concepts
11 questions (38 points)

Clearly Circle the correct answer.

1. (3 points) In ProModel, where do you dictate the frequency at which entities enter the system?
 - a. Locations
 - b. Entities
 - ☒ c. Arrivals
 - d. Processing

2. (3 points) In ProModel, you've created a seed of 1 with a stream 4. Select the distribution that correctly applies the stream/seed.
 - ☒ a. Wait N(3, 2) min
 - b. Wait N(3, 2, 1, 4) min
 - c. Wait N(3, 2, 1) min
 - d. Wait N(3, 2, 4) min

3. (3 points) What is the remainder for 17 mod (5)?
 - a. 1
 - ☒ b. 2
 - ~~c. 3.4~~
 - ~~d. 12~~

Handwritten work for Question 3:
17 divided by 5 equals 3 with a remainder of 2.
17 = 5 * 3 + 2

4. (3 points) **True or False:** In ProModel, a global variable can be updated in a location's operation logic and then decremented in the move logic when the entity leaves the location.
 - a. True
 - ☒ b. False

Show all work to receive full credit.

5. (3 points) **True or False:** Static resources generally do not appear graphically/visually in ProModel?
- ☒ a. True
 - b. False
6. (3 points) What is required for a dynamic resource?
- a. Attribute linked to User Distribution
 - b. Variables
 - c. Modules
 - ☒ d. Path Network
7. (3 points) **True or False:** After using the JOIN command to join entities, the entities can be separated later in the model.
- a. True
 - ☒ b. False
8. (3 points) **True or False:** The Group command requires an "If Group" request.
- a. True
 - ☒ b. False
9. (3 points) A _____ system starts and ends at a specified time or event; or at a time point specified by management.
- a. Continuous
 - b. Non-Continuous
 - ☒ c. Terminating
 - d. Non-Terminating

Show all work to receive full credit.

10. (3 points) Which of the following is the least useful when verifying a simulation?

- a. Output Viewer Results
- b. Trace Window
- c. Macro Scenario Parameters
- ☒ d. Display Command

11. (8 points) Review the logic below. Select all that HAVE an error or create an infinite loop. (Circle all that apply).

a)

```
Int lv_syst_check = 5
Do
{
    Wait 10 min
    Dec a_system_check
}
Until lv_syst_check = 0
```

☒ b)

```
a_rebake = 5
Do
{
    Wait 10 min
    Dec a_rebake
}
While a_rebake > 0
```

☒ c)

```
a_class = 1
a_toppings = 3
If a_class = 1 Then
{
    Do
    {
        Wait 10 min
        Dec a_toppings
    }
    Until a_toppings < 0
}
```

d)

```
a_class = 1
a_CT_A = 2
Int lv_CT_Y = 5

If a_class = 2 Or a_CT_A > 1 Then
{
    Do
    {
        Wait 10 min
        Dec lv_CT_Y, 2
    }
    Until lv_CT_Y = 0
}
```


Show all work to receive full credit.

Section B Output Analysis of a Single System
2 Questions (23 points)

You are simulating a T-Shirt Production Factory. You build a DES model for the factory and would like to determine a confidence interval for the average number of T-shirts produced daily. You run your simulation for twenty (20) replications and calculate the following point estimates:

Descriptive Statistics	
1005.7	Mean
125	STD DEV (Sample)
6750.326	Variance (Sample)
20	n (# of Repetitions)

12. (12 points) Compute a 95% confidence interval for the true value of the average number of T-shirts produced.

$\alpha = .05$
 $n = 20$
 $df = k - 1$

$$\bar{X} \pm t_{n-1, \frac{\alpha}{2}} \left(\frac{s}{\sqrt{n}} \right)$$

$$1005.7 \pm 2.086 \cdot \left(\frac{125}{\sqrt{20}} \right)$$

Confidence Interval: []
(Upper & Lower confidence limit format)

13. (11 points) If you wanted to achieve a relative error of five percent (5%), how many replications would you have to run?

$n' = \left(\frac{Z_{\alpha/2} s}{re \bar{x}} \right)^2$

$1.96 = \text{crit}$
 $\left(\frac{1.96 \cdot 125}{\frac{.05}{1.05} \cdot 1005.7} \right)^2$
 $\downarrow$
245

of Required Replications: 26.17

Show all work to receive full credit.

Section C Output Analysis of Two Systems
3 Questions (40 points)

You update your model and develop an alternative configuration for your T-Shift Factory. You run ten (10) replications of both the original configuration (Alternative 1) and your new alternative (Alternative 2). Using the data below, you will now determine if there is any difference in the average number of T-Shirts produced daily. A higher quantity is better.

14. (6 points) State your null and alternative hypotheses.

H₀: There is no difference between the 2 systems

H_a: There is a difference between the 2 systems

15. (12 points) Given the information below, calculate the degrees of freedom needed if you were to conduct a Welch CI Test.

Rep.	ALT 1	ALT 2
1	957.00	961.00
2	939.00	942.00
3	952.00	960.00
4	960.00	962.00
5	951.00	951.00
6	936.00	937.00
7	961.00	961.00
8	949.00	952.00
9	956.00	957.00
10	954.00	963.00

ALT 1	ALT 2	
951.5	954.6	Mean $\bar{x}$
8.317	9.009	STD DEV (Sample) s
69.167	81.156	Variance (Sample) s^2
10	10	n (# of Repetitions)

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{\left(\frac{s_1^2}{n_1} \right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2} \right)^2}{n_2 - 1}} \rightarrow \frac{\left(\frac{69.167}{10} + \frac{81.156}{10} \right)^2}{\frac{\left(\frac{69.167}{10} \right)^2}{9} + \frac{\left(\frac{81.156}{10} \right)^2}{9}}$$

Degrees of Freedom: 17.886

$$\begin{array}{r} 225.97 \\ 5.3156 + 7.318 \\ \hline 225.97 \\ \hline 12.63 \end{array}$$

Show all work to receive full credit.

16. (22 points) You receive new information so you update your model and rerun both alternatives for 15 replications each. You now have the following information. Complete your Welch CI Test and provide your determination by circling the appropriate answers below.

ALT 1	ALT 2	
794.8	798.07	Mean
7.37	9.208	STD DEV (Sample)
54.314	84.781	Variance (Sample)
15	15	n (# of Repetitions)

0.05	Significance Level
27	Degrees of Freedom

$$\bar{x}_1 - \bar{x}_2 \pm t_{df, \alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$> 2.052$$

$$\begin{aligned} & \bar{x}_1 \quad \bar{x}_2 \\ & (794.8 - 798.07) \pm (2.052) \cdot \sqrt{\frac{54.314}{15} + \frac{84.781}{15}} \\ & -3.27 \pm (2.052 \cdot 3.045) \\ & -9.518 \quad 2.978 \\ & [-9.518 \quad 2.978] \end{aligned}$$

Confidence Interval: -9.518 2.978

Determination:
(Circle One)

(Fail to Reject H_0)

(Reject the H_0)

Better Alternative:
(Circle One)

(ALT 1)

(ALT 2)

(Neither / Not Applicable)

Show all work to receive full credit.

Section D Output Analysis of Multiple Systems

3 Questions (51 points)

You developed three (3) alternatives for a widget assembly plant and ran each alternative for sixteen (16) replications each, recording the total widgets produced for each replication. You will now conduct an ANOVA to determine if there is any significant difference between the alternatives.

17. (6 points) State your null and alternative hypotheses. Use an $\alpha = 0.05$.

H_0 : There is no significant dif. between alts at $\alpha = 0.05$

H_a : there is significant dif. between alts. at $\alpha = 0.05$

18. (25 points) Use the information below to complete your ANOVA test. Fill out the associated table on the right and state your test determination.

	Alt 1	Alt 2	Alt 3	Source of Variation	Degrees of Freedom	Sum of Squares	Mean Square	Fcalc	Fcritical
$\bar{x}_i$	1129	1156.8	1247.4						
Variance	1.92	3.74	4.18						
$\bar{x}_i$	70.56	72.3	77.96						
$x_{..} = \sum_{i=1}^K x_i$	3533.20			Treatment	2	944.12	472.06	41.594	3.32
$\sum_{j=1}^n x_{ij}^2$	79693.88	83692.70	97313.08	Error	30	147.54	11.349		
SS_i	28.82	56.06	62.66	Total (corrected)					

$K=3$
 $n=16$

$SSE = 147.54$

$SST = \frac{1}{16} \left[\left(\sum_{i=1}^K 70.56^2 + 72.3^2 + 77.96^2 \right) - \frac{3533.2^2}{3} \right] = 944.1269$

$SSTC = 944.13 + 147.54 = 1091.66$

$MST = \frac{944.169}{3-1} = 472.06$

$MSE = \frac{147.54}{16-3} = 11.34923$

$F_{calc} = MST/MSE$

ANOVA Determination: Reject H_0 , significant because

F_{Calc} is greater than F_{Crit}

Show all work to receive full credit.

19. (20 points) You complete an ANOVA test on a different set of alternatives for a different factory, one that produces bottles. You run each of 3 alternatives for 11 replications. The owner wants to minimize daily costs. Use the information below to complete a Fisher Least Significance Difference (LSD) Test on two of the possible alternative pairings. Use a $\alpha = 0.05$. Write your conclusions in the table below.

Factor	Daily Cost (in Dollars)		
	Alternatives		
Replications	Alt 1	Alt 2	Alt 3
$\bar{x}_i$	14053.73	12672.23	13965.77
Variance	1087.88	575.87	1279.09
$\bar{x}_i$	1277.61	1152.02	1269.62

Source of Variation	Degrees of Freedom	Sum of Squares	Mean Square
Treatment	2	108773.5	54386.74
Error	30	29428.37	980.9457
Total (corrected)	32	138201.9	

$$LSD(\alpha) = t_{df(\alpha), \alpha/2} \sqrt{\frac{2MSE}{n}}$$

3.32

13.3549

3.32 • $\sqrt{\frac{2(980.9457)}{11}}$

LSD



44.338

A1 - A2 (125.59)

A1 - A3 (7.99)

A2 - A3 (-117.6)

LSD Value: 44.338

	Alt 2	Alt 3
Alt 1	125.59	7.99
Alt 3	-117.6	

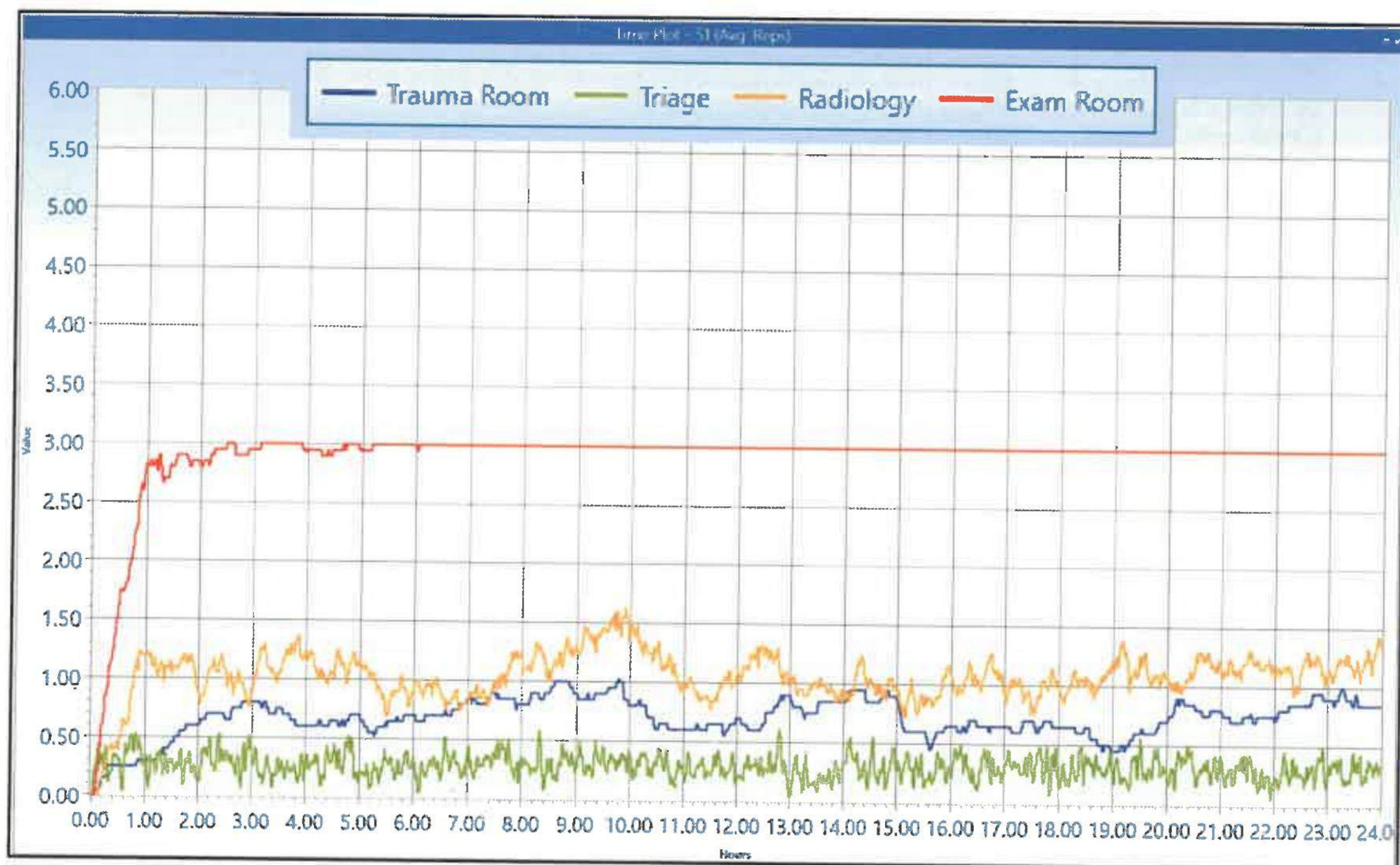
Show all work to receive full credit.

Section E Output Analysis Concepts
3 Questions (13 points)

You are modeling a Hospital Emergency Room. Answer the following questions. In the model, each location has its own respective queue. The location capacities are:

Service Location	Capacity
Trauma Room	3
Triage	1
Radiology	2
Exam Rooms	3

20. (5 points) The time plot below shows the average number of patients in each service location over 10 replications for 24 hours. Which service location(s) may have an unstable/runaway queue. (Circle all that apply).



- ☒ a. Trauma Room
- ☒ b. Triage
- ☐ c. Radiology
- ☐ d. Exam Rooms
- ☒ e. None of the locations

EM481 AT26-1
WPR #2

Last Name (Printed): Rambo

Show all work to receive full credit.

You update your model with new information and run it for 24 hours and 20 replications. The following chart and table show the average number of patients in the hospital ER at any point in time.



21. (4 points) If this is a non-terminating model, what needs to be done in order to calculate a more accurate average number of patients?

Calculate warm up period. This needs to be done to identify how long/how many patients enter at the start. This warm-up period will need to be excluded to find the accurate average. A moving test can help us find this.

22. (4 points) You create two alternative scenarios for the hospital simulation. You run each scenario for 20 replications. You don't use a set seed, so each replication (within and across scenarios) is independent. Which testing method should you use to see if there is a significant difference between the two alternatives.

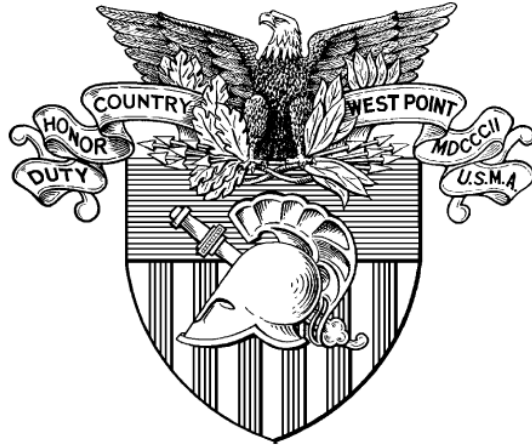
a) Welch CI Test

b) Paired-t Test



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB I FINAL COURSE GRADES



Department of Systems Engineering

***Final Grade Approval
AT26-1***



EM481 Systems Simulation

Department Head:

COL Coxen

Program Director:

COL Thompson

Course Director:

LTC Lainis

For Faculty Viewing Only

Secure When Not in Use

Caution: Not for Cadet Viewing!



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

MADN-SE

15 December 2025

MEMORANDUM THRU COL Brendan Thompson, Engineering Management Program Director
FOR COL Julia Coxen, Professor and Head, Department of Systems Engineering
SUBJECT: EM481 Final Grade Approval.

1. **RECOMMENDATION.** Recommend approval final course grades. This report includes the course order of merit list with recommended cut lines (Enclosure A).

2. **Course Information.**

- a. **Course Enrollment:** 75
- b. **Total Sections:** 4
- c. **Final Course Average:** 87.73%
- d. **Historical Course Averages:**

Academic Term	EM481 Course Average
19-1	89.61%
19-2	88.07%
20-1	86.40%
20-2	86.42%
21-1	89.41%
21-2	89.98%
22-1	87.05%
22-2	87.38%
23-1	89.12%
23-2	88.23%
24-1	86.83%
24-2	87.36%
25-1	85.96%
25-2	84.70%
26-1	87.73%

Table 1: Historical Course Averages

3. Final Course Grades.

a. The grade distribution table and histogram for the term are depicted below. Enclosure A contains the grade distribution and OML for the course, with two recommended changes to the grading scale cut lines.

Recommended Scale				Standard Scale			
Cutoff		Overall	% Overall	Overall	% Overall		Cutoff
94.50%	A+	2	2.7%	1	1.3%	A+	97.50%
91.80%	A	25	33.3%	24	32.0%	A	92.50%
88.90%	A-	9	12.0%	11	14.7%	A-	90.00%
87.00%	B+	8	10.7%	8	10.7%	B+	87.50%
81.70%	B	14	18.7%	14	18.7%	B	82.50%
79.80%	B-	5	6.7%	5	6.7%	B-	80.00%
77.50%	C+	4	5.3%	4	5.3%	C+	77.50%
72.50%	C	6	8.0%	6	8.0%	C	72.50%
70.00%	C-	2	2.7%	2	2.7%	C-	70.00%
65.00%	D	0	0.0%	0	0.0%	D	65.00%
0.00%	F	0	0.0%	0	0.0%	F	0.00%
		75	100.0%	75	100.0%		

Table 2: Recommended and Standard Grading Scale Distributions

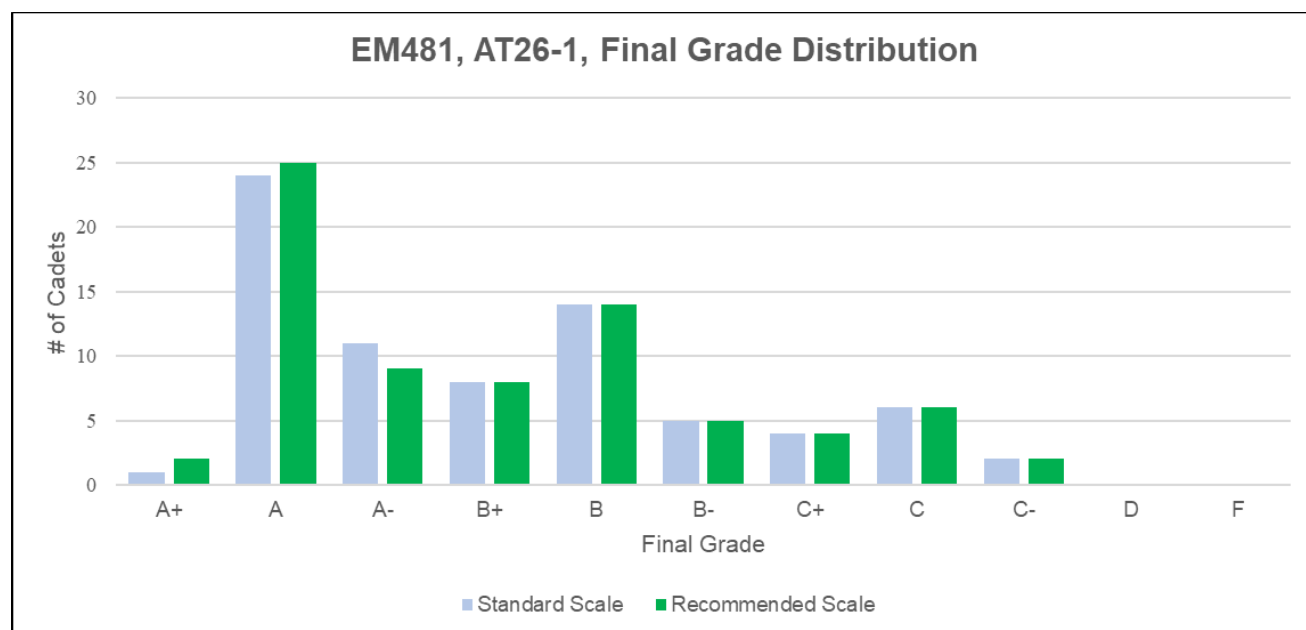


Figure 1: Histogram, Recommend vs. Standard Grade Distribution

4. **Failures.** There are no course failures at the completion of this course.

5. Performance by Major.

- a. There were 5 academic majors enrolled in EM481 this semester.

Major Code	Major	# Cadets
ORE2	Operations Research	33
SEN1	Systems Engineering	12
ENM1	Engineering Management	20
SDS0	Systems and Decision Sciences	9
EPS1	Engineering Psychology	1

Table 3: Number of Cadets, by Major, enrolled in EM481

- b. Similar to past semesters, ORE2 Cadets performed best, followed by SEN1 Cadets, ENM1 Cadets, and finally SDS0 Cadets.

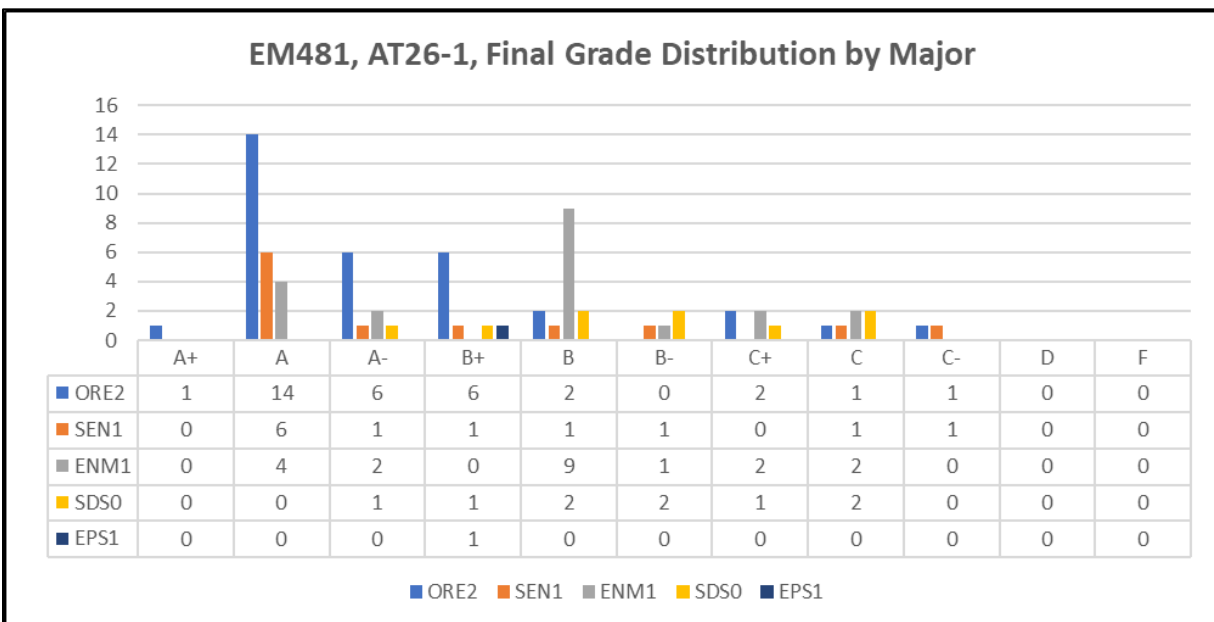


Figure 2: Grade Distribution by Cadet Major

	ORE2	SEN1	ENM1	SDS0	EPS1
AVG	90.3%	88.2%	85.8%	82.0%	88.4%
Max	98.0%	95.6%	94.7%	91.2%	88.4%
Min	71.2%	70.2%	75.0%	72.6%	88.4%

Table 4: Grade Statistics by Cadet Major

MADN-SE
SUBJECT: EM481 Final Grade Approval for AT26-1

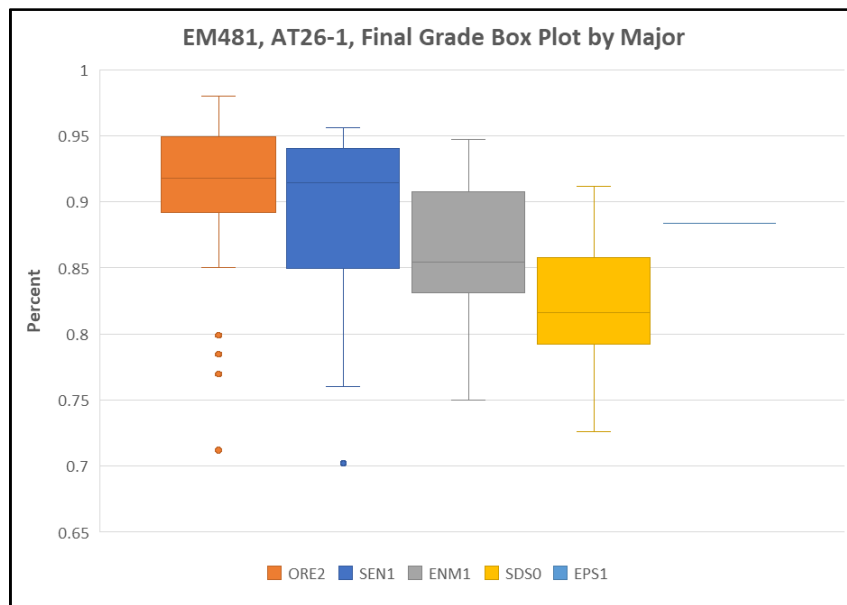


Figure 3: Final Grade Box Plot by Cadet Major

6. The point of contact for this memorandum is the undersigned at thomas.lainis@westpoint.edu.

Encls
Enclosure 1. Course OML with Cut Lines

TOM LAINIS
LTC, SF
SE385 Course Director

MADN-SE
SUBJECT: EM481 Final Grade Approval for AT26-1

Enclosure A. Course OML with Cut Lines

OML	Name	YG	Major	Section	Grade	Percent	Points Earned	Points Total	PS#1	Quiz #1	Quiz #2	Quiz #3	WPR #1	WPR #2	Graded	Proj. 1A	Proj. 1B	Proj. 1C	Proj. 2A	Proj. 2B	Proj. 2C
1	Ellis, Ethan	2026	ORE2	G2.1	A+	98.00%	980.0	1000	100.0%	100.0%	100.0%	100.0%	98.8%	98.2%	97.3%	98.8%	95.6%	100.0%	96.3%	98.8%	80
2	Clausing, John	2027	ORE2	H1.2	A+	96.30%	963.0	1000	97.5%	98.3%	90.0%	100.0%	95.0%	99.4%	97.3%	95.0%	98.8%	96.3%	92.5%	98.8%	80
3	Lynd, Benjamin	2027	SEN1	J1.2	A	95.60%	956.0	1000	100.0%	100.0%	90.0%	92.0%	91.9%	97.6%	91.3%	100.0%	100.0%	98.8%	91.3%	93.8%	91.3%
4	Yun, Grace	2027	ORE2	J1.2	A	95.40%	954.0	1000	100.0%	100.0%	90.0%	100.0%	96.3%	89.7%	94.7%	96.3%	97.5%	98.8%	95.0%	96.3%	96.3%
5	Meyer, Kathryn	2027	ORE2	G2.1	A	95.35%	953.5	1000	96.3%	100.0%	93.3%	92.0%	85.6%	97.6%	98.0%	101.3%	97.5%	102.5%	92.5%	93.8%	98.8%
6	West, Turner	2026	SEN1	J1.2	A	95.35%	953.5	1000	98.8%	98.3%	100.0%	94.0%	89.1%	97.0%	98.7%	85.0%	97.5%	93.8%	91.3%	93.8%	100.0%
7	Durand, Connor	2026	ORE2	J1.2	A	95.25%	952.5	1000	95.6%	100.0%	90.0%	92.0%	93.8%	93.6%	93.3%	101.3%	98.8%	100.0%	95.0%	96.3%	96.3%
8	Bender, Allison	2026	ORE2	J1.2	A	95.15%	951.5	1000	98.8%	100.0%	100.0%	80.0%	93.1%	94.5%	91.3%	100.0%	98.1%	100.0%	95.0%	96.3%	96.3%
9	Lu, Jerry	2027	ORE2	G2.1	A	95.05%	950.5	1000	98.1%	100.0%	90.0%	92.0%	95.0%	97.0%	90.0%	102.5%	91.3%	95.0%	96.3%	98.8%	94.4%
10	Gohlich, Tyler	2026	ORE2	J1.2	A	95.05%	950.5	1000	95.6%	95.0%	90.0%	100.0%	91.9%	96.4%	92.7%	92.5%	98.8%	88.8%	97.5%	97.5%	100.0%
11	Greenwood, Nolan	2028	ORE2	H1.2	A	94.85%	948.5	1000	96.9%	93.3%	90.0%	100.0%	93.4%	98.2%	92.7%	88.8%	98.1%	101.3%	92.5%	95.0%	91.3%
12	Schodrof, Cooper	2026	ENM1	H1.2	A	94.65%	946.5	1000	100.0%	100.0%	80.0%	100.0%	91.6%	92.7%	92.0%	101.3%	100.0%	98.8%	98.8%	98.8%	88.8%
13	Drabenstott, Jack	2026	SEN1	G2.1	A	94.20%	942.0	1000	97.5%	91.7%	100.0%	76.0%	92.8%	95.8%	94.0%	85.0%	95.6%	100.0%	92.5%	93.1%	97.5%
14	Reynolds, Zachary	2026	SEN1	I1.2	A	93.95%	939.5	1000	100.0%	100.0%	100.0%	70.0%	93.8%	92.1%	94.0%	92.5%	91.3%	100.0%	92.5%	92.5%	97.5%
15	Flamhaft, Magnolia	2026	ORE2	J1.2	A	93.95%	939.5	1000	98.8%	95.0%	80.0%	80.0%	89.7%	92.1%	93.3%	91.3%	98.8%	100.0%	97.5%	97.5%	100.0%
16	Karpinski, Cole	2026	ENM1	I1.2	A	93.80%	938.0	1000	98.1%	98.3%	100.0%	100.0%	92.8%	98.2%	85.3%	93.8%	91.3%	92.5%	92.5%	92.5%	97.5%
17	McCray, Jackson	2027	ORE2	I1.2	A	93.35%	933.5	1000	99.4%	100.0%	100.0%	100.0%	90.6%	94.5%	95.3%	93.8%	96.9%	85.0%	91.3%	91.3%	82.5%
18	In, Jonathan	2027	ORE2	J1.2	A	93.25%	932.5	1000	97.5%	100.0%	90.0%	78.0%	93.8%	93.3%	84.7%	101.3%	89.4%	95.0%	97.5%	97.5%	100.0%
19	Vollino, Roberto	2027	ORE2	G2.1	A	93.20%	932.0	1000	98.8%	100.0%	90.0%	88.0%	93.4%	89.1%	94.0%	90.0%	90.0%	88.8%	96.3%	98.8%	94.4%
20	Zhang, Alex	2026	ENM1	H1.2	A	93.15%	931.5	1000	95.0%	93.3%	90.0%	88.0%	93.8%	89.7%	90.3%	101.3%	97.5%	92.5%	98.8%	98.8%	88.8%
21	Kellogg, Carson	2026	SEN1	J1.2	A	93.15%	931.5	1000	96.9%	100.0%	100.0%	78.0%	89.7%	95.2%	94.7%	70.0%	95.0%	91.3%	91.3%	93.8%	100.0%
22	Cho, Ethan	2027	ORE2	G2.1	A	92.60%	926.0	1000	97.5%	100.0%	93.3%	88.0%	93.4%	91.2%	88.0%	95.0%	86.9%	93.8%	92.5%	93.8%	98.8%
23	Rybalowski, Mark	2026	SEN1	J1.2	A	92.60%	926.0	1000	96.9%	93.3%	90.0%	88.0%	87.5%	97.6%	88.7%	95.0%	93.1%	102.5%	90.0%	98.8%	96.3%
24	Miller, Andrew	2027	ENM1	G2.1	A	92.55%	925.5	1000	96.9%	100.0%	80.0%	92.0%	95.3%	89.1%	88.7%	82.5%	98.1%	93.8%	92.5%	93.1%	97.5%
25	Park, Yenna	2027	ORE2	G2.1	A	92.55%	925.5	1000	100.0%	80.0%	90.0%	88.0%	87.5%	95.2%	90.0%	92.5%	94.4%	92.5%	92.5%	93.8%	98.8%
26	Monagle, Ryan	2027	ORE2	G2.1	A	92.05%	920.5	1000	93.8%	80.0%	100.0%	100.0%	85.3%	97.6%	94.0%	100.0%	96.3%	91.3%	88.8%	85.0%	88.8%
27	Dorman, Lance	2026	ENM1	H1.2	A	92.05%	920.5	1000	98.1%	100.0%	90.0%	88.0%	91.9%	86.1%	92.0%	100.0%	90.0%	91.3%	98.8%	98.8%	90.0%
28	Lee, Grant	2027	ORE2	H1.2	A-	91.75%	917.5	1000	95.6%	100.0%	90.0%	80.0%	89.1%	87.3%	92.3%	95.0%	95.0%	92.5%	92.5%	95.0%	91.3%
29	Yhu, Logan	2026	ORE2	J1.2	A-	91.75%	917.5	1000	97.5%	100.0%	100.0%	86.0%	85.3%	93.3%	90.7%	81.3%	88.8%	95.0%	95.0%	92.5%	97.5%
30	Coffman, Carter	2026	ORE2	H1.2	A-	91.25%	912.5	1000	98.8%	100.0%	90.0%	100.0%	92.8%	96.4%	86.7%	60.0%	84.4%	98.8%	85.0%	93.8%	92.5%
31	Collins, Connor	2027	ORE2	J1.2	A-	91.20%	912.0	1000	92.5%	100.0%	90.0%	100.0%	89.4%	93.3%	86.7%	67.5%	93.8%	92.5%	95.0%	92.5%	97.5%
32	Brady, Ned	2026	SDSO	I1.2	A-	91.15%	911.5	1000	98.8%	100.0%	70.0%	88.0%	91.3%	86.7%	88.7%	82.5%	96.3%	96.3%	92.5%	92.5%	97.5%
33	Powers, Steven	2027	ORE2	J1.2	A-	90.80%	908.0	1000	91.3%	85.0%	80.0%	72.0%	86.9%	89.1%	94.7%	96.3%	96.3%	85.0%	95.0%	92.5%	97.5%
34	Uhorchuk, Caleb	2027	ENM1	I1.2	A-	90.35%	903.5	1000	91.9%	90.0%	90.0%	92.0%	90.3%	88.5%	91.3%	87.5%	95.0%	100.0%	86.3%	90.0%	85.0%
35	Velarde, Dan	2026	SEN1	H1.2	A-	90.25%	902.5	1000	97.5%	73.3%	100.0%	92.0%	92.8%	92.1%	88.0%	96.3%	96.3%	96.3%	90.0%	76.3%	82.5%
36	Adams, Aidan	2027	ORE2	I1.2	A-	90.00%	900.0	1000	90.6%	98.3%	90.0%	82.0%	85.9%	92.1%	94.7%	90.0%	88.8%	95.0%	85.0%	91.3%	83.8%
37	Myrick, Juliette	2027	ORE2	I1.2	B+	89.75%	897.5	1000	94.4%	100.0%	70.0%	80.0%	90.3%	89.7%	90.7%	86.3%	95.0%	93.8%	86.3%	90.0%	85.0%
38	Weeks, Quinten	2027	ORE2	I1.2	B+	89.45%	894.5	1000	93.1%	100.0%	90.0%	100.0%	83.8%	92.7%	87.3%	92.5%	91.3%	92.5%	85.0%	91.3%	82.5%
39	Perez, Gabriel	2027	ORE2	G2.1	B+	89.15%	891.5	1000	93.1%	78.3%	90.0%	100.0%	79.1%	92.7%	96.7%	82.5%	93.1%	87.5%	88.8%	85.0%	88.8%
40	Williamson, Ethan	2026	SDSO	G2.1	B+	88.70%	887.0	1000	96.3%	81.7%	80.0%	66.0%	94.4%	85.5%	88.7%	91.3%	75.6%	83.8%	92.5%	93.1%	97.5%
41	Bardak, Andrew	2026	SEN1	I1.2	B+	88.40%	884.0	1000	90.6%	96.7%	80.0%	86.0%	94.1%	92.1%	76.7%	85.0%	90.0%	97.5%	86.3%	90.0%	85.0%
42	Vorthmann, Benjamin	2026	EP51	J1.2	B+	88.35%	883.5	1000	94.4%	95.0%	93.3%	92.0%	82.2%	84.5%	88.7%	90.0%	92.5%	76.3%	90.0%	88.8%	96.3%
43	Orr, Marianne	2027	ORE2	H1.2	B+	87.95%	879.5	1000	92.5%	81.7%	93.3%	68.0%	83.8%	84.2%	88.0%	87.5%	97.5%	87.5%	85.0%	93.8%	92.5%
44	Harbers, Joshua	2027	ORE2	H1.2	B+	87.75%	877.5	1000	90.6%	85.0%	80.0%	66.0%	90.3%	91.5%	87.3%	75.0%	76.3%	96.3%	85.0%	93.8%	92.5%
45	Oakley, Thomas	2027	ENM1	G2.1	B	87.10%	871.0	1000	96.3%	100.0%	90.0%	86.0%	83.1%	96.1%	82.0%	60.0%	84.4%	88.8%	80.0%	91.3%	86.3%
46	Beck, Carter	2026	ENM1	I1.2	B	86.95%	869.5	1000	87.5%	88.3%	90.0%	98.0%	89.1%	88.5%	91.3%	60.0%	93.8%	96.3%	93.8%	76.3%	75.0%
47	Mellon, Connor	2027	ORE2	G2.1	B	86.40%	864.0	1000	93.1%	80.0%	80.0%	98.0%	92.5%	81.8%	78.0%	83.8%	88.8%	95.0%	88.8%	85.0%	88.8%
48	Foreman, Chadon	2026	ENM1	I1.2	B	85.85%	858.5	1000	90.0%	100.0%	80.0%	88.0%	84.4%	92.1%	84.7%	50.0%	92.5%	91.3%	92.5%	70.0%	91.3%
49	Anderson, Tyler	2026	SEN1	H1.2	B	85.80%	858.0	1000	95.0%	83.3%	70.0%	70.0%	81.9%	95.8%	79.0%	92.5%	91.3%	100.0%	91.3%	81.3%	73.8%
50	Smith, Trevor	2026	SDSO	H1.2	B	85.75%	857.5	1000	95.0%	93.3%	90.0%	82.0%	80.0%	83.6%	90.7%	75.0%	97.5%	88.8%	91.3%	81.3%	73.8%
51	Berginc, Ethan	2026	ENM1	G2.1	B	85.50%	855.0	1000	93.1%	86.7%	93.3%	72.0%	84.1%	87.3%	78.7%	80.0%	86.3%	92.5%	80.0%	91.3%	86.3%
52	McDaniel, Trae	2026	ENM1	H1.2	B	85.35%	853.5	1000	95.6%	88.3%	80.0%	72.0%	85.9%	82.1%	80.3%	91.3%	97.5%	93.8%	90.0%	76.3%	82.5%
53	Spoerl, James	2026	ENM1	I1.2	B	85.20%	852.0	1000	87.5%	91.7%	90.0%	72.0%	84.7%	80.0%	90.0%	90.0%	86.3%	90.0%	92.5%	70.0%	91.3%

MADN-SE

SUBJECT: EM481 Final Grade Approval for AT26-1

54	Gay, Stryker	2026	ENM1	I1 2	B	85.00%	850.0	1000	90.6%	75.0%	80.0%	72.0%	89.1%	85.5%	88.3%	70.0%	96.3%	83.8%	93.8%	76.3%	75.0%
55	Giarnello, Andrew	2027	ORE2	J1 2	B	84.95%	849.5	1000	95.0%	85.0%	90.0%	92.0%	87.2%	83.6%	83.3%	78.8%	91.9%	92.5%	83.8%	65.0%	85.0%
56	Ells, Alec	2026	ENM1	I1 2	B	84.55%	845.5	1000	90.6%	91.7%	80.0%	62.0%	86.9%	80.9%	90.0%	40.0%	97.5%	96.3%	92.5%	70.0%	91.3%
57	Hummel, Edward	2026	ENM1	I1 2	B	83.60%	836.0	1000	95.6%	91.7%	83.3%	54.0%	85.0%	78.2%	84.7%	93.8%	87.5%	88.8%	93.8%	76.3%	75.0%
58	Willey, Stephen	2027	SDS0	H1 2	B	83.30%	833.0	1000	93.8%	81.7%	90.0%	58.0%	87.2%	77.0%	76.0%	75.0%	97.5%	91.3%	90.0%	76.3%	87.5%
59	Miseli, Amy	2026	SEN1	H1 2	B-	82.45%	824.5	1000	92.5%	83.3%	90.0%	72.0%	77.8%	78.8%	87.3%	87.5%	77.5%	93.8%	91.3%	81.3%	73.8%
60	Coleman, Dewayne	2026	ENM1	G2 1	B-	81.70%	817.0	1000	92.5%	91.7%	73.3%	100.0%	81.9%	70.9%	73.3%	80.0%	90.0%	81.3%	80.0%	91.3%	86.3%
61	Britton, Charles	2026	SDS0	H1 2	B-	81.55%	815.5	1000	93.8%	85.0%	80.0%	74.0%	87.2%	86.7%	79.3%	85.0%	71.3%	92.5%	85.0%	68.8%	67.5%
62	Garcia, David	2025	SDS0	I1 2	B-	81.40%	814.0	1000	97.5%	100.0%	100.0%	100.0%	76.6%	89.7%	81.7%	0.0%	97.5%	95.0%	57.5%	77.5%	71.3%
63	Adusei-Poku, Kwame	2027	ORE2	J1 2	B-	80.00%	800.0	1000	83.8%	91.7%	80.0%	90.0%	81.3%	73.9%	77.3%	85.0%	82.5%	91.3%	83.8%	65.0%	86.3%
64	Rambo, Chloe	2027	SDS0	H1 2	C+	79.15%	791.5	1000	96.3%	71.7%	80.0%	56.0%	76.9%	73.0%	82.0%	77.5%	92.5%	93.8%	85.0%	68.8%	71.3%
65	Birl, Howard	2027	ORE2	I1 2	C+	78.50%	785.0	1000	92.5%	71.7%	90.0%	78.0%	73.8%	77.0%	82.7%	70.0%	85.0%	92.5%	57.5%	77.5%	70.0%
66	Flowers, Jack	2026	ENM1	I1 2	C+	78.50%	785.0	1000	94.4%	90.0%	90.0%	68.0%	73.4%	78.2%	74.7%	62.5%	93.8%	90.0%	57.5%	77.5%	73.8%
67	Franco, Ryan	2026	ENM1	I1 2	C+	77.80%	778.0	1000	85.0%	90.0%	90.0%	86.0%	70.9%	81.8%	74.7%	57.5%	90.0%	95.0%	57.5%	77.5%	70.0%
68	Hernandez, Austin	2026	ENM1	H1 2	C	77.25%	772.5	1000	86.9%	73.3%	80.0%	92.0%	66.6%	82.4%	78.7%	85.0%	77.5%	88.8%	85.0%	68.8%	66.3%
69	Nelson, Nathaniel	2027	ORE2	G2 1	C	76.95%	769.5	1000	87.5%	76.7%	90.0%	78.0%	69.4%	78.8%	64.7%	82.5%	76.3%	91.3%	76.3%	78.8%	85.0%
70	Pack, Elijah	2026	SEN1	J1 2	C	75.95%	759.5	1000	83.8%	81.7%	90.0%	60.0%	75.0%	67.0%	71.3%	65.0%	58.8%	91.3%	90.0%	88.8%	90.0%
71	Azizi, Jacob	2026	ENM1	H1 2	C	75.00%	750.0	1000	91.3%	88.3%	83.3%	52.0%	76.6%	68.8%	77.3%	82.5%	66.3%	78.8%	85.0%	68.8%	67.5%
72	Monturio, Christian	2026	SDS0	G2 1	C	74.26%	742.6	1000	96.9%	56.7%	63.3%	50.0%	78.4%	77.6%	57.3%	77.5%	70.6%	78.8%	76.3%	78.8%	80.8%
73	Harmon, Kennison	2026	SDS0	G2 1	C	72.55%	725.5	1000	93.1%	83.3%	60.0%	28.0%	68.1%	82.1%	69.7%	60.0%	35.0%	81.3%	76.3%	78.8%	92.5%
74	Kim, Nathan	2027	ORE2	G2 1	C-	71.15%	711.5	1000	96.9%	76.7%	80.0%	34.0%	62.8%	59.1%	62.7%	75.0%	81.3%	75.0%	76.3%	78.8%	85.0%
75	Lado, Loro	2026	SEN1	J1 2	C-	70.20%	702.0	1000	83.8%	96.7%	90.0%	88.0%	78.4%	89.1%	69.3%	0.0%	11.3%	45.0%	83.8%	65.0%	85.0%



UNITED STATES MILITARY ACADEMY
WEST POINT.

SECTION III

ASSIGNMENTS



UNITED STATES MILITARY ACADEMY
WEST POINT.

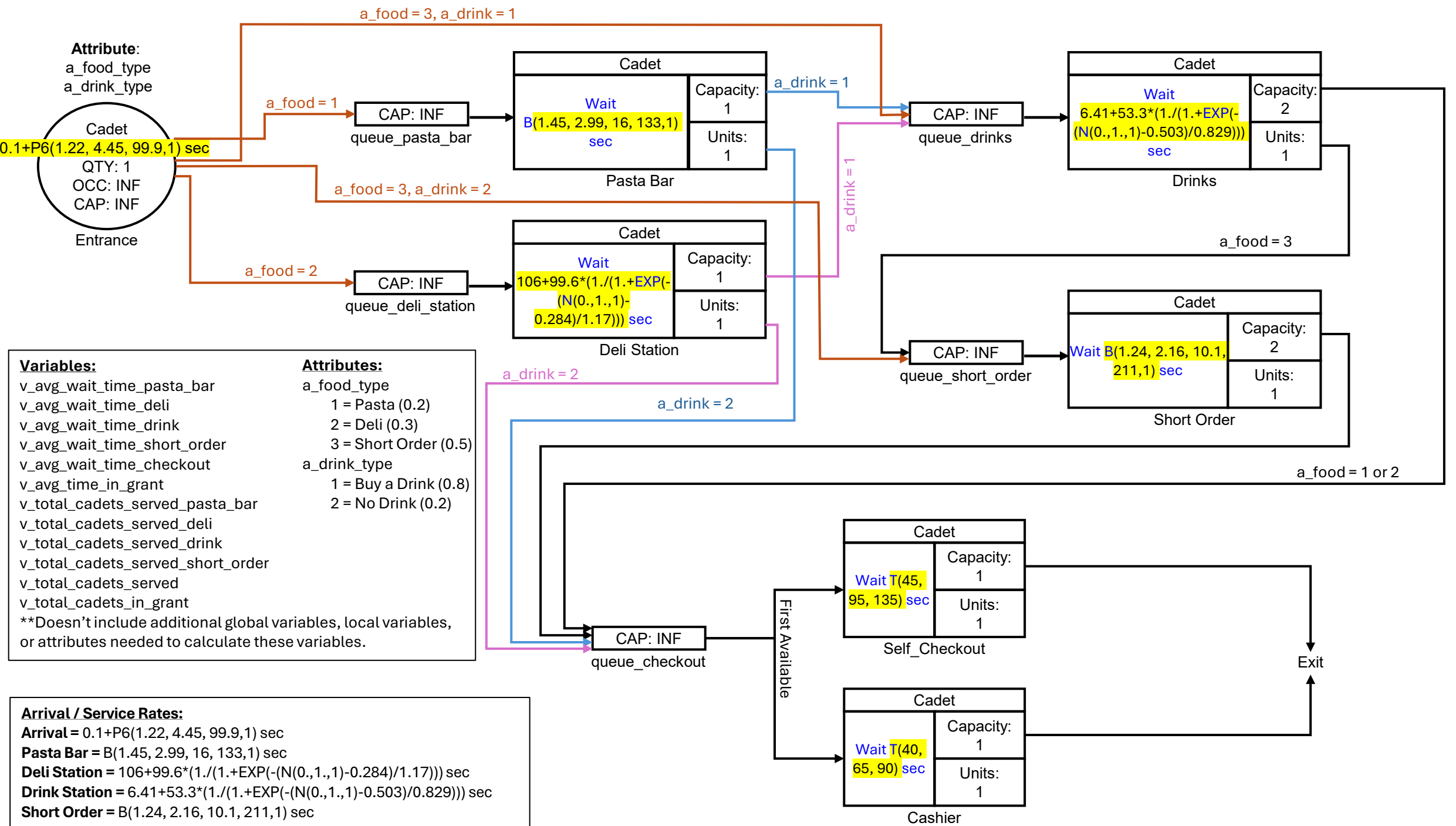
TAB J PROJECTS



UNITED STATES MILITARY ACADEMY
WEST POINT.

PROJECT #1

Instructor Solution



*
*
* Formatted Listing of Model: *
* C:\Users\thomas.lainis\OneDrive - West Point\EM481\AT26-1\3. Graded Events\Project #1\Solution\Final for Posting
to Canvas\AT26-1_EM481_Project#1_SOLUTION [Canvas].mod *
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Time Units: Minutes
Distance Units: Feet

* Locations *

Name	Cap	Units	Stats	Rules	Cost
Entrance	INF	1	Time Series Oldest, ,		
Pasta_Bar	1	1	Time Series Oldest, ,		
Deli_Station	1	1	Time Series Oldest, ,		
Drinks	2	1	Time Series Oldest, ,		
Short_Order	2	1	Time Series Oldest, ,		
Self_Checkout	1	1	Time Series Oldest, ,		
Cashier	1	1	Time Series Oldest, , First		
queue_pasta_bar	INF	1	Time Series Oldest, FIFO,		
queue_deli_station	INF	1	Time Series Oldest, FIFO,		
queue_drinks	INF	1	Time Series Oldest, FIFO,		
queue_short_order	INF	1	Time Series Oldest, FIFO,		
queue_checkout	INF	1	Time Series Oldest, FIFO,		

* Entities *

Name	Speed (fpm)	Stats	Cost
Cadet	150	Time Series	

* Processing *

Process		Routing			
Entity	Location	Operation	Blk	Output	Destination Rule Move Logic
Cadet	Entrance	a_food_type = u_food_type() // Type 1 = Pasta // Type 2 = Deli // Type 3 = Short Order			

```
a_drink_type = u_drink()
```

```
// Drink 1 = Drink
```

```
// Drink 2 = No Drink
```

```
a_time_in_grant = clock(min)
```

```
Inc v_total_cadets_in_Grant
```

```
If a_food_type = 1 Then Route 1
```

```
If a_food_type = 2 Then Route 2
```

```
If a_food_type = 3 Then
```

```
{
```

```
    If a_drink_type = 1 Then Route 3
```

```
    Else Route 4
```

```
}
```

```
1 Cadet queue_pasta_bar FIRST 1
```

```
2 Cadet queue_deli_station FIRST 1
```

```
3 Cadet queue_drinks FIRST 1
```

```
4 Cadet queue_short_order FIRST 1
```

```
Cadet queue_pasta_bar a_wait_time_pasta = clock(min)
```

```
1 Cadet Pasta_Bar FIRST 1 //Calculate AVG Wait Time in Pasta Queue
```

```
a_wait_time_pasta = clock(min) - a_wait_time_pasta
```

```
Inc v_queue_total_pasta
```

```
v_cuml_wait_time_pasta = v_cuml_wait_time_pasta +
```

```
a_wait_time_pasta
```

```
v_avg_wait_time_pasta = v_cuml_wait_time_pasta /
```

```
v_queue_total_pasta
```

```
Cadet Pasta_Bar Graphic 3
```

```
Wait B(1.45, 2.99, 16, 133,1) sec
```

```
Inc v_total_cadets_served_pasta
```

```
If a_drink_type = 1 Then Route 1 Else Route 2
```

```
// Drink 1 = Drink
```

```
// Drink 2 = No Drink
```

```
1 Cadet queue_drinks FIRST 1
```

```
2 Cadet queue_checkout FIRST 1
```

```
Cadet queue_deli_station a_wait_time_deli = clock(min)
```

```
1 Cadet Deli_Station FIRST 1 //Calculate AVG Wait Time in Deli Queue
```

```
a_wait_time_deli = clock(min) - a_wait_time_deli
```

```
Inc v_queue_total_deli
```

```
v_cuml_wait_time_deli = v_cuml_wait_time_deli +
```

```
a_wait_time_deli
```

```
v_avg_wait_time_deli = v_cuml_wait_time_deli /
```

```
v_queue_total_deli
```

```
Cadet Deli_Station Graphic 2
```

```
Wait 106+99.6*(1./(1.+EXP(-(N(0.,1.,1)-0.284)/1.17)))) sec
```

```
Inc v_total_cadets_served_deli
```

```
If a_drink_type = 1 Then Route 1 Else Route 2
```

```
// Drink 1 = Drink
```

```
// Drink 2 = No Drink
```

```
1 Cadet queue_drinks FIRST 1
```

```

                2 Cadet queue_checkout FIRST 1
Cadet queue_drinks a_wait_time_drinks = clock(min)
                1 Cadet Drinks FIRST 1 //Calculate AVG Wait Time in Drinks Queue
                a_wait_time_drinks = clock(min) - a_wait_time_drinks
                Inc v_queue_total_drinks
                v_cuml_wait_time_drinks = v_cuml_wait_time_drinks +
a_wait_time_drinks
                v_avg_wait_time_drinks = v_cuml_wait_time_drinks /
v_queue_total_drinks
Cadet Drinks Wait 6.41+53.3*(1./(1.+EXP(-(N(0,1,1)-0.503)/0.829)))) sec
Inc v_total_cadets_served_drinks

If a_food_type = 3 Then Route 1 Else Route 2
// Food Type 3 = Short Order
// Food Types 1 & 2 = Checkout
                1 Cadet queue_short_order FIRST 1
                2 Cadet queue_checkout FIRST 1
Cadet queue_short_order a_wait_time_short_order = clock(min)
                1 Cadet Short_Order FIRST 1 //Calculate AVG Wait Time in Short Order
Queue
                a_wait_time_short_order = clock(min) -
a_wait_time_short_order
                Inc v_queue_total_shortorder
                v_cuml_wait_time_shortorder = v_cuml_wait_time_shortorder
+ a_wait_time_short_order
                v_avg_wait_time_shortorder = v_cuml_wait_time_shortorder /
v_queue_total_shortorder
Cadet Short_Order Graphic 4
                Wait B(1.24, 2.16, 10.1, 211,1) sec
                Inc v_total_cadets_served_short_order

                1 Cadet queue_checkout FIRST 1
Cadet queue_checkout a_wait_time_checkout = clock(min)

                1 Cadet Self_Checkout FIRST 1 //Calculate AVG Wait Time in Checkout Queue
                a_wait_time_checkout = clock(min) - a_wait_time_checkout
                Inc v_queue_total_checkout
                v_cuml_wait_time_checkout = v_cuml_wait_time_checkout +
a_wait_time_checkout
                v_avg_wait_time_checkout = v_cuml_wait_time_checkout /
v_queue_total_checkout

                Cadet Cashier FIRST //Calculate AVG Wait Time in Checkout Queue
                a_wait_time_checkout = clock(min) - a_wait_time_checkout
                Inc v_queue_total_checkout
                v_cuml_wait_time_checkout = v_cuml_wait_time_checkout +
a_wait_time_checkout
                v_avg_wait_time_checkout = v_cuml_wait_time_checkout /
v_queue_total_checkout

Cadet Self_Checkout Wait T(45, 95, 135,1) sec
Inc v_total_cadets_served_checkout
                1 Cadet EXIT FIRST 1 //Calculate CDT Time in Grant
                a_time_in_grant = clock(min) - a_time_in_grant

```

```

//Reduce # Cadets in Grant.
Dec v_total_cadets_in_Grant

//Calculate AVG Time in Grant
Inc v_total_cadets_served_Grant
v_cuml_time_in_grant = v_cuml_time_in_grant +
a_time_in_grant
v_total_cadets_served_Grant

Cadet  Cashier      Wait T(40, 65, 90,1) sec
      Inc v_total_cadets_served_checkout
      1  Cadet  EXIT
```

FIRST 1 //Calculate CDT Time in Grant
a_time_in_grant = clock(min) - a_time_in_grant

```

//Reduce # Cadets in Grant.
Dec v_total_cadets_in_Grant

//Calculate AVG Time in Grant
Inc v_total_cadets_served_Grant
v_cuml_time_in_grant = v_cuml_time_in_grant +
a_time_in_grant
v_total_cadets_served_Grant
```

* Arrivals *

Entity	Location	Qty Each	First Time Occurrences	Frequency	Logic
Cadet	Entrance 1		INF	0.1+P6(1.22, 4.45, 99.9,1) sec	

* Attributes *

ID	Type	Classification
----	------	----------------

```
#
##// Type 1 = Pasta
##// Type 2 = Deli
##// Type 3 = Short Order
a_food_type      Integer  Entity
#
##// Drink 1 = Drink
##// Drink 2 = No Drink
a_drink_type      Integer  Entity
a_time_in_grant   Real     Entity
```

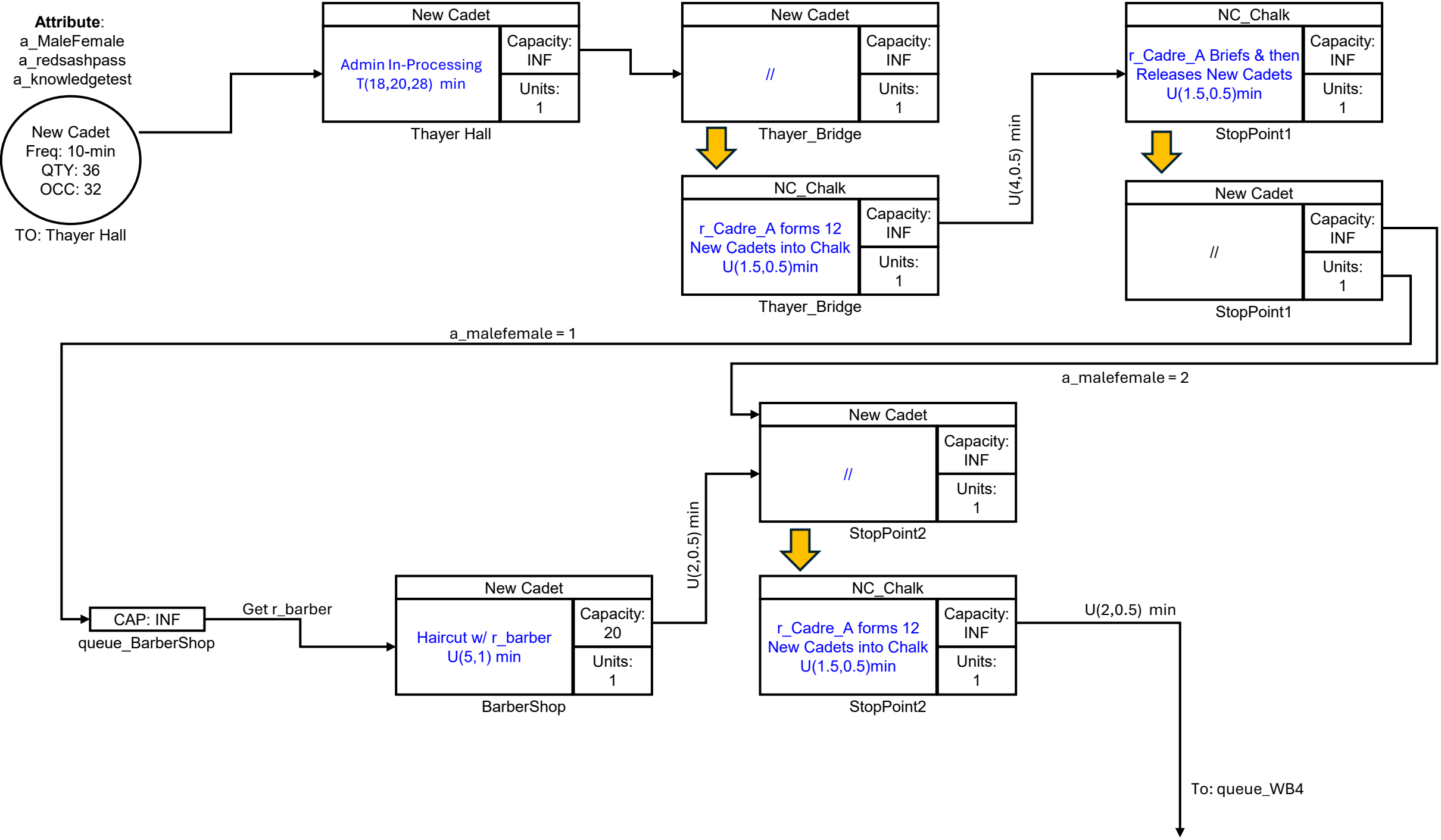
Stream #	Seed #	Reset
1	23	No

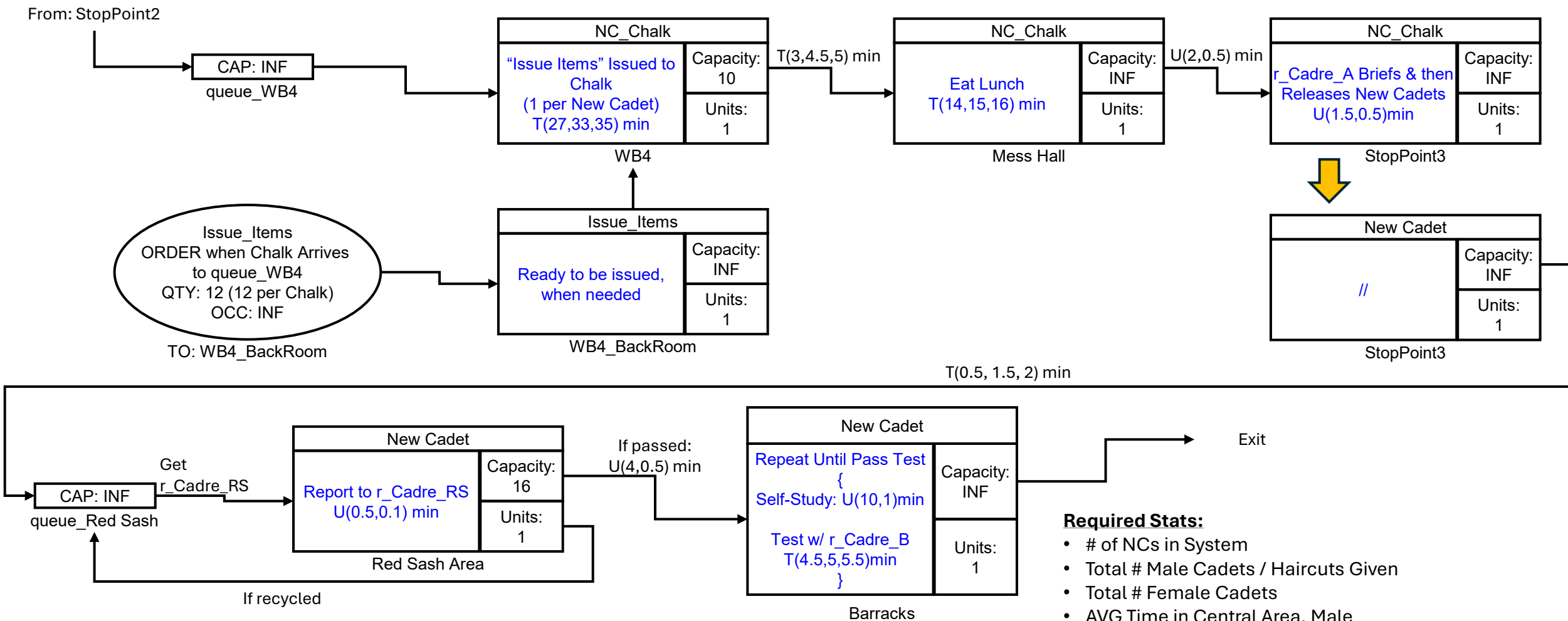


UNITED STATES MILITARY ACADEMY
WEST POINT.

PROJECT #2

Instructor Solution





Entities:

New_Cadet
NC_Chalk
Issue_Items

Attributes:

a_malefemale
a_redsashpass
a_knowledgetest

Resources:

r_Cadre_A
r_Cadre_B
r_Cadre_RS
r_Barber

System Requirements:

- Seed = 20, Reset = No
- Replications = 10
- Scenarios labeled as “COA 1”, “COA 2”, etc.

Required Stats:

- # of NCs in System
- Total # Male Cadets / Haircuts Given
- Total # Female Cadets
- AVG Time in Central Area, Male
- AVG Time in Central Area, Female
- AVG Wait Time in RedSash Queue
- AVG Wait Time at StopPoint1, StopPoint2, StopPoint3, Thayer Bridge, BarberShop Queue
- # of NCs in BarberShop Queue
- AVG Time for NC to Complete Phase 3
- Time to Complete Phase 3 (Start to Finish)
- **List doesn't include global variables, local variables, or attributes needed to calculate these required stats.

*
*
* Formatted Listing of Model: *
* C:\Users\thomas.lainis\OneDrive - West Point\EM481\AT26-1\3. Graded Events\Project #2\Solution\For Post to Canvas\EM481_AT26-1_Project#2 (Canvas 2B).mod *
*

Time Units: Minutes
Distance Units: Feet

* Locations *

Name	Cap	Units	Stats	Rules	Cost
Thayer	INF	1		Time Series Oldest, ,	
Thayer_Bridge	INF	1		Time Series Oldest, ,	
StopPoint_1	INF	1		Time Series Oldest, ,	
q_BarberShop	INF	1		Time Series Oldest, FIFO,	
BarberShop	20	1		Time Series Oldest, ,	
StopPoint_2	INF	1		Time Series Oldest, ,	
q_WB4	INF	1		Time Series Oldest, FIFO,	
WB4_BackRoom	INF	1		Time Series Oldest, ,	
WB4	10	1		Time Series Oldest, ,	
MessHall	INF	1		Time Series Oldest, ,	
StopPoint_3	INF	1		Time Series Oldest, ,	
q_Red_Sash	INF	1		Time Series Oldest, FIFO,	
Red_Sash_Area	16	1		Time Series Oldest, ,	
Barracks	INF	1		Time Series Oldest, ,	

* Entities *

Name	Speed (fpm)	Stats	Cost
New_Cadet	150		Time Series
NC_Chalk	150		Time Series
Issue_Items	150		Time Series

* Resources *

Name	Units	Res	Ent	Stats	Search	Search Path	Motion	Cost
r_cadre_A	m_cadre_A	By Unit	None	Oldest			Empty: 150 fpm	

Full: 150 fpm

r_cadre_B m_cadre_B By Unit None Oldest Empty: 150 fpm
Full: 150 fpm

r_cadre_RS m_cadre_RedSash By Unit None Oldest Empty: 150 fpm
Full: 150 fpm

r_barber m_barbers By Unit None Oldest Empty: 150 fpm
Full: 150 fpm

* Processing *

Process Routing

Entity Location Operation Blk Output Destination Rule Move Logic

New_Cadet Thayer // Assign NC attributes
a_redsashpass = u_redsash()
a_knowledge_test = u_knowledge()
a_malefemale = u_malefemale()

//Increment NCs into system & Mark NC Start Time (in hours).
Inc v_NC_in_System
a_NCRDay_time = clock(hr)
If a_malefemale = 1 Then Inc v_total_NC_male
If a_malefemale = 2 Then Inc v_total_NC_female

//NC move through Thayer Hall Stations
Wait T(18,20,28,1) min
1 New_Cadet Thayer_Bridge FIRST 1
New_Cadet Thayer_Bridge //Create chalks of 12x NCs
Group 12 As NC_Chalk
NC_Chalk Thayer_Bridge //Cadre forms & briefs chalk
Get r_cadre_A
Wait U(1.5,0.5,1) min
1 NC_Chalk StopPoint_1 FIRST 1 //Cadre moves NCs to Stop Point 1
Move For U(4,0.5,1) min

NC_Chalk StopPoint_1 //Cadre ungroups the chalk, then Freed.
Wait U(1.5,0.5,1) min
Free r_cadre_A
UnGroup

New_Cadet StopPoint_1 /* All NCs receive a Time Stamp to calculate time in Central Area.
Male NCs move to get Haircut (and then StopPoint2)
Female NCs move to StopPoint_2
*/
a_central_area_time = clock(min)

```

        If a_malefemale = 1 Then Route 1
        Else Route 2      1  New_Cadet  q_BarberShop  FIRST 1
                        2  New_Cadet  StopPoint_2  FIRST 1
New_Cadet  q_BarberShop      1  New_Cadet  BarberShop  FIRST 1 //Get Barber here, so NCs only
occupy chair with available barber.

                                Get r_barber
New_Cadet  BarberShop  //Male NC gets haircut
        Wait U(5,1,1) min
        Inc v_total_Haircuts
        Free r_barber
                        1  New_Cadet  StopPoint_2  FIRST 1 //Male NC moves to StopPoint2 to form new
chalk.

                                Move For U(2,0.5,1) min

New_Cadet  StopPoint_2  //Once 12 cadets available to form new chalk, calculate individual elapsed time in central
area

        Accum 12
        a_central_area_time = clock(min) - a_central_area_time

//Calculate average CA time by male & female.
If a_malefemale = 1 Then
{
        Inc v_total_CA_male
        v_cuml_CA_male_time = v_cuml_CA_male_time + a_central_area_time
        v_Avg_CA_male_time = v_cuml_CA_male_time / v_total_CA_male
}
If a_malefemale = 2 Then
{
        Inc v_total_CA_female
        v_cuml_CA_female_time = v_cuml_CA_female_time + a_central_area_time
        v_Avg_CA_female_time = v_cuml_CA_female_time / v_total_CA_female
}

//Form new Chalk
Group 12 As NC_Chalk

NC_Chalk  StopPoint_2  //Cadre forms & briefs chalk
        Get r_cadre_A
        Wait U(1.5,0.5,1) min
                        1  NC_Chalk  q_WB4      FIRST 1 //Cadre moves chalk to WB4 Queue
                                Move For U(2,0.5,1) min
NC_Chalk  q_WB4      //Cadre remains with Chalk
        //When Chalk enters WB4, 12x Issue Items Sets are ordered.
        Order 12 Issue_Items To WB4_BackRoom
                        1  NC_Chalk  WB4      FIRST 1
Issue_Items WB4_BackRoom      1  Issue_Items WB4      Join 1
NC_Chalk  WB4      //NCs receive their Issue Item Kits (1 per NC. 12 NCs per Chalk).
        //Cadre remains with Chalk
        Wait T(27, 33, 35,1) min
        Join 12 Issue_Items
        Inc v_total_NC_Issued_Equipment,12
                        1  NC_Chalk  MessHall  FIRST 1 //Chalk moves with Cadre to MessHall
                                Move For T(3,4.5,5,1) min
NC_Chalk  MessHall  //Chalk eats lunch

```



```

//Cadre remains with Chalk
Wait T(14,15,16,1) min
Inc v_total_Lunch_Complete,12
1 NC_Chalk StopPoint_3 FIRST 1 //Chalk moves with Cadre to StopPoint3
Move For U(2,0.5,1) min
NC_Chalk StopPoint_3 //Cadre finally released as Chalk is ungrouped
Wait U(1.5, 0.5,1) min
Free r_cadre_A
Ungroup
New_Cadet StopPoint_3 1 New_Cadet q_Red_Sash FIRST 1 //NCs enter RedSash Queue
Move For T(0.5,1.5,2,1) min
New_Cadet q_Red_Sash //Time Stamp to calculate AVG wait in RS Queue
a_RS_queue_wait_time = clock(min)
1 New_Cadet Red_Sash_Area FIRST 1 //NC moves forward only when called by
Cadre

Get r_cadre_RS

//Calculate AVG Wait Time in q_RedSash (For Visual Ticker)
a_RS_queue_wait_time = clock(min) - a_RS_queue_wait_time
Inc v_total_RS_queue_count
v_cuml_RS_queue_time = v_cuml_RS_queue_time +
a_RS_queue_wait_time
v_Avg_RS_queue_time = v_cuml_RS_queue_time /
v_total_RS_queue_count

New_Cadet Red_Sash_Area //Report to CDT in RedSash
Wait U(0.5,0.1,1) min
Free r_cadre_RS

//If Pass then move to Barracks. Recycles move to back of queue.
If a_redsashpass = 1 Then
{
Inc v_total_RedSash_Complete
Route 1
}
Else
{
Dec a_redsashpass //NC will repeat until a_redsashpass = 1
Route 2
}

1 New_Cadet Barracks FIRST 1 //NCs move into Barracks
Move For U(4,0.5,1) min
2 New_Cadet q_Red_Sash FIRST 1
New_Cadet Barracks //In Barracks, NCs must pass a knowledge test administered by Cadre.
Do
{
//Self Study
Wait U(10,1,1) min
//Knowledge Test with Cadre B
Get r_cadre_B
Wait T(4.5,5,5.5,1) min
Free r_cadre_B
Dec a_knowledge_test //NC will repeat until a_knowledge_test = 0
}

```

```

    Until a_knowledge_test = 0
        1 New_Cadet EXIT FIRST 1 //Tally NC complete
        //NC departs system
        Inc v_total_NC_Complete_with_RDay
        Dec v_NCs_in_System

        //Calculate AVG Time for NC (in hours) to complete all stations.
        a_NCRDay_time = clock(hr) - a_NCRDay_time
        v_cuml_RDay_Time = v_cuml_RDay_Time + a_NCRDay_time
        v_Avg_RDay_Time_per_NC = v_cuml_RDay_Time /

v_total_NC_Complete_with_RDay

//When all NCs finished, mark the total time (in hours) & Stop

Simulation.

If v_total_NC_Complete_with_RDay = 1152 then
{
v_RDay_Time_to_Complete = clock(hr)
Stop
}
```

* Arrivals *

Entity	Location	Qty	Each	First Time	Occurrences	Frequency	Logic

New_Cadet	Thayer	36	0 min	32	m	Interval min	

* Attributes *

ID	Type	Classification

a_malefemale	Integer	Entity
a_redsashpass	Integer	Entity
a_knowledge_test	Integer	Entity
a_NCRDay_time	Real	Entity
a_central_area_time	Real	Entity
a_RS_Report_Complete_Time	Real	Entity
a_RS_queue_wait_time	Real	Entity

* Variables (global) *

ID	Type	Initial value	Stats

v_NCs_in_System	Integer	0	Time Series
v_total_NC_male	Integer	0	Time Series
v_total_NC_female	Integer	0	Time Series

```
*****
*                                     *
*                               Macros                               *
*                                     *
*****
```

 * User Distributions *

```
*****  
*                               *  
Streams                        *  
*****
```

Stream #	Seed #	Reset

1	20	No



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB K QUIZZES



UNITED STATES MILITARY ACADEMY
WEST POINT.

QUIZ #1

Instructor Solution

EM481 QUIZ #1

(Show all work)

EM481 AT26-1
QUIZ #1 (30 Points)

Name _____
Hour _____

SOLUTION

1. This quiz is a 30-minute exam worth 30 points. It consists of two problems.
2. Write your name on each page. Disassemble the exam, as necessary. Be sure to reassemble the exam, cover page on top, when finished.
3. Authorized References: pencil/pen, calculator, EM481 Reference Sheet (handed out with exam). **You may not write on the EM481 Reference Sheet.**
4. Show all your work. Unsupported work earns little or no partial credit. Neatness will also help your cause.
5. Turn in your exam IAW the instructor's directions.
6. Grading Summary:

Section	Possible Points	Score
1: Simulation Basics	6	
2: Queuing Theory	24	
Total	30	

Section #1 Simulation Basics (6 points)

Consider the cadet barbershop as a system. In this processing system, cadets arrive during business hours, wait in line until a barber is available, receive a haircut and depart.

1) (2 points) List 2 factors that drive system complexity.

A) Variability

B) Interdependencies

2) (4 points) Listed below are four random variables. Indicate whether the random variable is discrete or continuous by placing an X in the appropriate column.

Variable	Discrete	Continuous
The number of haircuts completed in a day	X	
The time it takes a barber to cut a cadet's hair		X
The temperature inside the barbershop throughout the day		X
The amount of cadets in line at the barbershop	X	

Section #2 Queuing Theory (24 points)

On average, cadets arrive at the barbershop at a rate of 6 per hour according to a Poisson process. Due to limited funding, there is only one barber available to cut hair. This barber takes approximately 5 minutes for each haircut, exponentially distributed.

- 3) (1 point) Express this system using Kendall's Notation.

ANS: M | M | 1

- 4) (4 points) What are the arrival and service rates (in minutes)?

$$\lambda = \frac{6 \text{ cadets}}{1 \text{ hr}} \cdot \frac{1 \text{ hr}}{60 \text{ min}} = .1 \frac{\text{cadet}}{\text{min}} \quad \frac{1}{\mu} = \frac{5 \text{ min}}{\text{cadet}} : \mu = \frac{1 \text{ cadet}}{5 \text{ min}} = .2$$

Arrival: .1 cadet/min

Service: .2 cadet/min

- 5) (2 points) Calculate the barbershop's utilization. Will this system reach a steady state? Why or why not?

$$\rho = \frac{\lambda}{\mu} = \frac{.1}{.2} = .5$$

ANS: $\rho = .5$ yes; b/c $\rho < 1$

- 6) (4 points) What is the expected waiting time for cadets in the line at the barbershop?

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)} = \frac{.1}{.2(.2 - .1)} = 5$$

ANS: 5 min

- 7) (4 points) What is the probability that there is one cadet at the barbershop?

$$P_n = (1 - \rho) \rho^n$$

$$P_1 = (1 - .5) .5^1 = .25$$

ANS: 25%

For questions 8 and 9, the barbershop has hired one additional barber to support the increased rate of cadet arrivals.

- 8) (1 point) In addition to hiring one additional barber, the arrival rate is now determined to be 12 cadets per hour with a standard deviation of 3 minutes (normally distributed). The service rate remained the same. What would the Kendall's Notation be for the system?

$$\lambda = \frac{12 \text{ cadet}}{\text{hr}}$$

ANS: G|M|2

- 9) (8 points) Determine the total wait time in the system for cadets. (W)

$$W = W_q + \frac{1}{M}$$

$$\lambda = \frac{12 \text{ cadet}}{1 \text{ hr}} \bigg| \frac{1 \text{ hr}}{60 \text{ min}} = .2 \frac{\text{cadet}}{\text{min}}$$

$$W_q = \left(\frac{C_a^2 + C_s^2}{2} \right) \left(\frac{\rho \sqrt{2(s+1)} - 1}{s(1-\rho)} \right) \left(\frac{1}{M} \right)$$

$$M = .2 \text{ cadet/min}$$

$$C_a = \lambda \cdot \sigma_a = \frac{12 \text{ cadet}}{1 \text{ hr}} \bigg| \frac{1 \text{ hr}}{60 \text{ min}} = .2 (3 \text{ min}) = .6$$

$$C_s = 1$$

$$\rho = \frac{\lambda}{M \cdot s} = \frac{.2}{.2 \times 2} = .5$$

$$W_q = \left(\frac{.6^2 + 1^2}{2} \right) \left(\frac{.5 \sqrt{2(2+1)} - 1}{2(1-.5)} \right) \left(\frac{1}{.2} \right)$$

$$W_q = .68 \times .366 \times 5$$

$$W_q = 1.245 \text{ min}$$

$$W = W_q + \frac{1}{M} = 1.245 + \frac{1}{.2}$$

ANS: 6.245 min

$$W = \underline{\underline{6.245 \text{ min}}}$$



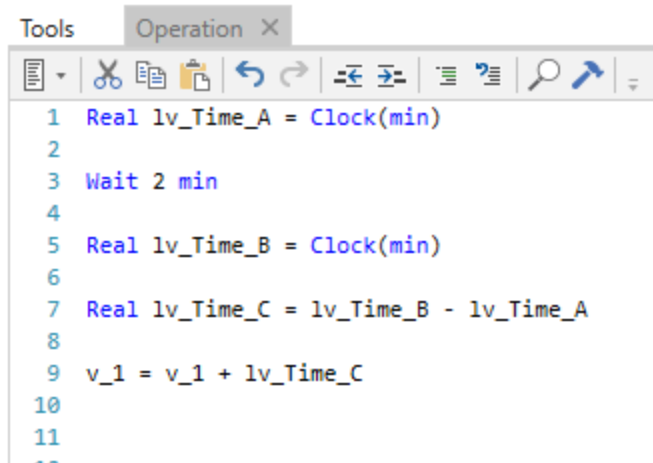
QUIZ #2

Instructor Solution

1

Multiple Choice 3 points Operation Logic

Use the following code to answer Questions 1-3:



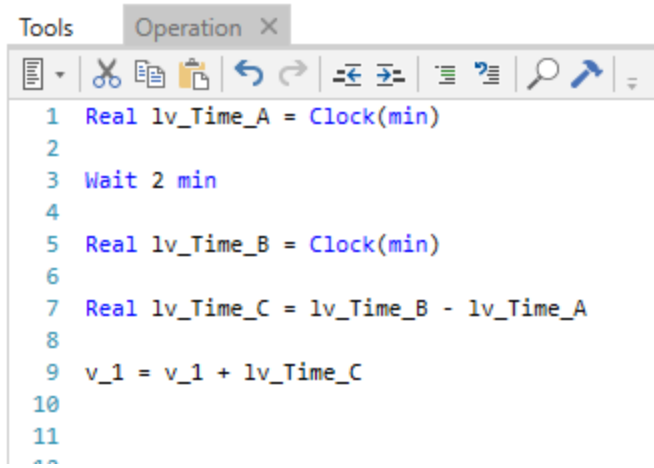
```
Tools Operation X
1 Real lv_Time_A = Clock(min)
2
3 Wait 2 min
4
5 Real lv_Time_B = Clock(min)
6
7 Real lv_Time_C = lv_Time_B - lv_Time_A
8
9 v_1 = v_1 + lv_Time_C
10
11
12
```

What is lv_Time_B?

- ☐ A global variable that captures total quantity in the system
- ☐ A local variable that captures quantity in the system
- ☐ A global variable that captures times
- ☒ A local variable that captures times

Use the following code to answer Questions 1-3:

Tools Operation X

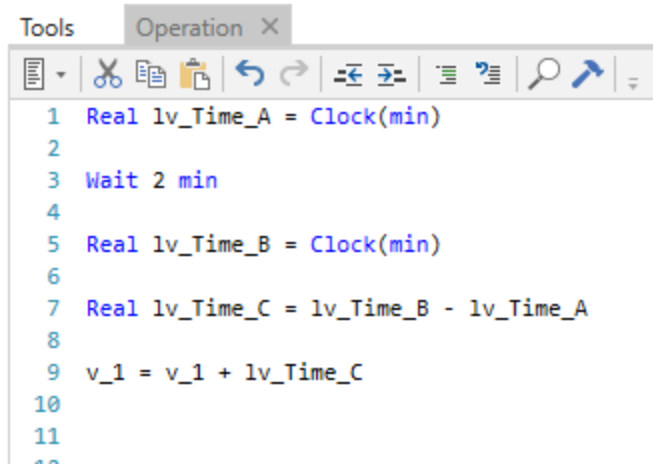


```
1 Real lv_Time_A = Clock(min)
2
3 Wait 2 min
4
5 Real lv_Time_B = Clock(min)
6
7 Real lv_Time_C = lv_Time_B - lv_Time_A
8
9 v_1 = v_1 + lv_Time_C
10
11
12
```

What is lv_Time_C collecting?

- ☐ How many entities are in the system
- ☐ The time the entity enters this location
- ☒ The total time the entity spent waiting at this location
- ☐ The time the entity completes the wait time at this location

Use the following code to answer Questions 1-3:



```
1 Real lv_Time_A = Clock(min)
2
3 Wait 2 min
4
5 Real lv_Time_B = Clock(min)
6
7 Real lv_Time_C = lv_Time_B - lv_Time_A
8
9 v_1 = v_1 + lv_Time_C
10
11
12
```

If the initial value of v_1 is 8 min, what will the value of V_1 be after one entity flows through this location?

- ☐ 4.22
- ☐ 6
- ☐ 8
- ☒ 10

Use the following Attribute Table, User Distribution Table, and Operation Logic to answer Questions 4-6:

ID	Type	Classification	Notes...
a_customer_type	Integer	Ent	

ID	Type	Cumulative	Table...
u_customer_type	Discrete	No	Defined

Percentage	Value
25	1
15	2
60	3


```
1 a_customer_type = u_customer_type()
2
3 If a_customer_type = 1 Then
4 {
5     Graphic 1
6     Route 1
7 }
8 Else If a_customer_type = 2 Then
9 {
10    Graphic 2
11    Route 1
12 }
13 Else
14 {
15    Graphic 3
16    Route 2
17 }
```

What percentage of entities will become Graphic 2?

- ☐ 2
- ☒ 15
- ☐ 25
- ☐ 60

Use the following Attribute Table, User Distribution Table, and Operation Logic to answer Questions 4-6

Attributes X

ID	Type	Classification	Notes...
a_customer_type	Integer	Ent	

User Distributions X

Layout Table X

ID	Type	Cumulative	Table...
u_customer_type	Discrete	No	Defined

Percentage	Value
25	1
15	2
60	3

Tools Operation X

```
1 a_customer_type = u_customer_type()
2
3 If a_customer_type = 1 Then
4 {
5     Graphic 1
6     Route 1
7 }
8 Else If a_customer_type = 2 Then
9 {
10    Graphic 2
11    Route 1
12 }
13 Else
14 {
15    Graphic 3
16    Route 2
17 }
```

Where will entities with an attribute of "3" be routed?

- ☐ Route 1
- ☒ Route 2
- ☐ Route 3
- ☐ Route 4

Use the following Attribute Table, User Distribution Table, and Operation Logic to answer Questions 4-6:

ID	Type	Classification	Notes...
a_customer_type	Integer	Ent	

ID	Type	Cumulative	Table...
u_customer_type	Discrete	No	Defined

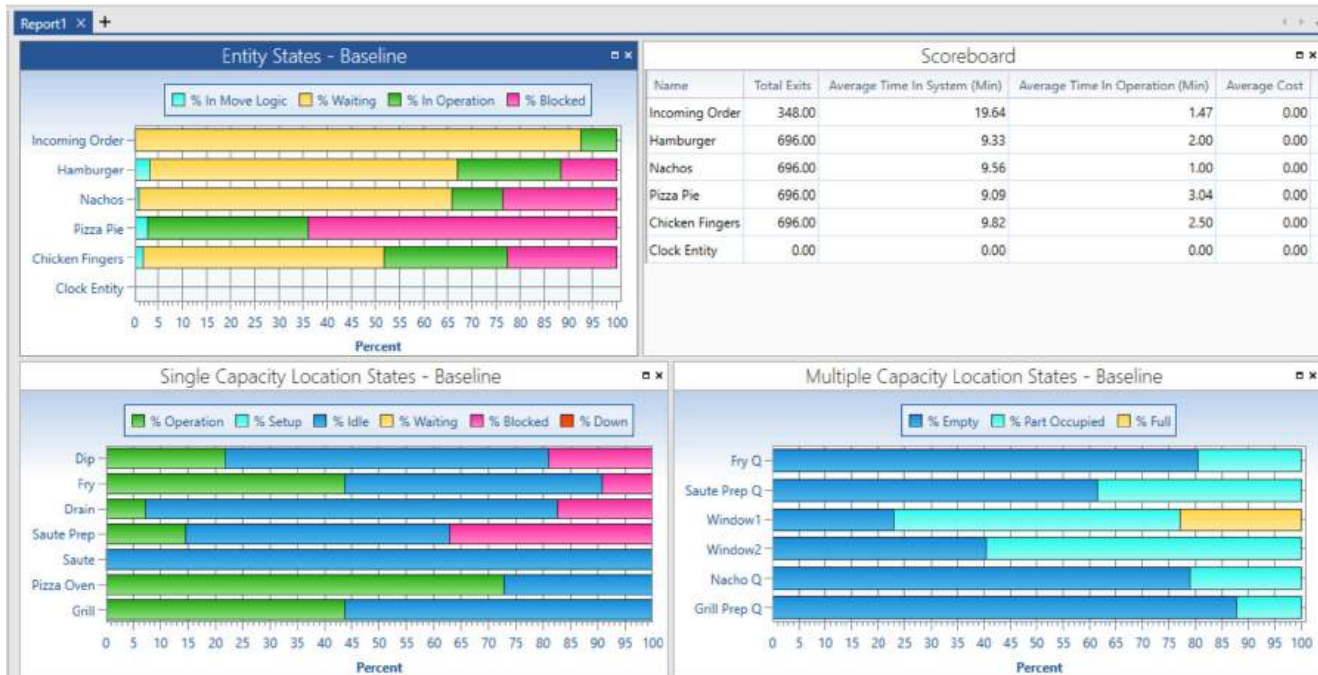
Percentage	Value
25	1
15	2
60	3


```
1 a_customer_type = u_customer_type()
2
3 If a_customer_type = 1 Then
4 {
5     Graphic 1
6     Route 1
7 }
8 Else If a_customer_type = 2 Then
9 {
10    Graphic 2
11    Route 1
12 }
13 Else
14 {
15    Graphic 3
16    Route 2
17 }
```

Which Graphic number will an entry with an attribute of "3" be assigned?

- ☐ Will not be assigned an attribute
- ☐ Graphic 1
- ☐ Graphic 2
- ☒ Graphic 3

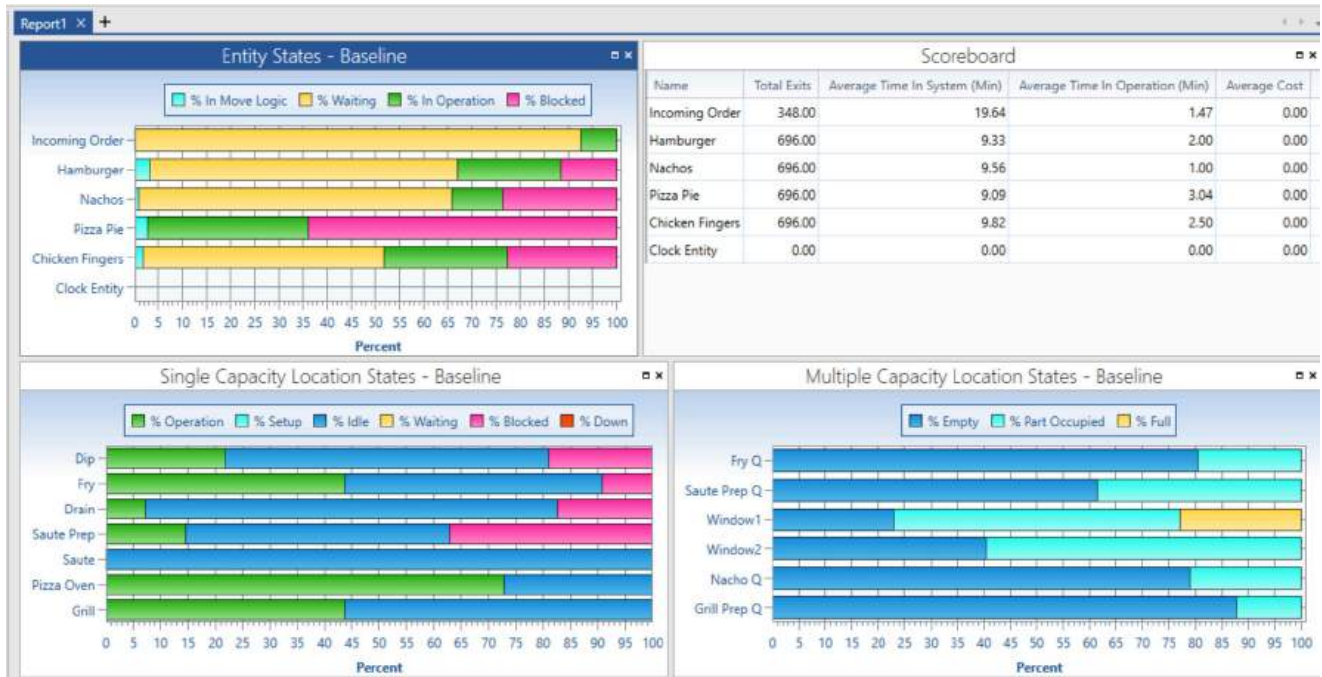
Use the following Output Viewer results to answer Questions 7-10:



Which entity spends the most time in operation?

- ☐ Hamburger
- ☐ Nachos
- ☒ Pizza Pie
- ☐ Chicken Fingers

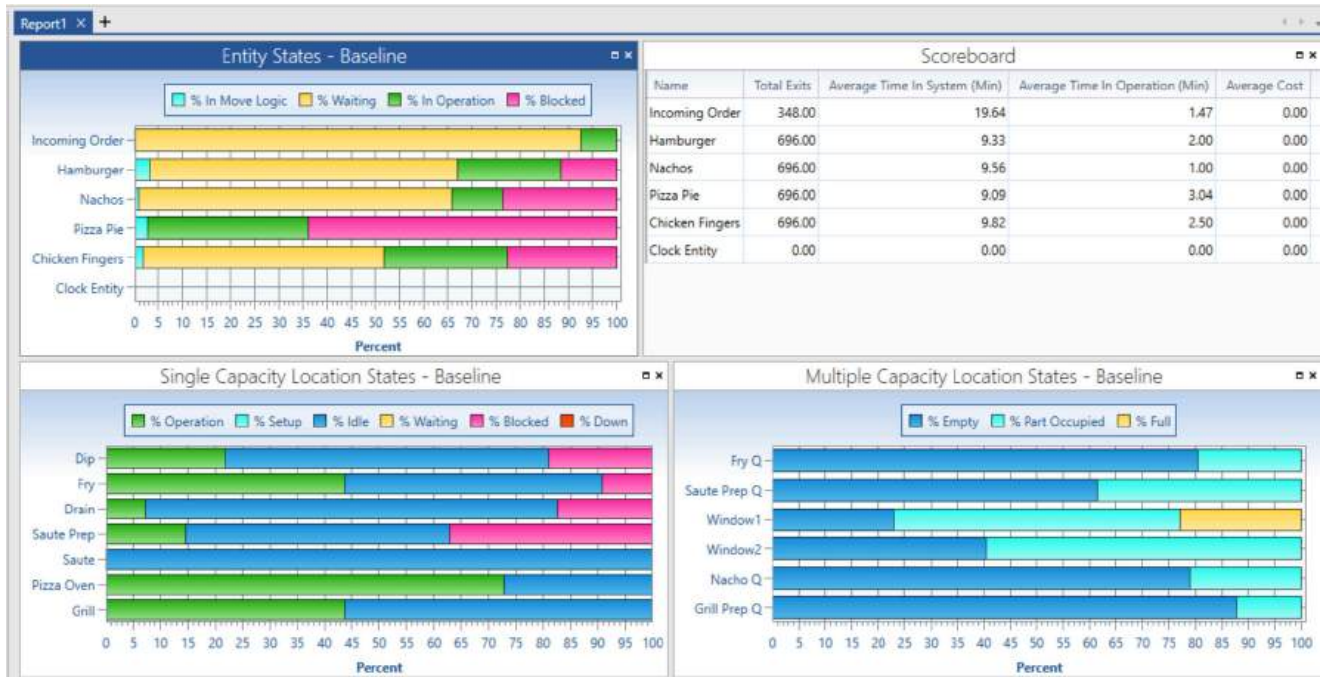
Use the following Output Viewer results to answer Questions 7-10:



On average, which food spends the most time in the system?

- ☐ Hamburger
- ☐ Nachos
- ☐ Pizza Pie
- ☒ Chicken Fingers

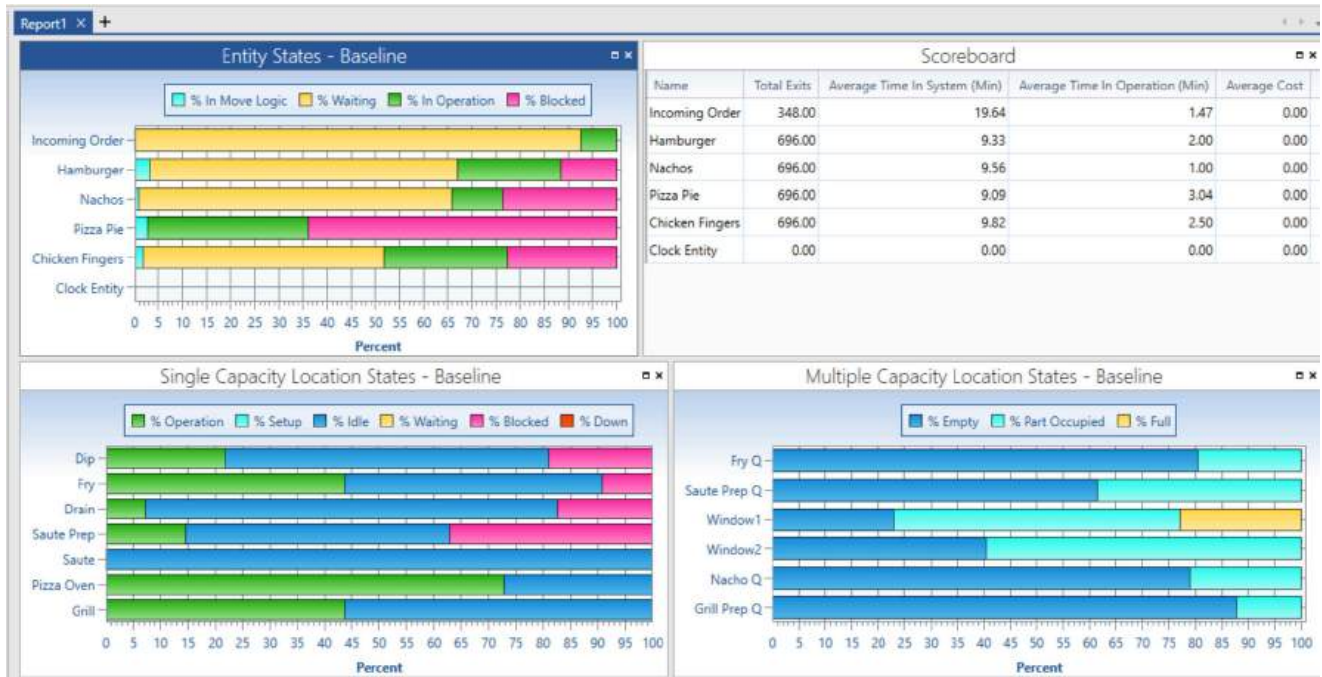
Use the following Output Viewer results to answer Questions 7-10:



Which single capacity location is the least busy?

- ☐ Dip
☐ Fry
☒ Saute
☐ Pizza Oven

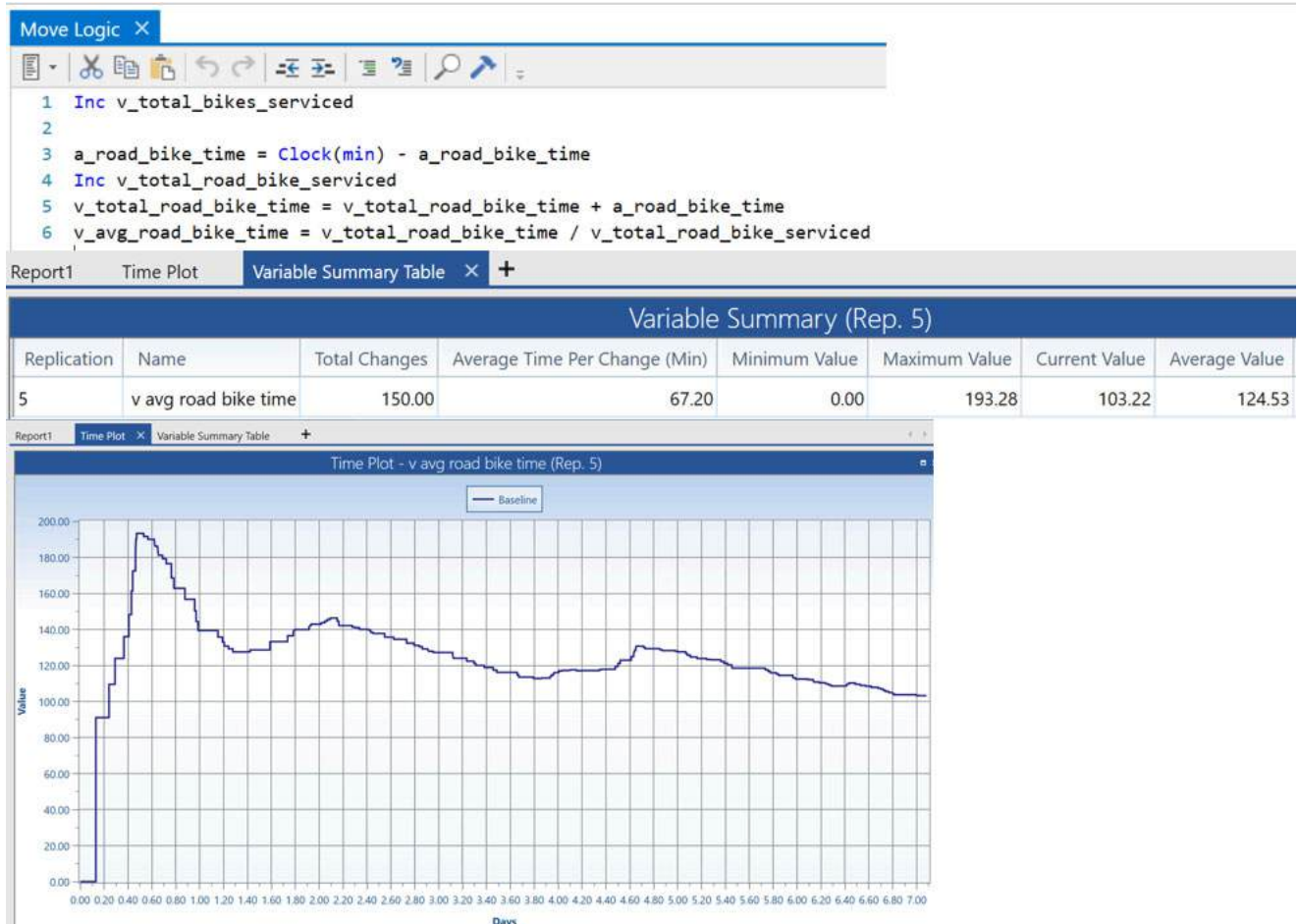
Use the following Output Viewer results to answer Questions 7-10:



Which multiple capacity location is experiencing the most backlog?

- ☐ Fry Q
- ☐ Saute Prep Q
- ☒ Window 1
- ☐ Nacho Q

Use the following Output Viewer results to answer Question 11:



On average, how long did a road bike spend in the system (calculated using v_avg_road_bike_time)?

- ☐ 67.2
- ☐ 193.28
- ☒ 103.22
- ☐ 124.53
- ☐ 150



UNITED STATES MILITARY ACADEMY
WEST POINT.

QUIZ #3

Instructor Solution

1

Multiple Choice 1.5 points

What command is used to permanently attach a specific quantity of one entity type onto a different entity type?

- ☐ Combine
- ☐ Group
- ☒ Join
- ☐ Load

2

Multiple Choice 1.5 points

What command is used to temporarily attach a specific quantity of one entity type onto a different entity type?

- ☐ Combine
- ☐ Group
- ☐ Join
- ☒ Load

3

True or False 2 points

Entities that are **COMBINED** together still retain their individual attributes and identities for use later on in the model.

☐ True

☒ False

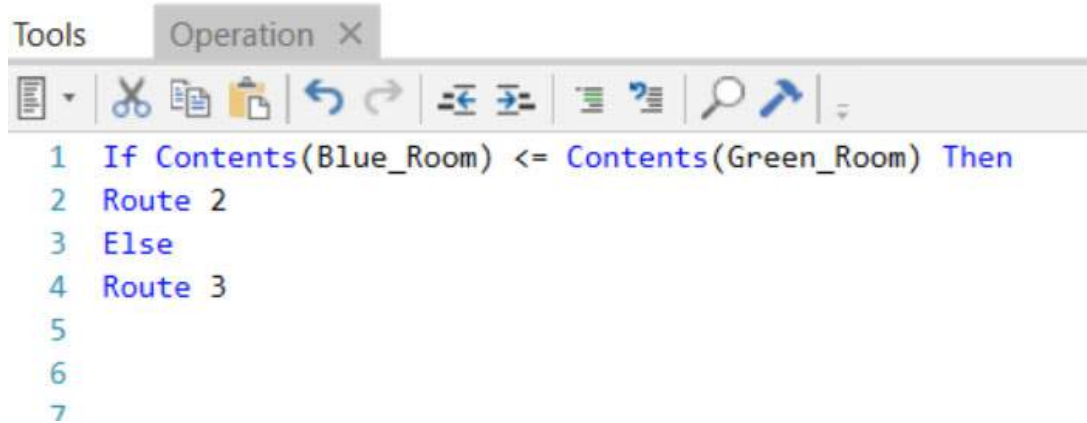
4

True or False 2 points

JOIN is a permanent attaching command and requires an If Join Request.

☒ True

☐ False



The screenshot shows a code editor window titled "Operation X". The toolbar includes icons for undo, redo, find, and other editing functions. The code is as follows:

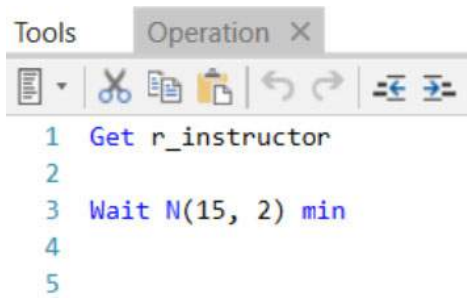
```
1 If Contents(Blue_Room) <= Contents(Green_Room) Then
2 Route 2
3 Else
4 Route 3
5
6
7
```

There are currently 14 Soldiers in the blue room and 12 Soldiers in the green room. Where will the next Soldier be routed?

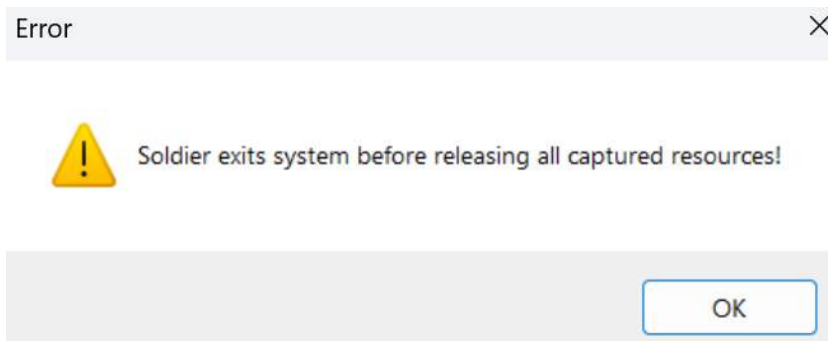
- ☐ Exit
- ☐ Route 2
- ☒ Route 3
- ☐ Soldier will not be routed out


```
16 Wait 30 min
17
18 Int lv_Score = 67
19
20 While lv_Score < 100
21 Do
22 {
23     Wait 30 min
24     lv_Score = lv_Score + 10
25 }
```

What is the total time that this Soldier must 'wait' in order to complete all lines of code shown? (Provide Answer in Minutes).



The logic above created the following error message when you ran the simulation.



What line of code must be added to line 5 to resolve the error and allow the model to run properly?

- ☐ Unload Soldier
- ☐ Unload r_instructor
- ☐ Free Solider
- ☒ Free r_instructor
- ☐ Ungroup Soldier
- ☐ Ungroup r_instructor

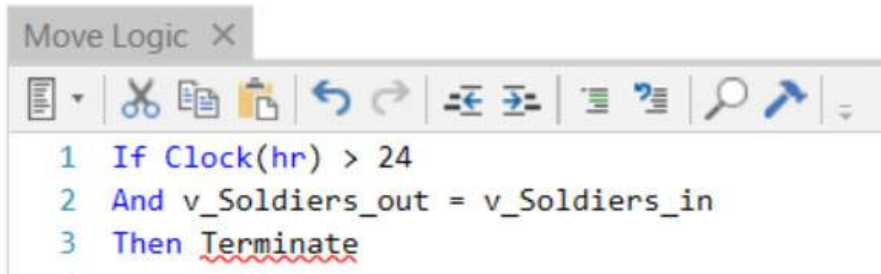
```
14 Int v_attempts = 3
15
16 While v_attempts > 0
17 Do
18 {
19     Dec v_attempts
20     Wait 30 min
21     Display "Current number of attempts is: " $ v_attempts $ " Try Again"
22 }
```

What message will pop up the first time this code is run?

- ☐ 2
- ☒ Current number of attempts is: 2 Try Again
- ☐ Current number of attempts is: 3 Try Again
- ☐ The simulation won't run and an error message will pop up

9

Multiple Choice 2 points



You're attempting to use termination logic at the end of your model. You type the code above, but get an error. What logic should replace line 3?

- ☐ Then End
- ☐ Else End
- ☒ Then Stop
- ☐ Else Stop
- ☐ Then Pause
- ☐ Else Pause

10

Multiple Choice 2 points

In ProModel, where do you go to set the number of Replications and Run Time for your model?

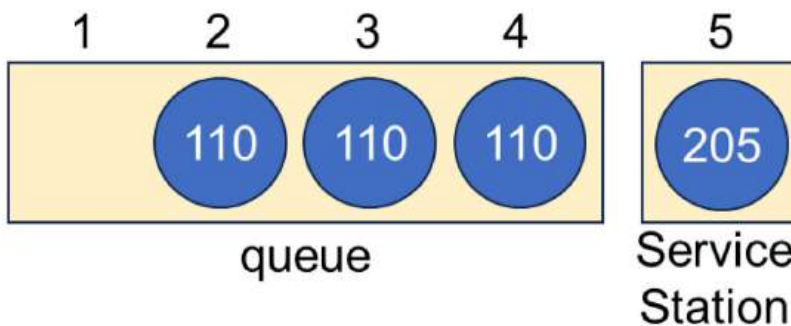
- ☐ Arrivals
- ☐ Processing Logic
- ☐ Scenario Manager
- ☒ Simulation Options

11 True or False 2 points

The **Order** command will bring entities from one location into another location.

- ☐ True
- ☒ False

12 Multiple Choice 3 points



You have a simple queuing system, consisting of a service station (Capacity: 1 / Units: 1) and a single queue (Capacity: INF). The entities (**in blue**) have their priority numbers shown (**white text**). A new entity enters the queue with a priority number of **225**. Where does this new entity end up?

- ☐ End of the Line/Queue (Position 1)
- ☒ In front of the entities in the queue (Position 4), and will enter the service station as soon as the "205" entity departs the service station.
- ☐ In the service station (Position 5) with the "205" entity, where both entities will be worked on.
- ☐ It will enter the service station, bumping out the current "205" entity.